RapidResponse

DEMAND PLANNING

Application Deployment Guide

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Kinaxis

RapidResponse®

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About Kinaxis

Kinaxis is a leading provider of cloud-based subscription software that enables our customers to improve and accelerate analysis and decision-making across their supply chain operations. The supply chain planning and analytics capabilities of our product, RapidResponse, create the foundation for managing multiple, interconnected supply chain management processes. By using the single RapidResponse product instead of combining individual disparate software solutions, our customers gain visibility across their supply chains, can respond quickly to changing conditions, and ultimately realize significant operating efficiencies.

Leaders across multiple industry verticals, including Aerospace & Defense, Automotive, High Tech, Industrial and Life Sciences, rely on RapidResponse for concurrent planning, continuous performance monitoring, and coordinated responses to plan variances across multiple areas of the business.

RapidResponse's configurable service solutions encompass a full spectrum of supply chain related business processes, including such functions as: S&OP, supply planning, capacity planning, demand planning, inventory management, MPS and order fulfillment. As a result, Kinaxis customers have replaced disparate planning and performance management tools and are realizing significant operations performance breakthroughs in planning cycles, supply chain response times and decision accuracy. From a single product, customers are able to easily model varying supply chain conditions to make both long-term and real-time demand and supply balancing decisions quickly, collaboratively, and in line with the shared business objectives of multiple stakeholders.

For more information, visit http://www.kinaxis.com or the Supply Chain Expert community at http://community.kinaxis.com.

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Document Revision History

Version	Changes	Author	Date
1	Initital Roll over from ASD to ADG. Includes updates for 2016.2 Event Planning.	Kerry Currier	12/20/16
1.1	Added changes to profile variables for RapidStart	Carmen Humplik	5/13/21
1.2	Updated the Closed Loop section	Carmen Humplik	28/10/21

Chapter 1

Introduction

1. Introduction

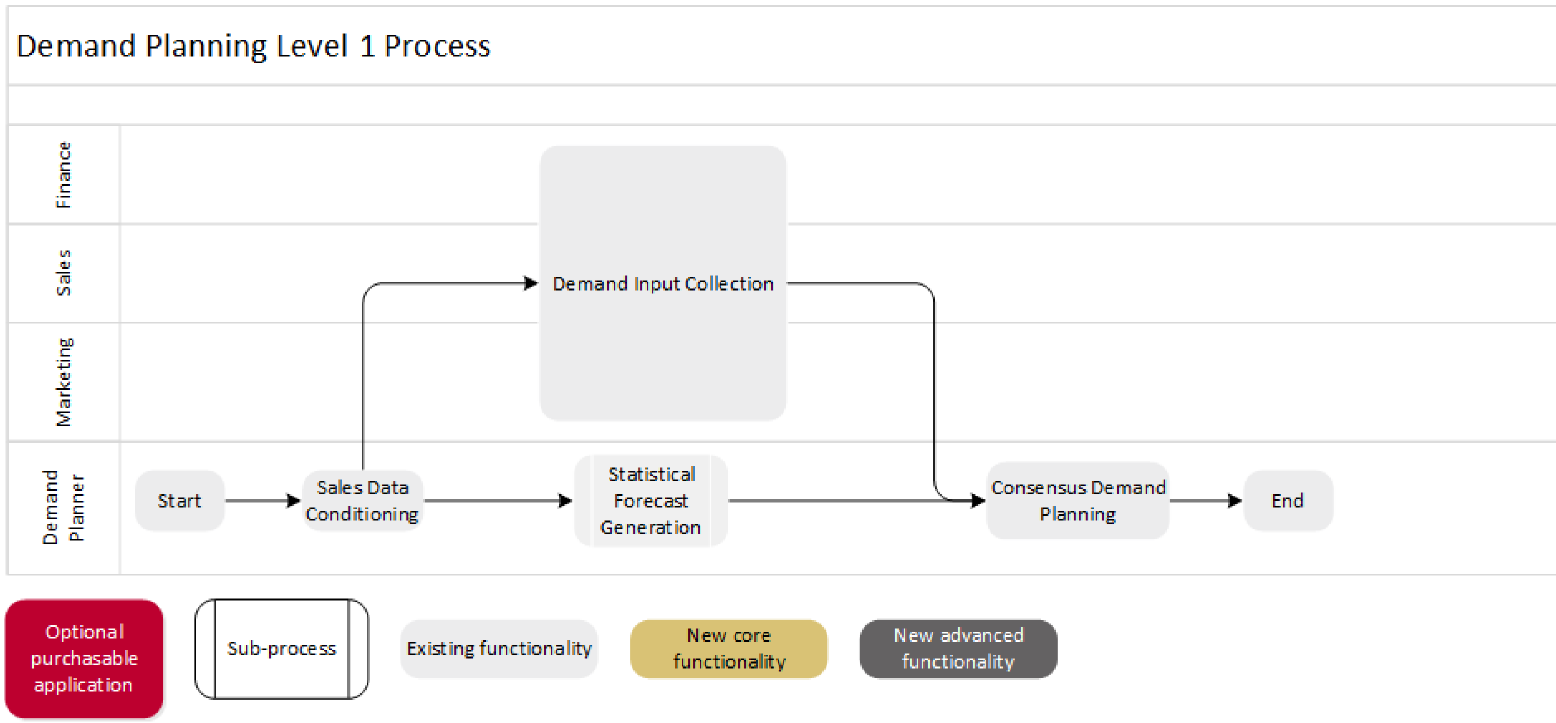
The Demand Planning Application is outlined below. If this application is purchased, the S&OP application will have also been purchased. The scope of this application includes the following processes and sub-processes: Sales Data Conditioning, Demand Input Collection, Consensus Demand Planning, and Adjust Demand Planning Parameters. The dashboards mentioned in this application are specific to the roles included in the processes for it and are also discussed as part of the S&OP application.

Chapter 2

Business Processes

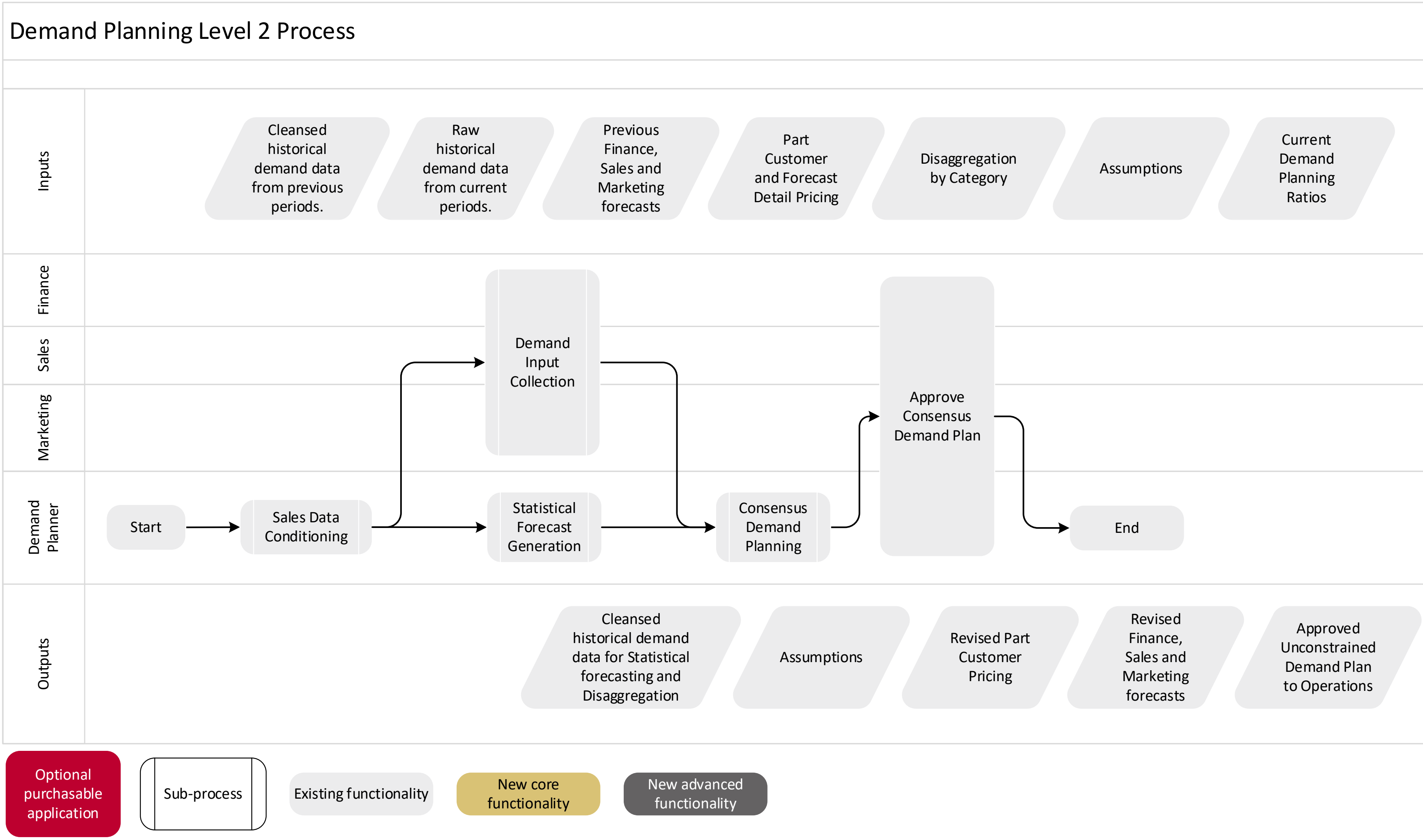
2. Business Process

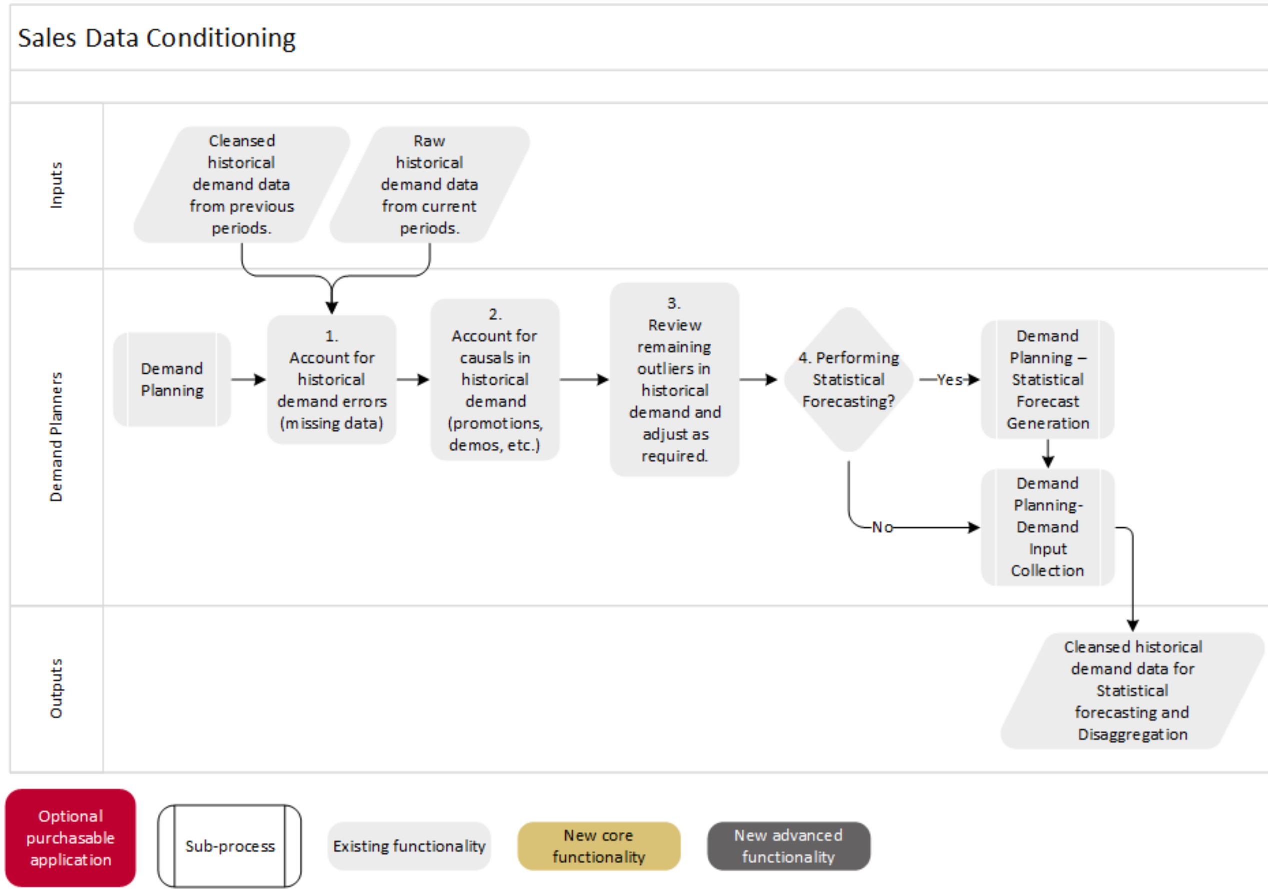
2.1. Business Process Flow Diagram



The overall Demand Planning process flow shown above includes various roles and functions. The "Sales Data Conditioning", "Demand Input Collection", and "Consensus Demand Planning" sub-processes are expressed in more detail below.

1. It should also be noted that, while the "Sales Data Conditioning" sub-process is not directly contributing to the development of the Consensus Forecast except through the creation of the statistical forecast, it is still an important input to the "Demand Input Collection" sub-process. It is the conditioned sales data that is used to define the default disaggregation of the various demand streams affected by the "Demand Input Collection" sub-process.


Demand Planning Level 2 Process
Inputs
Finance
Sales
Marketing
Demand Planner
Outputs
Cleansed historical demand data from previous periods.
Raw historical demand data from current periods.
Previous Finance, Sales and Marketing forecasts
Part Customer and Forecast Detail Pricing
Disaggregation by Category
Assumptions
Current Demand Planning Ratios
Demand Input Collection
Approve Consensus Demand Plan
Start
Sales Data Conditioning
Statistical Forecast Generation
Consensus Demand Planning
End
Cleansed historical demand data for Statistical forecasting and Disaggregation
Assumptions
Revised Part Customer Pricing
Revised Finance, Sales and Marketing forecasts
Approved Unconstrained Demand Plan to Operations
Optional purchasable application
Sub-process
Existing functionality
New core functionality
New advanced functionality

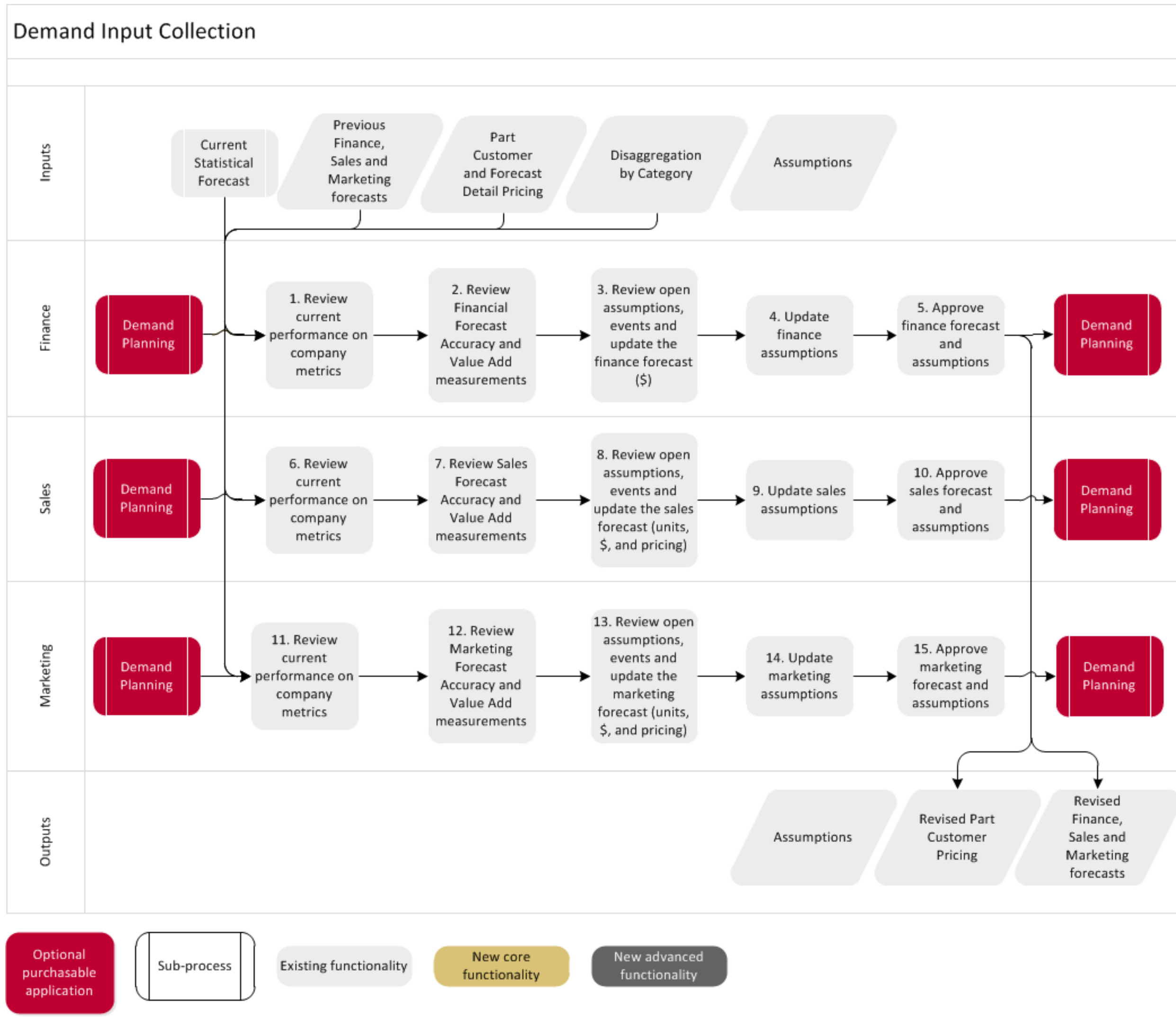


1. **Account for historical demand errors (missing data)**: Using the Demand Planner dashboard review and action Data Errors in the S&OP Data Cleansing workbook

2. **Account for causals in historical demand (promotions, demos, etc.)**: Using the Demand Planner dashboard review and action Causals in the S&OP Data Cleansing workbook

3. **Review remaining outliers in historical demand and adjust as required**: Using the Demand Planner dashboard review and action Outliers in the S&OP Data Cleansing workbook

4. **Performing Statistical Forecasting?:** If yes, move to Statistical Forecast Generation and Product Lifecycle Planning process, otherwise, move to Demand Input Collection process.



5. **Review current performance on company metrics**: Using the Finance dashboard Corporate Metrics tab

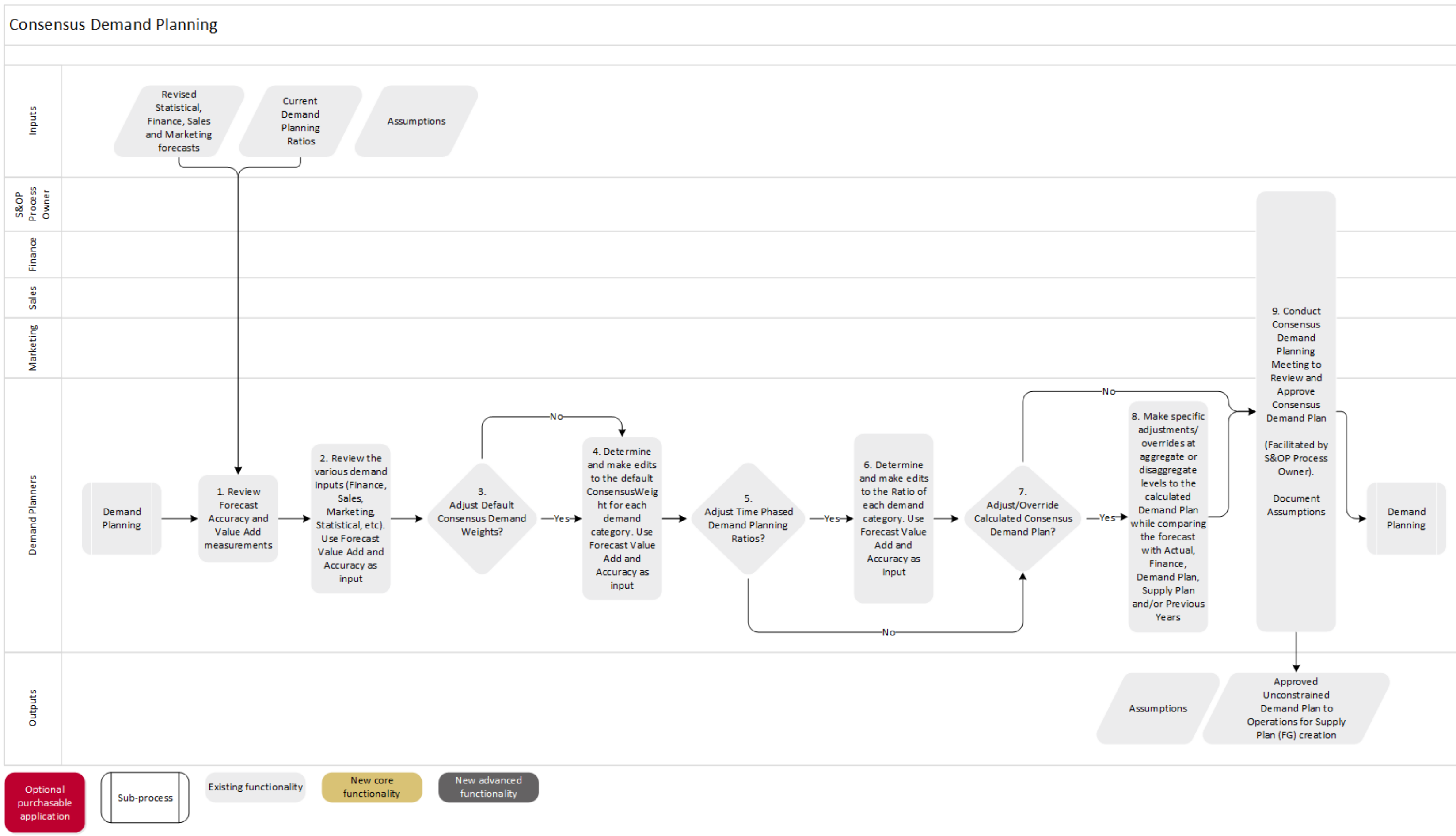
6. **Review Forecast Accuracy and Value Add measurements:** Using the **Finance** dashboard review the Finance Plan Forecast Accuracy (MAPE) widget.

7. **Review open assumptions, event and update the finance forecast ($)**: Using the S&OP Finance Operating Plan workbook

8. **Update finance assumptions**: Using the S&OP Finance Operating Plan workbook

- Existing assumptions can be edited at this point using the 'Edit Assumptions' icon
- Create any new assumptions by using the 'New Assumptions' icon found in the workbook toolbar

9. **Approve finance forecast and assumptions**: Using the S&OP Finance Operating Plan and S&OP Assumptions workbooks.
10. **Review current performance on company metrics**: Using the Sales dashboard Corporate Metrics tab
11. **Review Sales Forecast Accuracy and Value Add measurements:** Using the **Sales** dashboard review the Sales Forecast Accuracy (MAPE) widget.
12. **Review open assumptions, event and update the sales forecast (units, $, and pricing):** Using the **S&OP Sales Forecast** workbook. To update pricing information view the Forecast Summary worksheet in Revenue (not units) and drill to the Unit Pricing worksheet from the Proposed Plan row to override the effective unit price for a given record.
13. **Update sales assumptions**: Using the S&OP Sales Forecast workbook.
    - Existing assumptions can be edited at this point using the 'Edit Assumptions' icon
    - Create any new assumptions by using the 'New Assumptions' icon found in the workbook toolbar
14. **Approve sales forecast and assumptions**: Using the S&OP Sales Forecast and S&OP Assumptions workbooks.
15. **Review current performance on company metrics**: Using the Marketing dashboard Corporate Metrics tab
16. **Review Marketing Forecast Accuracy and Value Add measurements:** Using the **Marketing** dashboard review the Marketing Forecast Accuracy (MAPE) widget.
17. **Review open assumptions, event and update the marketing forecast (units, $, and pricing):** Using the **S&OP Marketing Forecast** workbook. To update pricing information view the Forecast Summary worksheet in Revenue (not units) and drill to the Unit Pricing worksheet from the Proposed Plan row to override the effective unit price for a given record.
18. **Update marketing assumptions**: Using the S&OP Marketing Forecast workbook.
    - Existing assumptions can be edited at this point using the 'Edit Assumptions' icon
    - Create any new assumptions by using the 'New Assumptions' icon found in the workbook toolbar
19. **Approve marketing forecast and assumptions**: Using the S&OP Marketing Forecast and S&OP Assumptions workbooks.



1. **Review Forecast Accuracy and Value Add measurements**: Using the Demand Planner dashboard link to S&OP Forecast Accuracy workbook from the Forecast Value Add widget.

2. **Review the various demand inputs (Finance, Sales, Marketing, Statistical, etc)**. Use Forecast Value Add and Accuracy as input: Using the Demand Planner dashboard review the Consensus Demand Plan widget. Note: If the optional Event Planning is configured, it will be part of the demand inputs and should be reviewed.

3. **Adjust Default Consensus Demand Weights?**: Using the S&OP Demand Planning Ratios workbook, Edit Default Ratio worksheet.

4. Determine and make edits to the default ConsensusWeight for each demand category. **Use Forecast Value Add and Accuracy as input:** Using the S&OP Demand Planning Ratios workbook. These adjustments are made to the category PartCustomer records.

5. **Adjust Time Phased Demand Planning Ratios?**: Using the S&OP Demand Planning Ratios workbook.

6. Determine and make edits to the Ratio of each demand category. **Use Forecast Value Add and Accuracy as input:**: Using the S&OP Demand Planning Ratios workbook, Edit Ratio by Period worksheet. These adjustments are made to the category PartCustomer records by date.

7. **Adjust/Override Calculated Consensus Demand Plan?**: Using the S&OP Consensus Demand Planning workbook.

8. Make specific adjustments/overrides at aggregate or disaggregate levels to the calculated Demand Plan while comparing the forecast with Actual, Finance, Demand Plan, Supply Plan and/or Previous Years: Using the S&OP Consensus Demand Planning and the S&OP Assumptions workbooks.

9. Conduct Consensus Demand Planning Meeting to Review and Approve Consensus Demand Plan (Facilitated by S&OP Process Owner). Document Assumptions.: Using the S&OP Consensus Demand Planning workbook click on ‘assumptions’ button to review and update the assumptions. Using the Demand Planner scorecard evaluate the impact of the proposed consensus demand plans on the demand planner metrics.

2.1.1. Inputs

The inputs for the Demand Input Collection sub-process include:

- Previous Finance forecasts
- Previous Sales forecasts
- Previous Marketing forecasts
- Part Customer and Forecast Detail Pricing
- Disaggregation by Category
- Cleansed Historical Actual data

The inputs for the Consensus Demand Planning sub-process include:

- Revised Statistical forecasts
- Revised Finance forecasts
- Revised Sales forecasts
- Revised Marketing forecasts
- Current Demand Planning Ratios

Out of the box a number of default forecast categories are available as profile variables which need to be linked to valid HistoricalDemandCategory values; however, the PDRs referenced in the processes above utilize only a subset of these variables: DP_BudgetForecastCategory, DP_MarketingForecastCategory, DP_SalesForecastCategory and DP_DemandPlanCategory. Refer to section Profile variables for the complete list.

2.1.2. Outputs

The Annual Plan targets are compared to each forecast category being evaluated when determining the final consensus forecast in the Consensus Demand Planning. It is also reported as the target "Annual Plan" in the various role based dashboards.

While there are no predefined alerts defined to automatically report on deviation from the above targets, they may be easily setup.

The outputs of the Sales Data Conditioning sub-process include:

- Cleansed historical actual data records

The outputs of the Demand Input Collection sub-process include:

- Revised Part Customer Pricing
- Disaggregation by Category
- Revised Finance forecasts
- Revised Sales forecasts
- Revised Marketing forecasts

The outputs of the Consensus Demand Planning sub-process include:

- Revised Demand Planning ratios
- Revised Demand Planning parameters
- Unconstrained Demand Plan

Chapter 3

Integration Requirements

3. Integration Requirements

3.1. Data and Integration Requirements

3.1.1. Database Scale

It is imperative that the consultant and customer both understand the database scale that is being created. This typically needs to be a concern when integrating large amounts of historical data or performing heavy disaggregation operations, both are items in the S&OP and Demand Planning Applications, but that does not mean it should not be thought about and considered when deploying other Applications.

For example, forecasting will create a series of detailed ForecastDetail records for eligible PartCustomer records from a very high-level summary perspective. The fact that we are forecasting at a high level does not mean that we are not creating these detailed records and the scope of this can be easily missed. If you are forecasting for 200K parts with an average of 200 customers each, then you are forecasting 40M PartCustomers. Further, if your DisaggregationCalendar is set to weekly and you are forecasting for 2 years, then this will result in 40M * 104 or greater than 4 billion ForecastDetail records. This can be supported in RapidResponse; however, the performance would be degraded.

With the introduction of vectors, tables that use this data type, don't have the same record limitations. For example, the HistoricalDemandSeriesDetail table is a vector on the HistoricalDemandSeries table. Each HistoricalDemandSeries record can have 4B HistoricalDemandSeriesDetails in its vector, so there is essentially no practical limit to the number of detail records contained in a vector table.

As of release 2016.2, RapidResponse supports the concept of *large* tables, which can hold over 4B records. This change will allow >4B records in all tables except control tables and the Site table.

With this concept of a *large* table, RapidResponse will use 64-bit addressing, instead of 32-bit addressing. Only tables that need to be *large* will switch to this model, since it requires more memory to store the address pointers. There is nothing for the user or any administrator to do, as this conversion will be automatic, when needed.

Although this *large table support* is available and automatic inside RapidResponse, care must be exercised as this limit is approached. As a *small* table grows in size (i.e., over time), once that table exceeds ~3B records, an internal flag is turned on and the table will be converted to a large table the next time RapidResponse is restarted. (This threshold was chosen to be far enough away from the 4B 'small table' record limit that a large surge of records would not be very likely. Exceeding 4B records while a table is still small would cause disruption to RapidResponse.) During an initial data load, if there is a chance that more than 4B records need to be loaded into a table, this process may need to be executed in multiple steps, being sure to restart RapidResponse after the table has exceeded 3.2B records, but before it exceeds 4B. Alternatively, if you need to start a table out as a 'large' table, contact Support and they will be able to force the internal flag prior to reaching the threshold.

3.1.2. Detailed Data Requirements

List all the important input tables that are required to support the application and point out mandatory or key fields that are necessary to populate for successful implementation of the application. Each table begins with a new 'Heading 4'.

3.1.2.1. AggregatePartCustomer

This table was added in 2014.4 SU2 and can be used to group part customers together to define the level at which forecast disaggregation occurs. Forecasts are generally created for part customers, however in some cases it might not be necessary to generate a forecast to such a detailed level. For example, a part's forecast might be made at an aggregate level for all customers within a given sales region, or for all parts belonging to a given product family. The AggregatePartCustomer table can be used to support generating forecasts at higher levels, and is used to link aggregate PartCustomer records with actual part and customer combinations that are also defined in the PartCustomer table. In cases where is it not necessary to disaggregate forecast values to the detailed part and customer level, this can result in fewer records being generated and improved database performance.

Generally, to model this situation, aggregate customer grouping values (eg. customer regions) should be added to the Customer table (or Part table, if grouping Parts into aggregate groups). Next, appropriate PartCustomer records should be added to identify the new levels at which the Part(s) should be aggregated. Finally, the AggregatePartCustomer records can be added to associate the detailed or component part-customer combinations with the aggregate part-customer combinations for which the forecast should be generated.

Fields with significance:

- **Aggregate:** Reference to an aggregate PartCustomer record
- **Component:** Reference to part and customer combination being aggregated under the PartCustomer value referenced in the Aggregate field
- Setting Aggregate=Component (rather than setting it to null) is the way to indicate that the component is no longer aggregated.
- **EffectiveInDate:** Date when the part customer aggregation defined on the record becomes effective
- **Type:** Reference to the control setting used to include or exclude aggregate part customers from disaggregation calculations.
    - Note that in order for a given AggregatePartCustomer value to be used in calculating forecast disaggregation rates, it must reference an AggregatePartCustomerType record that has a **ProcessingRule** set to **"Use".**

3.1.2.2. BillOfMaterial

If forecast and consumption is happening across "planning BOMs", then the MPSConfig type BillOfMaterial records must be defined.

3.1.2.3. CurveParameters

The table containing paramenters required for generarting the curves to be used for Event Planning adjustment type *Curve*

- **Type** – Enum Field
- LinearCurve
- ExponentialCurve
- DiffusionCurve
- **InitialValue -** The initial point on the curve for all types of curves, default 0.
- **MaximumValue -** Only relevant to DiffusionCurve, as time progresses the curve will be asymptotically reaching this value. Defaults to InitialValue if not specified or if lower than InitialValue.
- **LinearRate -** The slope of the Linear curve
- **ExponentialRate -** The rate of the exponent, relevant to ExponentialCurve. The rate is not allowed to be negative. If specified negative then automatically corrected to be zero.
- **InnovationRate -** Only relevant to DiffusionCurve: The proportion of innovators in the consumers' population (the ones buying new products for the first time without depending on the prior adoptions of other consumers), default 0. If not specified or equals zero and the resulting diffusion curve will be flat and equeal to the Initial Value. Valid range of values is [0.0, 1.0]. In the case user specifies values outside the range, they are forced by the calculation to be at least 0.0 and at most 1.0.
- **ImitationRate -** Only relevant to DiffusionCurve: The proportion of imitators in the consumers' population (the ones buying new products for the first time only if other consumers already bought the product), default 0. Valid range of values is [0.0, 1.0]. In the case user specifies values outside the range, they are forced by the calculation to be at least 0.0 and at most 1.0.

3.1.2.4. Customer

The base table for the Demand Planning process is the PartCustomer table. As such, customers must be defined as well. While we really don't need to know much about the customers, at least one aspect of hierarchies is usually associated with customer characteristics. As such, custom fields or references from either Customer or CustomerGroup are often required.

It should be noted here that summary records in the PartCustomer table are typically handled by creating a summary or decorated Customer Id. These special customers are usually maintained in RapidResponse and not imported.

Also note that Customer has a Site which is normally setup as a key reference. This may be disabled but is disabled for the entire database then. That is, if you setup the Customer table to be not site-specific, then ALL Customer records MUST reference the blank Site. Otherwise, none of the Customer records should reference the blank Site. It's all-or-nothing.

It is recommended that this table be set to not allow data update to delete records (found in the table properties of the data model) in order to avoid unintentional cascading deletes in the PartCustomer, and Historical tables.

3.1.2.5. CustomerPrice

If prices are defined specific to customers for a part and/or time-phased, then this information must be provided in the CustomerPrice table. Note that if the price is not to be associated with a specific customer (should be applied to all customers) but is still time-phased, then the customer reference on this table should be blank.

3.1.2.6. ForecastDetail

This table holds current forecast data at the detail level (after any disaggregation calculations have been performed). Each record pertains to a given part, customer, and category combination, and shows details such as the forecast quantity and date. Different types of forecasts can be stored in this table such as the statistical forecast, sales forecast, marketing forecast, and so on.

The values shown in this table are based on aggregate values entered or maintained through various resources in RapidResponse. For example, details of the statistical forecast are based on values generated by the Save Forecast command in the S&OP Statistical Forecast workbook. Details of other types of forecast are based on values entered in workbooks belonging to the related constituent group (for example, values might be provided through the S&OP Marketing Forecast workbook, S&OP Sales Forecast workbook, S&OP Finance Operating Plan workbook, and so on).

The ForecastDetail table is also used to store targets when the **Header.Category.Type.ProcessingRule** is set to “Target”.

In 2014.4, this table contains vector data (if configured). It contains vector set data for records in the HistoricalDemandHeader table.

Unlike the HistoricalDemandActual table, this table is keyed and we expect records to be created, modified and deleted. We need to be able to uniquely identify records. The three keys defined are:

- **Header**: This is the key reference to the HistoricalDemandHeader which defines the Part/Customer/Category combination.
- **Id**: This is a base key string that uniquely identifies this record within the set of records referencing the same header. The value is not relevant but must be unique within the header. As such, it is generally set to automatically generate a unique value on record creation.
- **Date**: The date of this forecast item.

There are two money fields that are normally expected to be provided in the currency of the Header.PartCustomer.Part.Site.Currency reference. The other (non-key) fields that need to be populated on this table include:

- **Quantity**: The amount of this forecast. This is a Quantity field (obviously). For Target records, this is the field that will be used if the **Header.Category.Type.UnitType** is “Quantity”.
- **Value**: The monetary value associated with this forecast. For Target records, this is the field that will be used if the **Header.Category.Type.UnitType** is “Value”.
- **UnitPrice**: Allows for a unique unit price to be specified for this forecast order. For example, this might be used to reflect the price effective during a sales promotion.
  If a non-negative value is provided here, it is always reported in the EffectiveUnitPrice field (typically used for revenue calculations). If a negative value is provided here, the unit price is calculated based on matching records in the CustomerPrice or Part tables.
- **ProtectQuantity**: Indicates whether the value in the Quantity field is modified when data is edited in a grouped worksheet. This field is used in the Advanced Data Editing dialog box as part of the expression that determines which records are not edited when a grouped worksheet is edited. Valid values are:
  - Y: the Quantity cannot be modified in summarized worksheets.
  - N: the Quantity can be modified.

3.1.2.7. Event

New table added for Event Planning to group event phases.

- **Name -** Event Name
- **CreationDate** - Date when Event was created (optional, for user's reference)
- **Description** - Event description
- **Type** - Reference to EventType should be user extendable
- **PlannerCode** - Reference to PlannerCode

3.1.2.8. Event Type

New table added for Event Planning. Used to capture the various promotion tactics or event types.

- **Value** - String value i.e. TradePromotion, ConsumerPromotion
- **Description** - User description

3.1.2.9. EventPhase

New table added for Event Planning used to make adjustments to a forecast.

Event Phase is a single item that can adjust a forecast (increase/decrease forecast by a certain number or percentage, or apply a more complex curve adjustment, within a certain date range). In general, a single EventPhase may be applied to several forecast streams grouped according to hierarchy chosen by the user.

- **Name** – The name of the event phase
- **Event** - Reference to Event table
- AdjustmentType – Enum field used to make adjustments. The options are as follows:
    - Quantity
    - UnitPrice
- **CreationDate** - Date when EventPhase was created (optional, for user's reference)
- **Description** - Event Phase description
- **Category** - Reference to HistoricalDemandCategory. This field is nullable.
    - If the field is non-null, the EventPhase applies to HD Headers (via associated EventPhaseHeaders) whose category matches the category referenced by this field.
    - If the field is null, then the EventPhase applies to all associated HD Headers (grouped by HD Category).
- **Calendar** - Reference to Calendar.
    - Calendar to be used when applying adjustment specified in this EventPhase, its input parameters, such as Quantity, Percent and CurveParameters, are specified in terms of this Calendar. Input data is rebucketed to this calendar (if necessary) before adjustment is applied.
- **StartDate** - Date (inclusive) used in the calculation of FirstEffectiveDate of this EventPhase
- **EndDate** - Date (exclusive) used in the calculation of LastEffectiveDate of this EventPhase
- **ActionType** - Enum field used to define adjustedments to be made to the forecast. The options are as follows:

- **Quantity (default):** Add a fixed amount to the Forecast Quantity or Unit Price
- **CompoundedQuantity:** Add a compounding amount (based on compounding calendar) to the Forecast Quantity or Unit Price
- **Percentage:** Add a fixed percentage to the Forecast Quantity or Unit Price
- **CompoundedPercentage:** Add a compounding percentage (based on compounding calendar) to the Forecast Quantity or Unit Price
- **Constant:** Make the Forecast Quantity or Unit Price equal to a constant value for the duration of the phase
- **Curve**: Apply a more complex curve to the forecast (linear, exponential or diffusion)

- **Quantity** - Amount to adjust Forecast. Applicable to ActionTypes Quantity, CompoundedQuantity and Constant.
- **Percent Quantity** - Percent to adjust by, scaling 1 to 100% (applicable to ActionTypes Percentage and CompoundedPercentage.
- **CompoundingCalendar** - Reference to Calendar
- Calendar used to determine compounding intervals (within compounding buckets). Default is Everyday (if an invalid calendar is used (example: single date calendar), it will revert to default)
- **CurveParameters** - Refrerence to CurveParameters (null by default). Applicable to ActionType Curve
- **CompoundingIntervalCount** - Integer Number of intervals in a single compounding bucket. Default: 1. If a negative value specified, the logic will use 1 as well.
- **UsageRule** - Use or Ignore this phase when calculationg Event Adjustments
- Use (default)
- Ignore
- **FirstEffectiveDate** - Calculated first effective date for the EventPhase (inclusive), converted to the Calendar specified by this record.
- **LastEffectiveDate** - Calculated last effective date for the EventPhase (inclusive), converted to the Calendar specified by this record.

3.1.2.10. EventPhaseHeader

New table added for Event Planning used to make link events and Demand headers. An event is considered to be applied to a header is there is an esxisiting record linking it to a phase.

- **EventPhase** – Reference to EventPhase table
- **Header** – Reference to HistoricalDemandHeader
- **Id** – String field relevant only if Attribute support is required. It’s recommended to be left blank if ABP is not being used. Note: ABP is not supported by the OOTB resources.

3.1.2.11. HistoricalDemandActual

These records are date-effective records associated with a particular HistoricalDemandHeader and they represent some form of historical demand data. One category of these records is the normal source of information for history in order to calculate the statistical forecast. The same or another category may be used to disaggregate input when there is no pre-existing data.

In 2014.4, this table contains vector data (if configured). HistoricalDemandActual records are stored as a set of vector data on the HistoricalDemandActual table. Also note that the currency of these records must be defined through the Header record and cannot be defined on the specific HistoricalDemandActual record.

- **Line**: This is an optional string field that may be populated if required by the customer for reporting. It represents the order line of the sales order and it is not required for calculating on time delivery as it is assumed the historical demand actual records are already the line level detail.
- **LineCreatedDate**: This is an optional date field that may be populated if required by the customer for reporting. It represents the date the sales order line was created and can be used for calculating sales order lead times if required.
- **Date**: Key field when converted to vector table. This is the primary date field effective for this record. It is the date associated with this actual shipment. The date used should be consistent with the historical demand series dates. For example, if the historical demand series dates represent the dates that parts ship from the dock, then this date should be a dock departure date.
- **CommitDate**: This is the date the order is committed to ship. It is currently information-only but is available for determining some metrics.
- **RequestDate**: This is the date the customer requested the order to ship. It is currently information-only but is available for determining some metrics.
- **Quantity**: This is the number of parts in this actual shipment.
- **Line Quantity**: This is an optional field that may be populated if required by the customer for reporting and it represents the sales order line quantity for the associated sales order line, it does not always equal the Quantity value on the historical demand actual record.
- **UnitPrice**: The unit price for the historical demand. This value can be used to calculate revenue associated with historical actual demand.
- **UnitCost**: The unit cost for the historical demand. This value can be used in calculating margins where the margin per unit is calculated as the difference between UnitPrice and UnitCost.
- **Route**: This is an optional string field that may be populated if required by the customer for reporting and represents the route the order took during shipping.
- **Shipset**: This is a string field indicating if the record was part of a shipset and is used in calculating order level on time delivery.
- **Order**: This is a reference field to the HistoricalDemandOrder table described in the previous section.

Note: If no HistoricalDemandActual records are included in the data, the Forecast Value Add widgets need to be removed from the S&OP Dashboards as they will not render any data. This widget is found on the Monitoring tab of each dashboard.

3.1.2.12. HistoricalDemandCategory

The categories of demand must be defined here. There are a standard set of categories defined with a variety of types, which would be defined in the previous section. Different types have differing **AggregationRule**, **ProcessingRule** and **UnitType** settings. It is the **ProcessingRule** that typically defines the purpose of the category. It can be set to:

- “None”: Use this to ignore standard Target categories that you are not using. This will keep them out of workbooks and drop-down lists.
- “Actual”: This is used for storing historical demand actuals. Values in this category can be used in calculating the statistical forecast.
- “Target”: This is used for defining target metric values against which the S&OP annual plan is measured.
- “Forecast”: This is used for storing streams of forecast demands or forecast adjustments. Values in this category can be used in calculating the consensus forecast.
- “ForecastOverride”: This is used for specifying values to override the calculated consensus forecast.
- “ReBalancingForecastOverride”: This is used for specifying values to override the calculated consensus forecast based on demand and supply balancing.

Only the “Actual”, “Forecast” and two override rules are of interest to Demand Planning.

The HistoricalDemandCategory also defines a default **ConsensusForecastWeight** number. This field must be set to the default weight that this forecast category should contribute to the calculated consensus forecast. It should be a number from 0 to 100%. Zero means that, by default, this category does not contribute to the calculated consensus forecast. Specific PartCustomer/Category values can override this in the HistoricalDemandHeader.

First, you define the categories in the Historical Demand Category worksheet of the Control Tables workbook. You need to define all your required categories for actuals and forecasts including the forecast overrides under the “Value” column.

Make sure that you have set the Type appropriately for each one in order to get the appropriate “Processing Rule”, “Aggregation Rule” and “Unit Type”.

Also, be sure to set the Consensus Forecast Weight appropriately for each of the “Forecast” and “Forecast Override” types. In the SampleMS, the consensus forecast is driven by the Statistical forecast combined with the ForecastAdjustments and FcstOverride forecast streams. Since the Statistical and ForecastAdjustments have **Type.ProcessingRule** of ”Forecast”, these values are added together unless there is a FcstOverride record for the same date. The FcstOverride has a **Type.ProcessingRule** of “ForecastOverride” which means that when one exists on a date, the Statistical and ForecastAdjustments will be ignored and the ForecastOverride will be used. You can see this in the Control Tables by looking at the Historical Demand Category worksheet. Also, bear in mind that these are the default consensus forecast weights. This can be overridden. See the section on “Consensus Demand Planning”.

Note: If “Forecast Override” types will be used, the Forecast Override column must be un-hidden in the S&OP Consensus Demand Planning workbook.

3.1.2.13. HistoricalDemandHeader

This represents a combination of each PartCustomer and Category required. In fact, the keys are the references to PartCustomer and to Category (HistoricalDemandCategory).

Again, though, there is a hidden BaseKeyfield. If you can even see this LEAVE IT BLANK! ALWAYS!!!

The one non-key field on this record is the ConsensusForecastWeight field. This is the override value for the HistoricalDemandCategory.ConsensusForecastWeight. If this is set to anything other than -1.0, then the category value has been overridden by this. This field should be defaulted to -1.0 and only overridden for specific headers (Part/Customer/Category). Note that this field indicates the override for the header for all dates (past to future). If a specific date range for the override is required, then use the HistoricalDemandHeaderTimePhasedAttributes table instead.

As of 2014.4, two fields support vector data sets, and if this table has been configured to contain vector data, the following is applicable:

- **Actuals**: Vector Set data type; represents the HistoricalDemandActual records associated with the historical demand header
- **ForecastDetails**: Vector Set data type; represents the ForecastDetail records associated with the historical demand header
- Note that currency for money fields on the vectors MUST be defined through the header record and never directly on the Actuals or ForecastDetails vectors themselves.

3.1.2.14. HistoricalDemandHeaderTimePhasedAttributes

The purpose of this table is to hold date-effective overrides for the ConsensusForecastWeight for specific Part/Customer/Category combinations (HistoricalDemandHeader). There are two keys on this table along with 3 data fields. Note that this table is generally populated by the “S&OP Demand Planning Ratios” workbook. Also note that, generally, a record is created for every DisaggregationCalendar (pre-2014.4 was $DP_BaseCalender) in the worksheet bucket.

- **Header**: This is the key reference to the HistoricalDemandHeader that we are providing date-effective overrides for.
- **Id**: This is a base key string that uniquely identifies this record within the set of records referencing the same header. The value is not relevant but must be unique within the header. As such, it is generally set to automatically generate a unique value on record creation.
- **EffectiveInDate**: The date on which this records ConsensusForecastWeight value becomes effective. A string version of this might be a good choice for the Id field if you are importing these records rather than just maintaining them in the “S&OP Demand Planning Ratios” worbook.
- **EffectiveOutDate**: The on which this records ConsensusForecastWeight value is no longer effective.
- **ConsensusForecastWeight**: This is the final override value and is expected to be somewhere from 0.0 to 1.0. Zero is a legitimate override to say that this header will not produce consensus demand for this period. If you don’t want to see the override any longer, then delete the record(s).

3.1.2.15. HistoricalDemandOrder

This is a table introduced in the Solutions Namespace in 2014.2 and contains the following fields. The purpose of this table is to capture Order information for calculating Order level on time delivery metrics.

- **Id:** String field containing the sales order number or order id typically stored in the IndependentDemand table.
- **Site:** Reference to Site table to indicate what site the order belongs to.
- **CreatedDate:** Date field for capturing the date the order was created in the ERP system.

3.1.2.16. IndependentDemand and DemandOrder

New to 2015.3. Prior to 2015.3, the **IndependentDemand.RequestDate** input field, which is intended to hold the date on which a customer originally requested *completion and shipment* of the order, was included for reporting and analysis purposes and not included in any analytic calculations. As of 2015.3, the **RequestDate** field is used to support the new two-date planning analytic, and is now included in the prioritized list of criteria used to sort demands during the supply allotment process. On upgrade, if the **RequestDate** had previously been populated on independent demands having the same due date, then slight supply-demand allotments might now be reported. Note that the change in the **RequestDate** definition will only affect how the forecast gets consumed in Demand Planning, along with potential available date calculation when two-date planning is enabled.

It is expected that all (or most) forecast will be generated from the ForecastDetails. However, backlog (current customer orders) that consumes this forecast must be defined in the IndependentDemand table with an **Order.Type.ProcessingRule** of “SalesActual”.

Other demands that should not consume forecast may be loaded here as well with an **Order.Type.ProcessingRule** of “Regular”.

Occasionally, one DemandOrder is required to have lines (IndependentDemand records) that are a mix of backlog that can consume forecast and others that may not. When this is true, define a set of DemandStatus values that can be set on the IndependentDemand records with **ForecastConsumptionOverride** set to either “Normal” or “DoNotConsume”, as required.

If the backlog is past due, then the date where it consumes forecast from can be either the DueDate of the backlog or the DataDate depending on the **PartType.ForecastConsumptionDateRule** and the **Status.ForecastConsumptionDateRule**.

3.1.2.17. OnHand

OnHand is only required if you are doing supply planning.

3.1.2.18. Part

If the parts are going to include supply planning of any sort then all the supply policy fields must also be populated. Reference the Aggregate Supply Planning and Master Production Scheduling applications for those requirements.

- All Part Types loaded for this must have a **ProcessingRule** of either “MPS” or “MPSConfig” in order for the consensus forecast to be generated.
- In order to use any of the currency disaggregation or reporting, it is important to populate the AverageSellingPrice.
- If the part is also going to load backlog (SalesActuals) with forecast consumption of the consensus forecast, then the following fields need to be populated:
    - DemandTimeFence
    - BeforeForecastInterval
    - AfterForecastInterval
    - SpreadForecastInterval
- **PartSolution.Part Class**: This field is used to classify a part. The part classification is used to help determine if the proper planning, safety stock and replenishment strategies are set based on the category of part. It is an enum list field found on the PartSolution table referenced from the Part table. Valid values are: None, FinishedGoods, RawMaterial, WorkInProgress. It is used to classify inventory in the ‘Part Class Value’ widget found on the Inventory Planner dashboard.
- **PartSolution.Part Strategy**: This field is used to identify the manufacturing strategy of a part (eg. Make-To-Stock, Make-To-Order, Assemble-To-Order). The strategy is used to help determine if the proper planning, inventory, sales, and supply chain strategies are set based on the manufacturing strategy of the part. It is a string field found on the PartSolution table referenced from the Part table. If the customer ERP system does not contain this information it is expected that it would be maintained by users directly in RapidResponse.
- **PartSolution.KeyPart**: This is a Boolean field to identify a Part as being a Key or Critical part within a BillofMaterial. Key Parts are focused on in the S&OP analysis as they have pertinent limitations (long lead times, single sourced, etc.) to the long term plan.

It is recommended that this table be set to not allow data update to delete records (found in the table properties of the data model) in order to avoid unintentional cascading deleted in the PartCustomer, and Historical tables.

3.1.2.19. PartCustomer

PartCustomers must be defined for every required Part and Customer combination. This is the base table for the S&OP process. It is not enough to just have the Part and Customer records defined separately.

This applies also to any Parts that are going to be used as part of the **Inventory Management** blueprint within the S&OP Process context. If PartCustomer records do not exist, the **High Inventory Violations** and **Low Inventory Violations** widgets on the **Supply Planner** dashboard will not render data.

There are a few fields on the PartCustomer table with significance.

- **ForecastItem**: This is a specific reference to the ForecastItem that will manage the statistical forecast for this PartCustomer. This is normally maintained by the “Statistical Forecasting Setup” dialog (or equivalent) and is left pointing to the blank ForecastItem on import.
- **BaseKey**: If you can even see this LEAVE IT BLANK! ALWAYS!!!
- **DemandType**: This reference to the DemandType may be left out and set to null. It is nullable. However, if the calculated Consensus Forecast is expected to be spread from the DisaggregationCalendar (pre-2014.4 was DP_BaseCalendar) to something smaller (Month to Week or Day), then you can set this to a DemandType with the SpreadRule set to Spread and a defined SpreadProfile provided.
- **DisaggregationCalendar**: Defines the periods that the forecast quantities can be disaggregated for a part customer. If the field is left null, the disaggregation calendar specified in the SOPAnalyticsConfiguration table is used for disaggregation.
- **Pool**: If forecast consumption by Pool is required, then you can set the Pool for this PartCustomer on this reference. Usually, we expect the Pool to be set to the same value as the Customer.Id. If the Pool is not equal to the Customer and one customer should belong to more than one Pool, then you will need to create new Customer records with the Id decorated to include the Pool. This is because the Pool reference in not a key reference on the PartCustomer table.
- **OrderPriority**: You can override the Part.DefaultPriority for forecasts from this PartCustomer here.
- **MinimumShelfLife**: You can override the Part.MinimumShelfLife for forecasts from this PartCustomer here.

3.1.2.20. PartSource, Source and Supplier

Although supply planning is optional for this process, the Parts should all have valid PartSources defined in order for the consensus forecast to be generated.

As mentioned in the section “Pre-Defined Filters”, the effectivity date range of PartSource records can be significant if the “***NPI Parts***” filter is to be used. In particular, this filter assumes that NPI Part records can be identified by the first effective PartSource (PrimaryPartSource) having an EffectiveInDate value that is on or after the system MRPDate minus 1 EffectiveDisaggregationCalendar.

Beyond effectivity dates, the other PartSource fields are really only of significance to the supply planning process.

3.1.2.21. ScheduledReceipt and SupplyOrder

Scheduled Receipts are only required if you are doing supply planning.

3.1.3. Control Table Configuration

List all the important control tables that are required to support the application and point out mandatory or key settings that are necessary to populate for successful implementation of the application. Each table begins with a new 'Heading 4'. Table and field names should be **Bold** and control table settings should be in double quotes "xxx".

3.1.3.1. AggregatePartCustomerType

Control table added in 2014.4 SU2 to enable or disable records in the AggregatePartCustomer table for processing when determining disaggregation rates.

Fields of significance include:

- **ProcessingRule:** indicates whether AggregatePartCustomer records that belong to this type are used in generating disaggregation rates; can be set to "Ignore" or "Use"

In order for a given AggregatePartCustomer value to be used in calculating forecast disaggregation rates, it must reference an AggregatePartCustomerType record that has a **ProcessingRule** set to **"Use".**

3.1.3.2. AssumptionType and AssumptionStatus

The required Assumption Types (Categories) and Statuses are listed below.

Assumption Types (labeled Category in the dialogs and reports):

- **Finance**: Finance Assumption
- **Marketing**: Marketing Assumption
- **Operations**: Operations Assumption
- **Sales**: Sales Assumption
- **Inventory**: Inventory Assumption

Assumption Statuses:

- Closed
- Open

There is no pre-defined application resource installed that allows you to edit, delete or add these values and they are not auto-created. If you need to add/change them, then you will need to create a worksheet on the AssumptionType and AssumptionStatus tables.

3.1.3.3. BOMType

For MPSConfig parts it is important to have valid BOMTypes where the MPSConfigDemandSource is defined appropriately to ensure that component actual sales consume assembly forecast within a PlanningBOM relationship if required.

3.1.3.4. CausalFactorCategory

This table stores categories defined for grouping causal factors. The **ProcessingRule** field indicates whether the causal factor details associated with the causal factor category should be included in S&OP analytics calculations, inventory planning & optimization (safety stock) calculations, or both. Set to “SOP” to include the causal factor details in S&OP calculations only. If required for safety stock calculations as well, set to “All”.

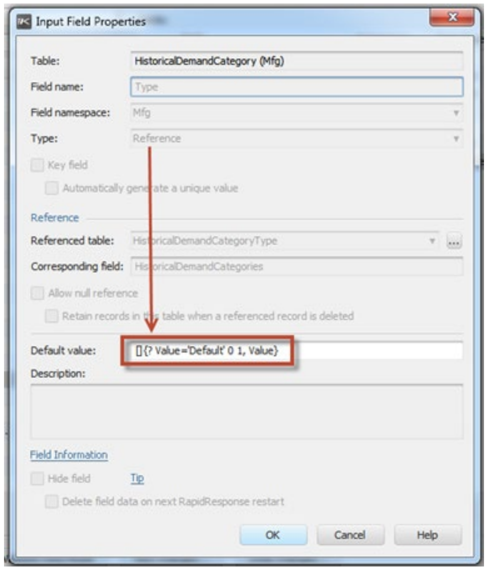
The installed list of Causal Factor Categories is given below. There is no PDR installed, outside of Control Tables, that allows you to edit, delete or add these values however it may be auto-created. If you need to change them, then you will need to create a worksheet on the CausalFactorCategory table.

Causal Factor Categories:

- <blank>
- DemandAdjustment
- Demo/Test
- ExtraCausal
- Outlier
- Promotion

If you do add or change any of these Causal Factor categories, though, then you will need to work them into the “S&OP Data Cleansing” workbook. Look at how the $ExtraCausal variables is used in this workbook to see how this is done.

3.1.3.5. Data Model Defaults



3.1.3.6. DemandType

This table defines values for the types of demands. It's used in the IndependentDemand table to identify the processing rules for different types of demands, and in other demand tables to specify demand-processing behavior. At least one DemandType must be defined, but there is typically one defined for each of customer or sales orders, forecasts, shipments, and dependent demand.

Configuration Notes: If the consensus forecast generated as part of the S&OP planning process will be copied to non-S&OP scenarios (eg. cross-scenario update to Baseline and child scenarios), it is important that the DemandType.ProcessingRule = 'Ignore' in the S&OP Intermediate scenario, in order to avoid driving duplicate demand in the S&OP scenario tree. For additional configuration points related to DemandType and ConsensusForecast, see Configuring for Consensus Forecast and DemandType.

3.1.3.7. Forecast Spreading Parameters

- The PartCustomer record has a DemandType reference (nullable) that can be set to a specific type that can define further spreading of the ConsensusForecast.
- The DemandType control record for forecast record types that need to be further spread will need to have the following fields set:
    - **ProcessingRule** should be set to “SalesForecast”.
    - **SpreadRule** should be set to “Spread” if you want spreading to occur.
    - **SpreadProfile** should be set to reference a defined SpreadProfile.
- If you are spreading, you need to define the appropriate spreading profiles in the SpreadProfile control table. This is a series of 1 to 13 points that describe the shape of the spreading curve.
- For the parts where this spreading will occur, the Part.PlanningCalendar will need to have the ForecastCalendar set to the forecasting (outer) interval for spreading while the SecondCalendar will need to be set to the interval that we spread it to (inner). For more details on the spread logic and options, refer to the RapidResponse Data Model and Analytic guide.

*It should be noted that all calculated ConsensusForecast is deemed to have a **ProcessingRule** of “SalesForecast” regardless of the DemandType referenced in the PartCustomer. That DemandType is only used to determine spreading.*

3.1.3.8. HistoricalDemandCategoryType

This table supports S&OP and is used to define how demand data associated with a given HistoricalDemandCategory is processed.

Note that the **AggregationRule** and **UnitType** fields only function if the **ProcessingRule** is “Target”.

DisaggregationQuantityRule determines whether historical actuals or the consensus forecast should be used to calculate forecast disaggregation rates, and can be set to either “Actuals” or “ConsensusForecast”.

	Value	Number of Historical Demand Categories	Description	Aggregation Rule	Disaggregation Quantity Rule	Processing Rule	Unit Type
1	Actual	5		None	Actuals	Actual	Quantity
2	Default			None	Actuals	None	Quantity
3	Forecast	22		None	Actuals	Forecast	Quantity
4	ForecastOverride	1		None	Actuals	ForecastOverride	Quantity
5	None	38		None	Actuals	None	Quantity
6	ReBalancingForecas	1		None	Actuals	RebalancingForecastOverride	Quantity
7	Target			None	Actuals	Target	Quantity
8	TargetAvg	9		Average	Actuals	Target	Quantity
9	TargetSum	3		Sum	Actuals	Target	Quantity
10	TargetValue	16		Sum	Actuals	Target	Value
11	**Grand Sum**	**95**					

EventAllowNegative is used n calculating the value of the EffectiveAdjustment field for Quantity adjustments (EventStatisticalForecastDetailAdjustment and EventForecastDetailAdjustment tables)

3.1.3.9. PartType

All Parts for which you require consensus forecast to be generated in S&OP, the Part Types loaded must have a **ProcessingRule** of either “MPS” or “MPSConfig” otherwise the consensus forecast cannot be generated.

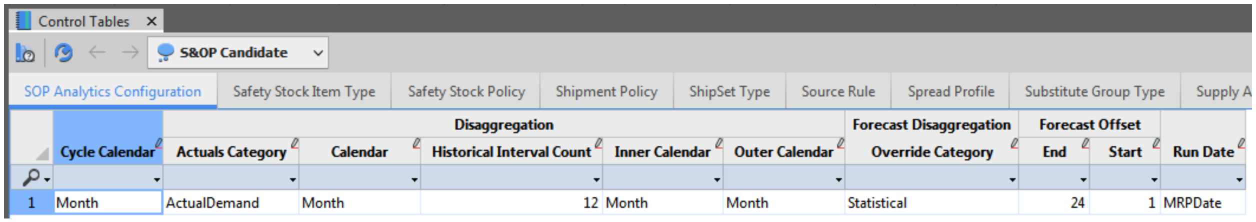
3.1.3.10. SOPAnalyticsConfiguration

This table supports some of the key S&OP and Demand Planning disaggregation and forecasting analytics, and is used to set up S&OP and Demand Planning parameters for disaggregation and statistical forecasting. It was introduced to replace some key statistical forecasting and disaggregation pluggable functions. As a result, some profile variables that were used prior to 2014.4 are no longer used, and are replaced by fields in this table. Note however, that the "old" behavior is still available to existing customers for backward compatibility.

No	Field Name	Replaces Profile Variable	Description
1	RunDate	N/A	Current system date for S&OP calculations
2	CycleCalendar	$DP_CycleForecastCalendar	Represents the calendar reflecting the S&OP cycle (eg. Month)
3	ForecastStartOffset	N/A	Number of CycleCalendar dates from historical end date to start forecast disaggregation
4	ForecastEndOffset	N/A	Number of CycleCalendar dates from forecast start date to stop forecast disaggregation
5	CalcHistoricalEndDate	$DP_HistoricalEndDate	Calculated field representing the date to stop collecting historical data
6	CalcHistoricalStartDate	$DP_ForecastStartDate	Calculated field representing the date to start reporting disaggregation rates
7	CalcForecastEndDate	N/A	Calculated field representing the date to stop reporting disaggregation rates
8	DisaggregationCalendar	$DP_BaseCalendar	Base calendar for storing forecast disaggregation
9	DisaggregationHistorical IntervalCount	$DP_HistoricalIntervalCount	Number of DisaggregationCalendar periods to collect history
10	DisaggregationInnerCalendar	$DP_InnerCalendar	Default inner calendar
11	DisaggregationOuterCalendar	$DP_OuterCalendar	Default outer calendar
Unless there is seasonality in the disaggregation ratios, these two intervals (inner and outer) should be set to the same calendar and should never be smaller than the base calendar. If there is seasonality in the disaggregation ratios, the Inner Calendar should never be larger than the Outer Calendar.			
12	DisaggregationActualsCategory	$DP_DefaultDisaggregation Actualls	Default disaggregation actuals category (eg. HistoricalDemandActuals)

No	Field Name	Replaces Profile Variable	Description
13	ForecastDisaggregation OverrideCategory	N/A	Default category that contains any disaggregation overrides regardless of the category being disaggregated

There is no Control Set reference, therefore there is no corresponding worksheet in the Control Sets workbook, but it can be accessed in the Control Tables workbook.



3.1.4. Configuration Points

3.1.4.1. Profile Variables

List of all profile variables used in the application.

Many profile variables have been used to configure the Demand Planning solution. The general criterion for deciding whether a configuration point would be a profile variable or workbook variables was whether the variable was needed by more than one workbook. If it is yes, then the configuration point became a profile variable. If it is no, then the configuration point became a workbook variable in the associated workbook.

In 2014.4, many profile variables used in disaggregation and statistical forecasting were replaced by fields in a new table, SOPAnalyticsConfiguration, in order to support S&OP analytic disaggregation and forecasting (vs. pluggable functions). Details are in the table below .

- All profile variables have Variable Name beginning with "DP_". This prefix comes from "demand planning".
- The Application Settings workbook has a worksheet, entitled "Application Variables", for managing these profile variables. This workbook can be accessed through Administration pane, System Settings, Application Settings (or through the Explorer).
- Profile variables are used to set target categories. These all have names ending in "Target" and the details are outlined in the S&OP Application.
- Profile variables are used to set demand categories. These all have names ending in either “Category”, “Optimistic” or “Pessimistic” except for one called DP_DemandTypeForConsensus.
- 5 profile variables that **were** used for Disaggregation Settings, have been replaced by fields in the SOPAnalyticsConfiguration table. They have an AttributeType = Disaggregation (see table below). For details on the SOPAnalyticsConfiguration control table, see Section 3.1.3.10 SOPAnalyticsConfiguration.
- Profile variables are used for Other settings. (See table below)

Below is a list of the profile variables used in the Demand Planning Application. All of these are utilized in the workbooks outlined in this application, but may also be used in additional Pre-Defined Resources within RapidResponse. For a complete list of which resources, you can use the **Tools** > **Search Resources** feature to perform a search.

No	Variable Name	Description	Default value
	The following variables are all “Category” names. They all refer to HistoricalDemandCategory record values where specific types of forecast are recorded and managed. The HistoricalDemandCategory values shown in the Default value column require set up on install and should match the default value exactly. Details for how to set up these values can be found in section 3.1.3 Control Table Configuration.		

No	Variable Name	Description	Default value
	The DP_DemandTypeForConsensus category is located in the "Miscellaneous" type section of the Application Variables worksheet. It really is a category and not a DemandType (as it was for release 10.0). It is used in 7 of the S&OP workbooks to select historical versions of the ConsensusForecast captured as snapshots.		
1	DP_ActualsCategory	Actuals Category	Shipment
2	DP_AdjustmentsCategory	Adjustments Category	ForecastAdjustments
3	DP_BaselineForecastCategory	Baseline Forecast Category	Statistical
4	DP_BudgetForecastCategory	Budget Forecast Category	FinanceOperatingPlan
5	DP_BudgetForecastOptimistic	Budget Forecast Optimistic Category	FinanceOperatingPlanOptimistic
6	DP_BudgetForecastPessimistic	Budget Forecast Pessimistic Category	FinanceOperatingPlanPessimistic
7	DP_DemandPlanCategory	Demand Plan Category	Statistical
8	DP_DemandTypeForConsensus (see disclaimer in "Consensus Demand Planning")	Demand Type for Consensus	DemandPlan
9	DP_MarginPercentCategory	Margin Percent Category	ActualDemand
10	DP_MarketingForecastCategory	Marketing Forecast Category	MarketingForecast
11	DP_MarketingForecastOptimistic	Marketing Forecast Optimistic Category	MarketingForecastOptimistic
12	DP_MarketingForecastPessimistic	Marketing Forecast Pessimistic Category	MarketingForecastPessimistic
13	DP_OnTimeDeliveryToCommitCategory	On Time Delivery to Commit Category	ActualDemand
14	DP_OnTimeDeliveryToRequestCategory	On Time Delivery to Request Category	ActualDemand
15	DP_PastDueBacklogCategory	Past Due Backlog Category	PastDue

No	Variable Name	Description	Default value
16	DP_PromotionalAdjustmentsCategory	Promotional Adjustments Category	PromotionalAdjustments
17	DP_SalesForecastCategory	Sales Forecast Category	SalesForecast
18	DP_SalesForecastOptimistic	Sales Forecast Optimistic Category	SalesForecastOptimistic
19	DP_SalesForecastPessimistic	Sales Forecast Pessimistic Category	SalesForecastPessimistic
20	DP_StatisticalForecastCategory	Statistical Forecast Category	Statistical
21	DP_SupplyPlanCategory	Supply Plan Category	SupplyPlan
The following variables are related to Event Planning			
22	DP_BudgetForecastEventCategory	Budget Forecast Event Category	FinanceOperatingPlanEvent
23	DP_BudgetForecastOptimisticEvent	Budget Forecast Optimistic Event Category	FinanceOperatingPlanOptimisticEvent
24	DP_BudgetForecastPessimisticEvent	Budget Forecast Pessimistic Event Category	FinanceOperatingPlanPessimisticEvent
25	DP_ConsensusEventAdjustment	Consensus Event Adjustments Category	EventAdjustment
26	DP_MarketingForecastEventCategory	Marketing Forecast Event Category	MarketingForecastEvent
27	DP_MarketingForecastOptimisticEvent	Marketing Forecast Optimistic Event Category	MarketingForecastOptimisticEvent
28	DP_MarketingForecastPessimisticEvent	Marketing Forecast Pessimistic Event Category	MarketingForecastPessimisticEvent
29	DP_SalesForecastEventCategory	Sales Forecast Event Category	SalesForecastEvent

No	Variable Name	Description	Default value
30	DP_SalesForecastOptimisticEvent	Sales Forecast Optimistic Event Category	SalesForecastOptimisticEvent
31	DP_SalesForecastPessimisticEvent	Sales Forecast Pessimistic Event Category	SalesForecastPessimisticEvent
32	DP_HideEventPlanning	Hide Event Planning	Yes
	The following variables are all related to disaggregation and/or forecasting, BUT WERE REPLACED BY FIELDS ON THE SOPANALYTICSCONFIGURATION TABLE IN 2014.4. They are no longer used in standard application resources, but are still available for backward compatibility. See Section 3.1.3.10 SOPAnalyticsConfiguration for details regarding the SOPAnalyticsConfiguration control table.		
1	DP_BaseCalendar	Base Calendar	Month
2	DP_DefaultDisaggregationActuals	Default Actuals	ActualDemand
3	DP_HistoricalIntervalCount	Historical Interval Count	12
4	DP_InnerCalendar	Inner Calendar	Month
5	DP_OuterCalendar	Outer Calendar	Month
		Unless there is seasonality in the disaggregation ratios, these two intervals (inner and outer) should be set to the same calendar and should never be smaller than the base calendar. If there is seasonality in the disaggregation ratios, the Inner Calendar should never be larger than the Outer Calendar. For more detailed information refer to the "RapidResponse Data Model and Analytic Guide".	
6	DP_CycleForecastCalendar	Cycle Forecast Calendar	Month
7	DP_ForecastStartDate	Forecast Start Date	MRPDate + 1 $DP_CycleForecastCalendar
	The following variables are miscellaneous "Other" S&OP configuration variables. Each needs to be configured.		
1	DP_AssumptionComponentsLimit	Assumption Component Limit	100

No	Variable Name	Description	Default value
2	DP_ConsensusDemandOrderNumber	Consensus Demand Order Number	Demand Plan
5	DP_GatingSuppliesLimit	Gating Supplies Limit	150

3.1.4.2. Macros

List of all macros used in the application.

The following macros are indicated for use in the Master Production Scheduling application, and can be configured in the Macros and Profile Variables workbook:

No macros are used in the various S&OP and Demand Planning resources.

3.1.4.3. Filters (Pre-defined)

Any complex or important filters or filters requiring modification used in the application.

There is a standard filter that is shipped with RapidResponse called "***NPI Items***". It is described as follows:

- This filter can be used to focus on items that are newly introduced, or are going to be introduced.

The filter expression is defined as:

- HAS PartCustomers [Part.PrimaryPartSource.FirstEffectiveDate >= MRPDate - 1 EffectiveDisaggregationCalendar]

The idea of this filter is to list Part (or PartCustomer) records that have not yet made it into production. The filter expression is designed to look for parts where the sourcing data has not yet come into effect. This may not be an appropriate test. For example, if the process for defining these new parts is to create PartSource records with effectivity that runs from past to future, then the filter expression may need some other means to identify these parts.

Careful thought needs to be put into what the expression will look like if it needs to be redefined to meet the customers need. Ideally, NPI parts should be identified by a discrete code value in a field (no wildcards), preferably in a referenced table.

3.1.4.4. Automation

Detailed information on any out of the box automation provided to support the application, include screen shots when applicable.

The following Automation resources are provided with this application:

Name	Type	Description	Expected Frequency
S&OP Insert Demand Planning History	Scheduled Task	This scheduled task writes records from the ForecastDetail table into the HistoricalDemandSeriesDetail table for all active forecast streams.	Just prior to Publish S&OP cycle is performed
S&OP Delete Adjustments and Overrides	Scheduled Task	This scheduled task deletes the Adjustments and Overrides Forecast Detail records that were created in the S&OP Consensus Demand Planning workbook.	Just prior to Publish S&OP cycle is performed but after inserting history - Optional task depending on customer preference to delete or not delete these records prior to their next S&OP cycle.
S&OP Record Demand Adjustments	Scheduled Task	This scheduled task runs the “Record Demand Adjustments” command in the S&OP Data Cleansing workbook, and adjusts the ForecastItemParametersOutlier. ProcessingRule to ‘All’ in order for demand adjustments to be included in further outlier calculations	Once per cycle

The following Automation tasks might need to be created to support this application:

Name	Type	Description	Expected Frequency
Delete Historical Demand Series	Scheduled System Task	Defined for a rolling period of time it prunes data in the HistoricalDemandSeries and through cascading deletes the HistoricalDemandSeriesDetail tables for data older than ‘n’ calendar periods, typically 12-24 months. Run in S&OP Intermediate.	Daily
Delete Historical Demand Actuals	Scheduled System Task	Defined for a rolling period of time it prunes data in the HistoricalDemandActuals table for data older than ‘n’ calendar periods, typically 12-24 months. Run in Enterprise Data using DeleteEnterpriseData command	Daily
Delete ForecastDetail records	Scheduled System Task	For ForecastDetail (forecast) records that are contained within RapidResponse as the system of record, in order to regularly prune unneeded records. Defined for a rolling period of time it prunes data in the ForecastDetail table for any data meeting agreed-upon business conditions (eg. created greater than ‘n’ calendar periods ago for relevant HistoricalDemandCategory values such as SalesForecast, MarketingForecast, etc.) Run in S&OP Intermediate using the DeleteData command. Note that for forecasts that are imported, the import process would take care of pruning.	Daily
Delete Assumptions	Scheduled System Task	Defined for a rolling period of time it prunes data in the Assumption table for any data meeting agreed-upon business conditions (eg. created greater than ‘n’ calendar periods ago and no longer effective). Actual deletion rules to be determined by business rules at deployment. Run in S&OP Intermediate using the DeleteData command	Daily
Write Event Causals	Scheduled Task	This scheduled task would write Forecast Event adjustment to the Causal Factor Detail table as Promotional adjustments. A command is available in the S&OP Write History Records.	Once per cycle, as part of the Sales Data Conditioning sub-Process

3.1.4.5. Miscellaneous

Below is a list of other configuration points to be aware of for this application.

3.1.4.5.1. Configuring for Consensus Forecast and DemandType

DemandType

It is critically important that the DemandType on the PartCustomer be set to something other than null for all PartCustomer records that are going to calculate a consensus forecast. If this is not done, then the data change in the “Closed Loop Within Cycle” will fail.

It is equally important that the DemandType used for these PartCustomer records not be used for imported DemandOrders (imported IndependentDemand records) nor used as any of the AllocationDemandType values in the SupplyType table. That is, you should see no usage of these PartCustomer DemandTypes in any scenario above the Baseline scenario for anything except PartCustomers. In the Baseline scenario and any descendents, you will see them also used on DemandOrders (IndependentDemand) but only after the data change in the “Closed Loop Within Cycle” has run and not for any other IndependentDemands. That is, these demand types should only ever occur in DemandOrders where the Order.Id starts with “CF: “. In no scenario at all should these demand types show up as dependent demand. The reason there must not be any overlap in the use of these demand types is that we need to ensure that double-driving the demand does not happen in either Baseline nor in the S&OP scenarios by turning them on and off appropriately. All of the other demand types used must not be touched between the scenarios (they should either drive demand or not, the same way in both scenario trees).

Also related to avoiding the double-driving of demand, it is vital that in the S&OP Intermediate scenario and all descendent scenarios, these PartCustomer DemandType controls all have the ProcessingRule set to Ignore. In this scenario tree, we want all of the consensus forecast to drive from ForecastDetails and not from the IndependentDemands that were written to Baseline and subsequently updated into S&OP Intermediate. For exactly the same reason, however, it is required that these same PartCustomer DemandType controls all have the ProcessingRule set to SalesForecast in all of the other (non-S&OP) scenarios [at least in Baseline down excluding the S&OP branch of scenarios].

It is important to understand that the PartCustomer DemandTypes are always treated as if the DemandType.ProcessingRule = ‘SalesForecast’ when consensus forecasts are calculated. Therefore, setting the ProcessingRule to Ignore in the S&OP scenarios for these PartTypes will not prevent them from driving consensus forecast and real forecasted demands there.

In any of the cases where the consensus forecast is expected to be spread differently than the applicable DisaggregationCalendar (either defined in the PartCustomer or, if null there, defined in the SOPAnalyticsConfiguration), it is important that the DemandType.SpreadRule is set to None in the Baseline scenarios and set to Spread in the S&OP scenarios. This is required because the data change in the “Closed Loop Within Cycle” will create the IndependentDemand records in the Baseline scenario already spread and we don’t want to try to spread them again in the Baseline. The selected SpreadProfile does not need to be different in the scenarios.

HistoricalDemandCategory

The double-driving forecast demand problem just discussed combined with the fact that PartCustomer DemandTypes are always treated as if the DemandType.ProcessingRule = ‘SalesForecast’ when consensus forecasts are calculated, has an implication on the settings of the HistoricalDemandCategory between Baseline scenarios and S&OP scenarios. We need to ensure that calculated consensus forecast does not occur in the Baseline scenarios but that it does occur in the S&OP scenarios.

The normal assumption is that there will be no non-Target ForecastDetail records in any non-S&OP scenario. If this assumption is true, then nothing more needs to be done. However, if ForecastDetail records are brought into RapidResponse at any scenario above the S&OP scenario and the Header.Category.Type.ProcessingRule for these records is one of Forecast, ForecastOverride, RebalancingAdjustment or RebalancingForecastOverride, then steps must be taken to prevent them from producing consensus forecast in the Baseline scenarios but allowing them in the S&OP scenarios.

If the ForecastDetail records in the non-S&OP scenarios have a Header.Category.Type.ProcessingRule of only Forecast, you can effectively disable them in the non-S&OP scenario by setting the ConsensusForecastWeight to zero and greater than zero in the S&OP scenarios as long as there are no overrides provided in the HistoricalDemandHeader or HistoricalDemandHeaderTimePhasedAttributes.

However, a safer approach is to completely disable these categories outside of the S&OP scenarios by changing the HistoricalDemandCategoryType ProcessingRule from anything with a value of Forecast, ForecastOverride, RebalancingAdjustment or RebalancingForecastOverride to None (only in the non-S&OP scenarios). In the S&OP scenarios these HistoricalDemandCategoryType records will be the correct ProcessingRule of Forecast, ForecastOverride, RebalancingAdjustment or RebalancingForecastOverride.

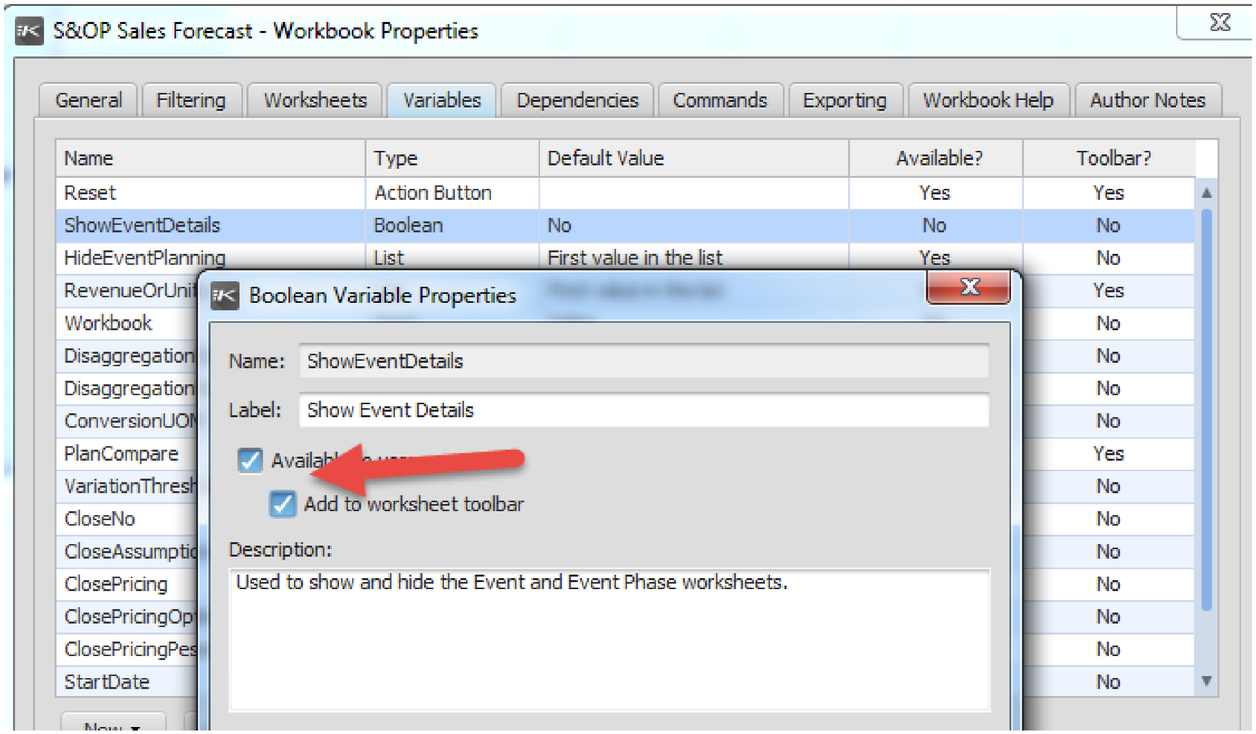
The following table describes these setting as required:

Table and Field	In scenarios above S&OP Intermediate (**Enterprise Data, Baseline,** etc.)	In **S&OP Intermediate** and all descendant scenarios (**Current S&OP, S&OP Candidate**, etc.)
For all DemandType records used by PartCustomer records (should never include DemandType records used by DemandOrders where the Id does not start with “CF:” nor any that have SupplyTypes)		
ProcessingRule	SalesForecast	Ignore
SpreadRule	None	Set as required for the Consensus Forecast spreading beyond the disaggregation calendar.
For all HistoricalDemandCategory records that could be used to produce a Consensus Forecast in the S&OP scenarios but **have ForecastDetail records in the Baseline scenario**. The Type ProcessingRule in the S&OP scenarios would be one of Forecast, ForecastOverride, RebalancingAdjustment or RebalancingForecastOverride but there are ForecastDetail records being imported above the S&OP Intermediate scenario.		
Type.ProcessingRule	None	Forecast, ForecastOverride, RebalancingAdjustment or RebalancingForecastOverride as required.
Or, alternatively… **ConsensusForecastWeight** if the Type.ProcessingRule is Forecast	0.0%	> 0.0% (but only for those where the Type.ProcessingRule is Forecast).

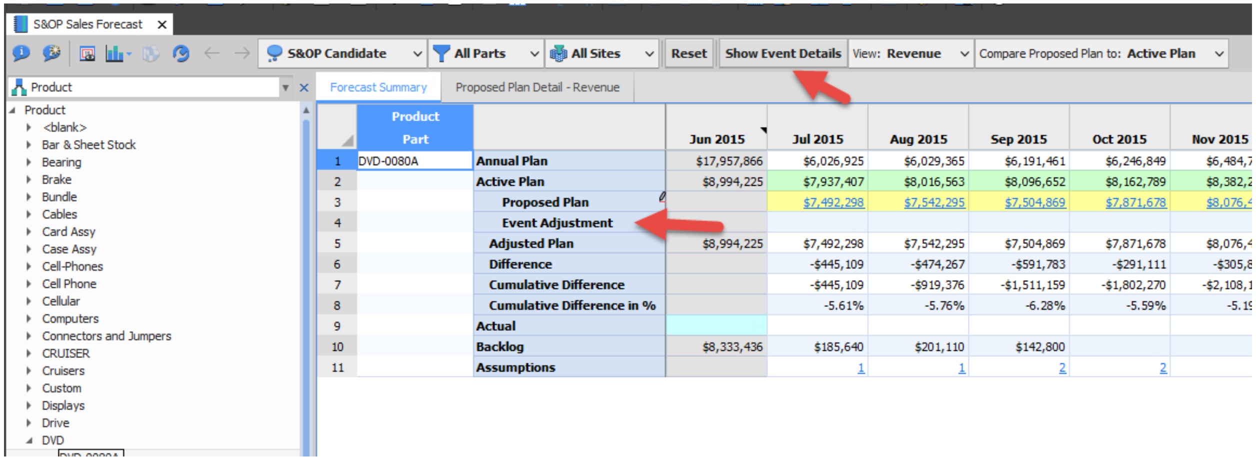
3.1.4.5.2. Configuring for Event Planning

The consultant will need to manually modify the following three resources in order to make the ShowEventDetails Boolean available to the users:

- S&OP Sales Forecast
- S&OP Marketing Forecast
- S&OP Finance Operating Plan



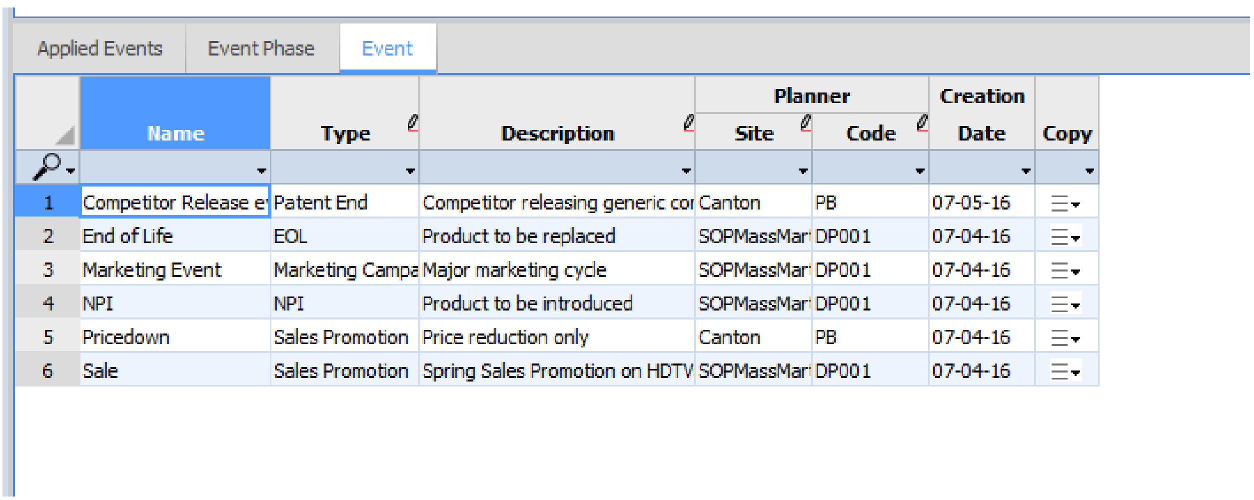
Also, the profile variable DP_HideEventPlanning needs to be set equal to “No” in order for Event Adjustment lines to become visible.



	Product Part		Jun 2015	Jul 2015	Aug 2015	Sep 2015	Oct 2015	Nov 2015
1	DVD-0080A	Annual Plan	$17,957,866	$6,026,925	$6,029,365	$6,191,461	$6,246,849	$6,484,7
2		Active Plan	$8,994,225	$7,937,407	$8,016,563	$8,096,652	$8,162,789	$8,382,2
3		Proposed Plan		$7,492,298	$7,542,295	$7,504,869	$7,871,678	$8,076,4
4		Event Adjustment						
5		Adjusted Plan	$8,994,225	$7,492,298	$7,542,295	$7,504,869	$7,871,678	$8,076,4
6		Difference		-$445,109	-$474,267	-$591,783	-$291,111	-$305,8
7		Cumulative Difference		-$445,109	-$919,376	-$1,511,159	-$1,802,270	-$2,108,1
8		Cumulative Difference in %		-5.61%	-5.76%	-6.28%	-5.59%	-5.1
9		Actual						
10		Backlog	$8,333,436	$185,640	$201,110	$142,800		
11		Assumptions		1	1	2	2	

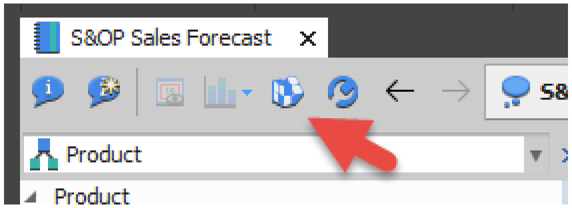
After clicking on Show Event Details, three worksheets should becom available in the lower pane:

- Applied Events
- Event Phase
- Event



	Name	Type	Description	Planner Site	Planner Code	Creation Date	Copy
1	Competitor Release e	Patent End	Competitor releasing generic cor	Canton	PB	07-05-16	
2	End of Life	EOL	Product to be replaced	SOPMassMar	DP001	07-04-16	
3	Marketing Event	Marketing Campa	Major marketing cycle	SOPMassMar	DP001	07-04-16	
4	NPI	NPI	Product to be introduced	SOPMassMar	DP001	07-04-16	
5	Pricedown	Sales Promotion	Price reduction only	Canton	PB	07-04-16	
6	Sale	Sales Promotion	Spring Sales Promotion on HDTV	SOPMassMar	DP001	07-04-16	

Events Phases can be applied from either the Event Phase or Event worksheet only. The Apply Event icon in the toolbar will become enabled when the cursos is in any cell of these worksheets. Note: This will only happened if the current user is a member of the appropriate user group (sales, marketing or finance depending on which workbook is being used)



3.1.4.5.3. S&OP Dashboards

The consultant will need to remove certain widgets for the role based dashboards if the customer has not purchased the related application, otherwise the widget will not render any data. For example, if the Capacity Planning application is not purchased the Key Constraint Utilization widget on the Corporate Metrics tab of the role based dashboards

3.1.4.5.4. Workbook Links

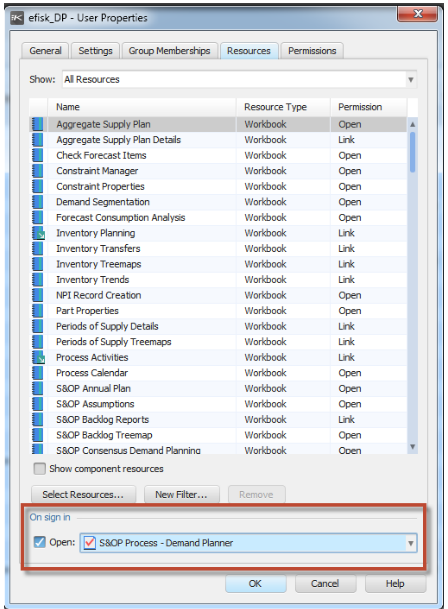
On install the workbook links off of the Part and Reference Part fields are no longer set up. This should be reviewed with the customer to come to an agreed upon list of links to be made available for these fields.

Reference Authoring Guide section on Creating drill dependencies and links.

3.1.4.5.5. User Set Up

The user experience for each Application is that upon login the user's relevant Task Flow and Dashboard is opened. In order to enable this, when performing the user set-up in the deployment each user needs to have the "On sign in" checkbox enabled (found at the bottom of the Resources tab of the User Properties dialog) and the appropriate task flow selected. The first step in each task flow is to open the appropriate dashboard.

For example, users in the Demand Planner role would have the S&OP Process – Demand Planner task flow selected to open on sign in as shown below:



In support of the Demand Planning Application, Users in the following groups should have the task flow noted below set to open on sign in:

- Demand Planners: S&OP Process – Demand Planner
- Finance: S&OP Process – Finance
- Sales: S&OP Process – Sales
- Marketing: S&OP Process – Marketing
- S&OP Process Owners: S&OP High Level Process – Process Owners

3.1.5. Closed Loop and Data Reconciliation

Each application has the ability to 'close the loop' on data to the Host system (ERP), i.e. Send data that is created/edited within back to the host system so that a user is not updating two systems. The below table outlines the out of the box supported transactions by application, those that are identified with an M indicates this is a transaction that must be manually entered into the ERP; those marked with a C indicate that the data is used in different scenarios in the tree and updated through a cross scenario update:

Transaction Type	S&OP	DP	ASP	CP	IM	IPO	MPS	OF	SAM	SC	IPM	DRP
Purchase Req Create												
Purchase Req/Order Change												
Purchase Req/Order Cancel												
Firm Planned Order Create												
Work Order Change												
Work Order Cancel												
Sales Order Date, Qty Change												
Consensus Demand Plan		C										
Inventory Transfer					M	M	M					
Fixed Safety Stock Qty Change												
Time Phased Safety Stock Qty Change					M	M	M					
Inventory Disposition					M	M						
Transfer Req Create												
Transfer Req/Order Change												
Transfer Req/Order Cancel												
Constraint Property Change				M			M					
Master Data Changes (Safety Stock Policy)					M	M						
Master Data Changes (Planning Time Fence)												
Master Data Changes (Planning BOM, Part Number of Days Supply, Part Source Properties, Order Priority)							M					

Master Data Changes (Sales Order Date & Qty, Order Priority, Delivery Route, Delivery Destination, Ship Set, Customer Request Date)								M				
Part UOM Conversion												M
Part Source (Routing ID, Routing Site, Supplier UOM, Supplier Control Set)												M
Source (Carrier, Transportation Mode, Ship Calendar, Transit Calendar, Transit Time, Pre Ship Lead Time)												M

NOTES:

1. If integrating with SAP, Supply Types of ‘FPO’ and ‘NB’ must be created to support Firm Planned Order create and Purchase Requisition create transactions.

2. In order to update Unit Price on ScheduledReceipts that are created and sent back to the host system (Firm Planned Order create and Purchase Requisition create transactions above) the value needs to be provided in the RAW value in the EBM. As such the consultant needs to create a custom field to hold this information until such time as the ability to generate transactions from worksheet payloads is available. In conjunction with this the two Business Message definitions will need to be updated to use the custom field rather than UnitPrice as they are defined out of the box.

Those transactions marked as 'M' or 'Manual' are identified to the user through the following workbooks:

- **Manual Closed Loop Constraint**: identifies constraint property edits
- **Manual Closed Loop Demand Planning**: identifies planning BoM edits
- **Manual Closed Loop DRP**: identifies part source and unit of measure edits
- **Manual Closed Loop for Supply**: identifies supply order edits
- **Manual Closed Loop Inventory Planning**: identifies order policy, inventory disposition and safety stock policy edits
- **Manual Closed Loop MPS**: identifies planning BoM, Number of Days Supply, Part Source properties, Order Priority edits
- **Manual Closed Loop OF**: identifies independent demand record edits. If the edited records identified in these workbooks are not manually entered in to the ERP system prior to the next Data Update, the changes will be lost.

For newly created records that are sent back to the ERP system, the records created in RapidResponse do not exist in the host system; therefore, there needs to be a way to know which records those are once inserted into the ERP System so duplicate records do not result on the next Data Update in to

RapidResponse. In order to identify the record a concatenation of the RapidResponse Record Id + Date is copied into a field called OriginalRecordId on the applicable table and is sent on the outbound transaction.

This OriginalRecordId needs to be kept on the record when it is imported into the ERP system therefore it needs to be defined as a custom field in the host ERP system in every table that will have new records created via closed loop feeds. Also in the case of creating a new Purchase Requisition or Firm Planned Order, the host system needs to carry this OriginalRecordId info to the Purchase Order or Work Order if it is being auto-generated. Then the OriginalRecordId needs to be included in the data extract for the inbound transaction. Once the data update is performed reconciliation within RapidResponse is completed using either the **Data Reconciliation Within Cycle** or **Data Reconciliation Outside Cycle** workbooks.

Within an S&OP cycle this reconciliation is done through a multi-scenario compare between the Baseline scenario and the S&OP Intermediate scenario. When the duplicate record is identified a command runs to delete the record from the S&OP Intermediate scenario, then the S&OP Intermediate scenario can be updated.

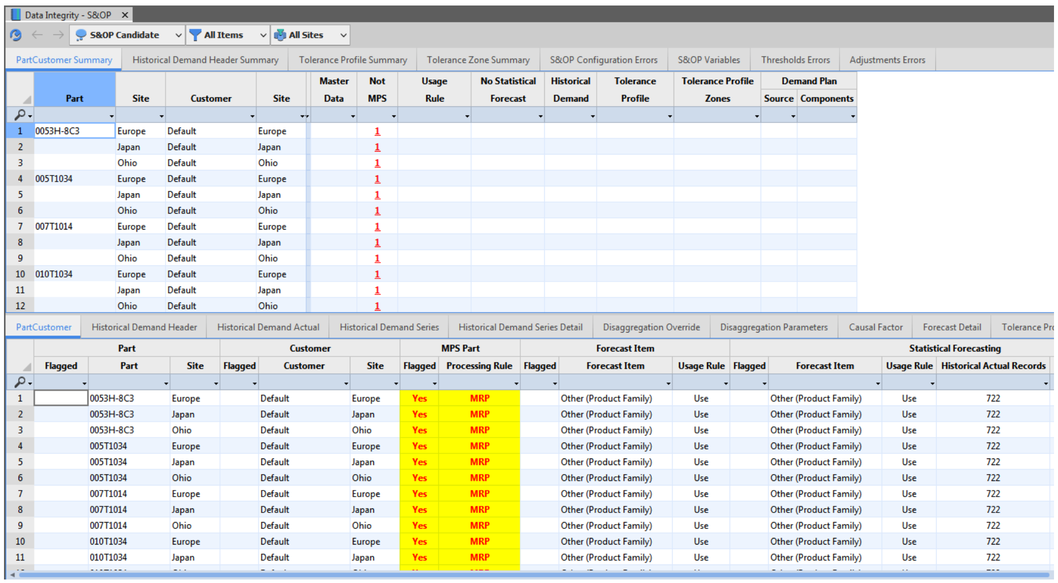
Outside an S&OP cycle this reconciliation is done through a multi-scenario compare between the Approved Actions scenario and the Baseline scenario. When the duplicate record is identified a command runs to delete the record from the Baseline scenario, then the Baseline scenario can be updated.

An important note to make about the closed loop and reconciliation processes is that there needs to be clear definition of where planning/execution is done, RapidResponse or ERP system, otherwise differences in planning policies, calendars etc. could cause unexpected results from closed loop transactions.

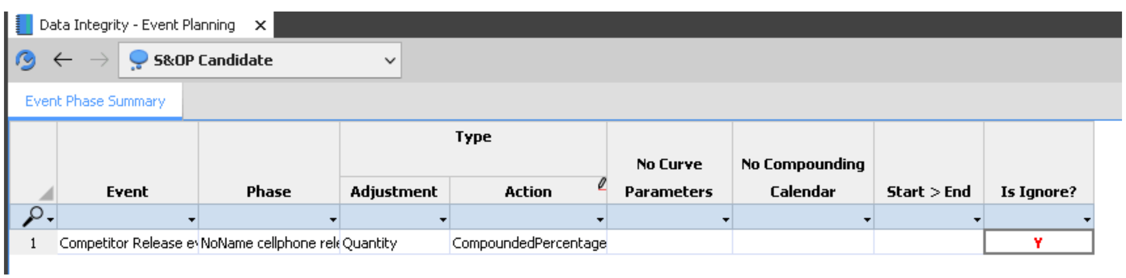
3.2. Data Integrity Requirements

Outline of any Data Integrity issues the user needs to be aware along with any applicable data integrity workbooks that are available in support of the application.

The Data Integrity-S&OP and Data Integrity-Event Planning workbooks can be used to monitor the accuracy of data that is supporting the Demand Planning blueprint. Review the workbooks and worksheet help for more detailed information on the data integrity checks these resource performs.



Data integrity - Event Planning



Chapter 4

Elaborated Resources

4. Elaborated Resources

4.1. Dashboards and Widgets

The following roles have Dashboards available to them as they relate to the Demand Planning application; however, these dashboards are installed through purchasing the S&OP application. By default the dashboards are not set to refresh automatically. If the customer would like auto-refresh turned on it can be done through the dashboard properties.

- Finance
- Sales
- Marketing
- Demand Planner

The Corporate Metrics tab of the dashboard for each role reports:

- Revenue
- Ending Inventory Value
- Margin %
- On Time Delivery to Request
- Key Constraint Utilization

The Finance Metrics tab of the Finance dashboard reports:

- Spend
- Backlog Value
- Finance Plan Forecast Accuracy (MAPE)

The Sales Metrics tab of the Sales dashboard reports:

- Revenue by Product Family
- Sales Forecast
- Forecast Accuracy Exceptions
- Sales Forecast Accuracy (MAPE)

The Marketing Metrics tab of the Marketing dashboard reports:

- Revenue by Product Family
- Marketing Forecast
- Forecast Accuracy Exceptions
- Marketing Forecast Accuracy (MAPE)

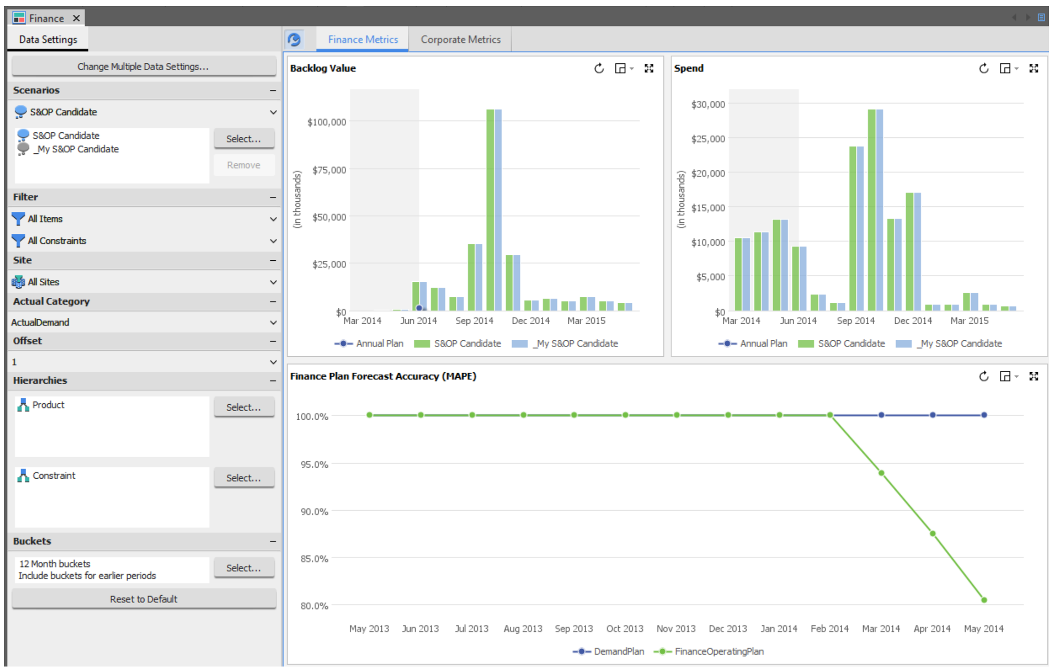
The Demand Planner Metrics tab of the Demand Planner dashboard reports:

- Consensus Demand Plan
- Forecast Accuracy Exceptions
- Data Errors / Outliers
- Forecast Value Add

4.1.1. Finance Dashboard

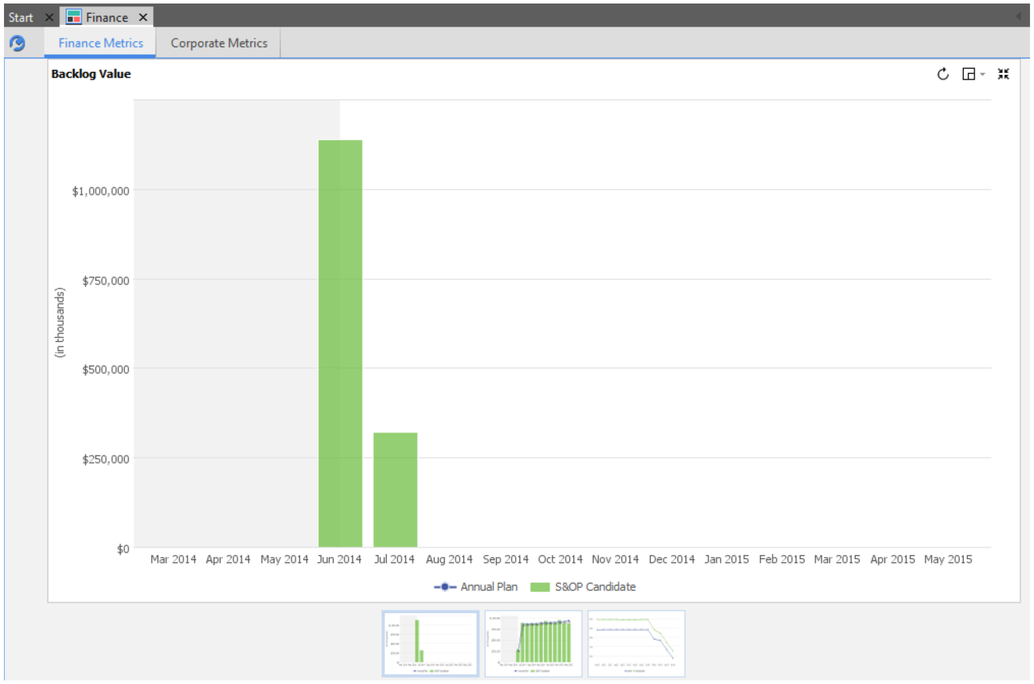
Dashboard Name	Finance
Purpose	Contains a variety of graphs and links that are of interest to Finance during the S&OP Demand Planning process. Provides the “jumping off” point for the Finance role in the overall process and is opened at the start of the **S&OP Process – Finance** task flow.

4.1.1.1. Finance Metrics sheet



This is the first tab available on the Finance dashboard. It is specific to the Finance role.

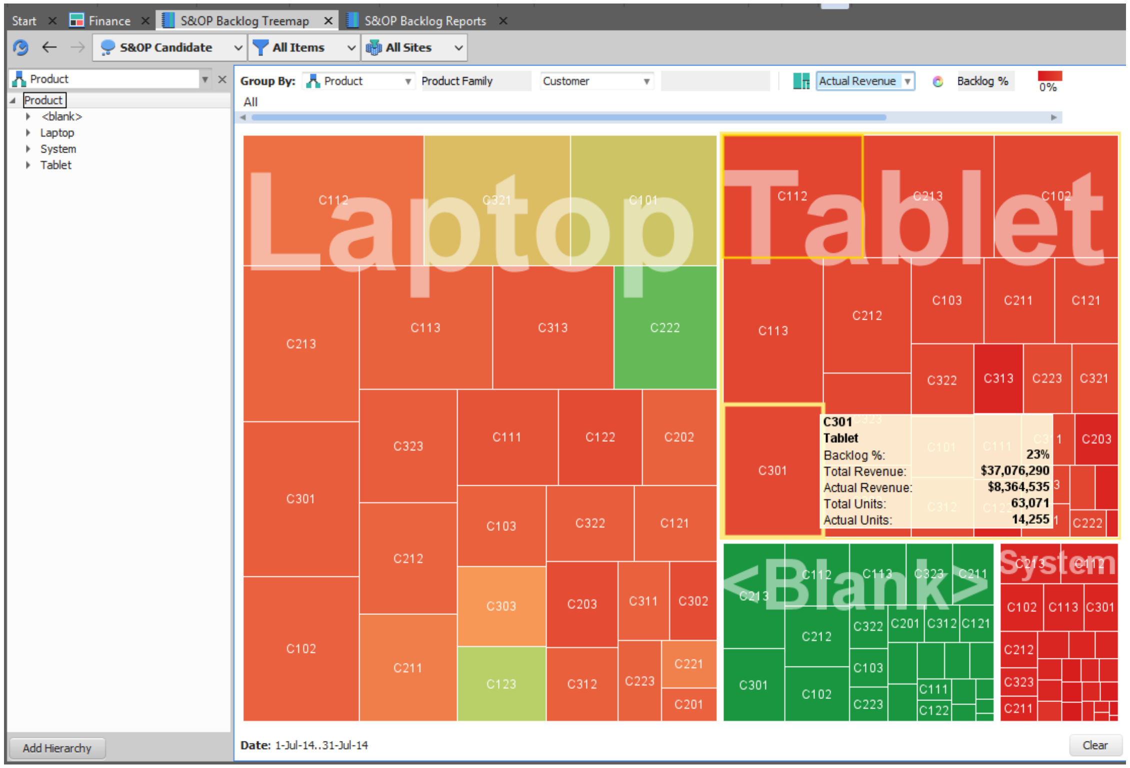
4.1.1.1.1. Backlog Value



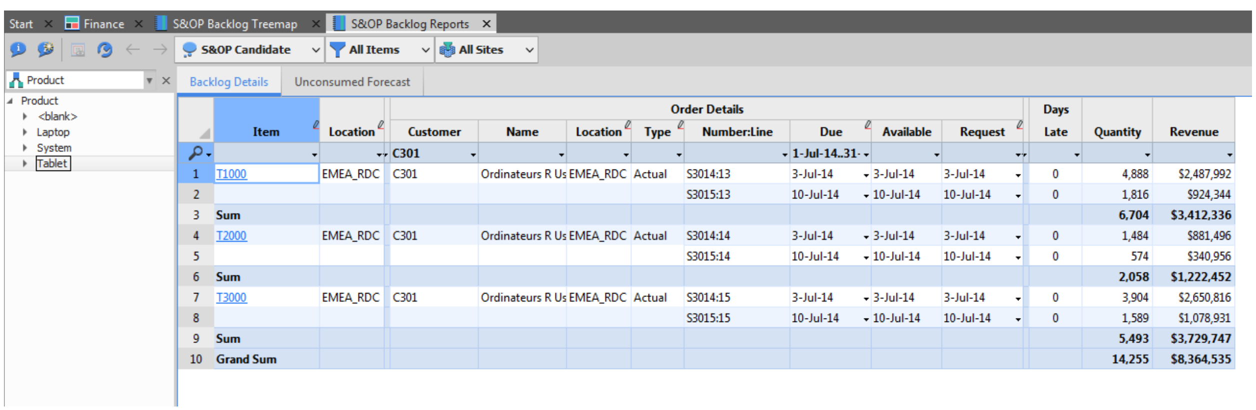
Summarizes the revenue on open sales orders by period based on the order due date. Past due periods have a gray background.

This widget is derived from the **S&OP Widgets Workbook** and **Backlog Value Chart** worksheet. The Annual Plan is derived from ForecastDetail records with a category set to $DP_BacklogValueTarget (or 'BacklogValueTarget' by default). Only independent demands with an effective quantity (still due to ship) with processing rules of 'SalesActual' and 'Regular' are included.

When you click on one of the columns, the following treemap is opened that summarizes the revenue for actual orders for drill date month, and expresses it as a percentage of the forecast revenue.

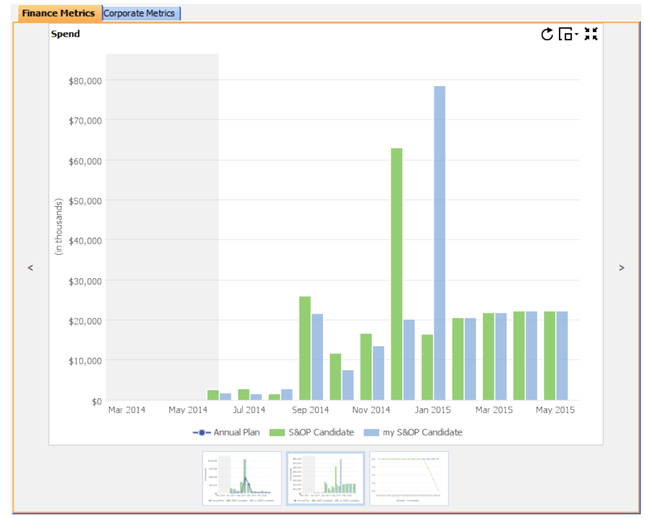


Double-clicking on any of the segments can drill to further detail showing the specific backlog order details from the Backlog Details worksheet in the S&OP Backlog Reports workbook.



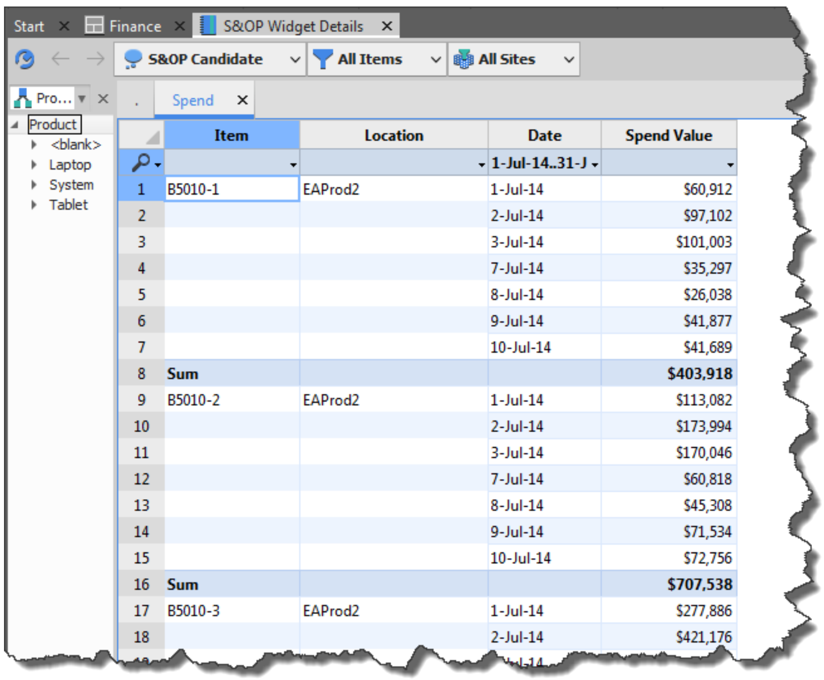
4.1.1.1.2. Spend

Summarizes the material cost of buy part purchases, by expected ship period, for the unconsumed forecast plus open sales orders. Historical data is displayed with a gray background.



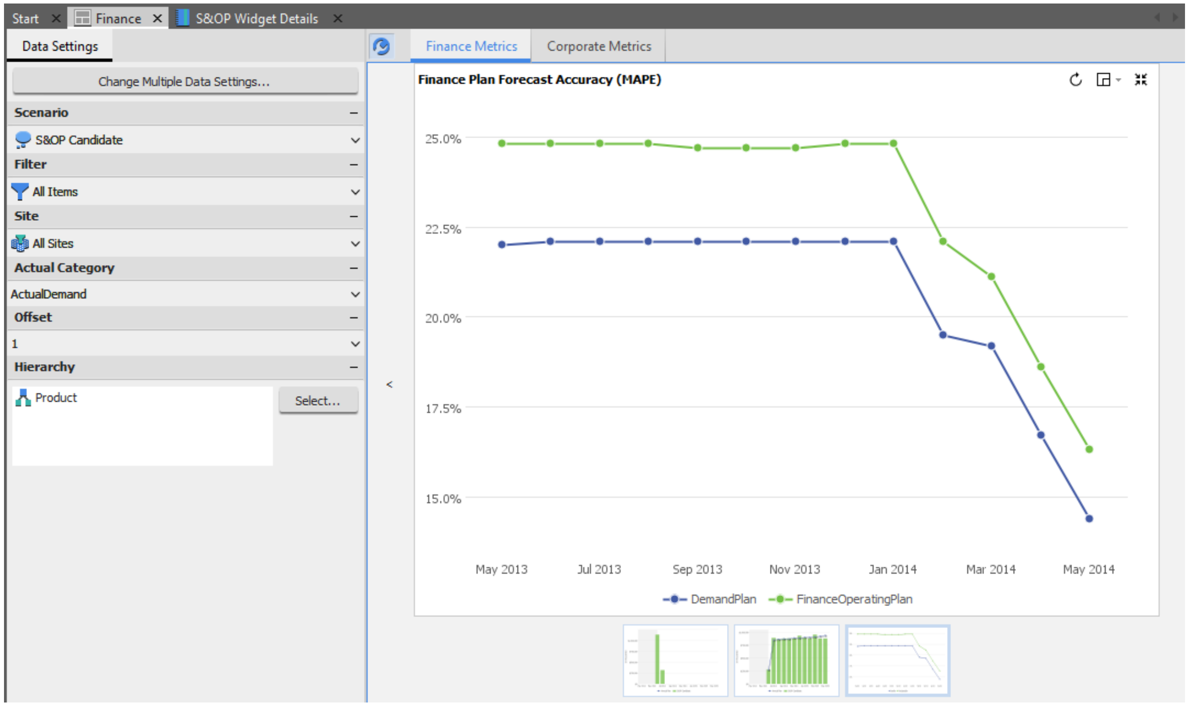
This widget is derived from the **S&OP Widgets Workbook** and **Spend Value Chart** worksheet. The Annual Plan is derived from ForecastDetail records with a category set to $DP_SpendTarget ('SpendTarget' by default). For open orders and forecasts the period is based on available date of the end item demand. Note that spend is bucketed based on the end item demand's available date, which may be later than the period in which purchased components will be received. This makes it possible to tie spend to end product and/or customer selections in a hierarchy. History is based on historical supply actual for the 'Shipment' category (configurable) and respects the same hierarchy settings.

When you click on one of the columns, the **S&OP Widget Details** workbook is opened that reports the material spend (purchase costs) that go into the independent demands and forecasts that comprise the bucket.



	Item	Location	Date	Spend Value
			1-Jul-14..31-J	
1	B5010-1	EAProd2	1-Jul-14	$60,912
2			2-Jul-14	$97,102
3			3-Jul-14	$101,003
4			7-Jul-14	$35,297
5			8-Jul-14	$26,038
6			9-Jul-14	$41,877
7			10-Jul-14	$41,689
8	**Sum**			**$403,918**
9	B5010-2	EAProd2	1-Jul-14	$113,082
10			2-Jul-14	$173,994
11			3-Jul-14	$170,046
12			7-Jul-14	$60,818
13			8-Jul-14	$45,308
14			9-Jul-14	$71,534
15			10-Jul-14	$72,756
16	**Sum**			**$707,538**
17	B5010-3	EAProd2	1-Jul-14	$277,886
18			2-Jul-14	$421,176

4.1.1.1.3. Finance Plan Forecast Accuracy (MAPE)



Displays the mean absolute percent error for the forecast by period and the actual demand for the period. The mean absolute percent error for each time period is the average of the absolute value of the percent difference between the forecast and actual demand for the given time period, and the eleven periods before it. You can select a number of periods offset in the data settings.

This widget is derived from the **S&OP Forecast Accuracy Widgets** workbook and **Mean Absolute Percent Error Finance** worksheet. This makes use of the *Statistical Errors By Date* function setting the number of samples to 12 (the eleven periods before it using CycleCalendar).

There is no drill-to-details available for this graph.

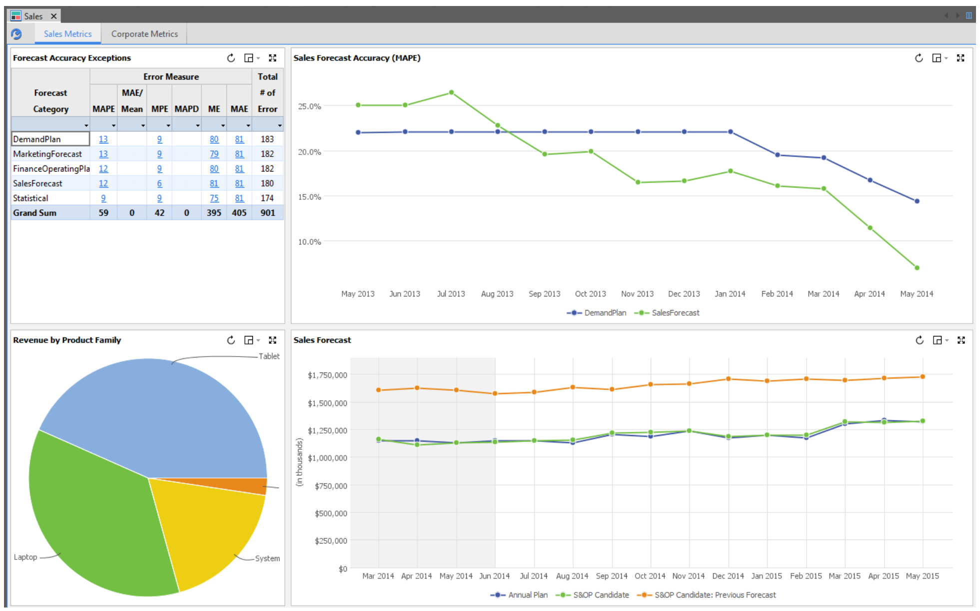
4.1.1.2. Corporate Metrics Sheet

See the shared resources section earlier in this document for a description of Corporate Metrics.

4.1.2. Sales Dashboard

Dashboard Name	Sales
Purpose	Contains a variety of graphs and links that are of interest to Sales during the S&OP Demand Planning process. Provides the “jumping off” point for the Sales role in the overall process and is opened at the start of the **S&OP Process – Sales** task flow.

4.1.2.1. Sales Metrics Tab



4.1.2.1.1. Forecast Accuracy Exceptions

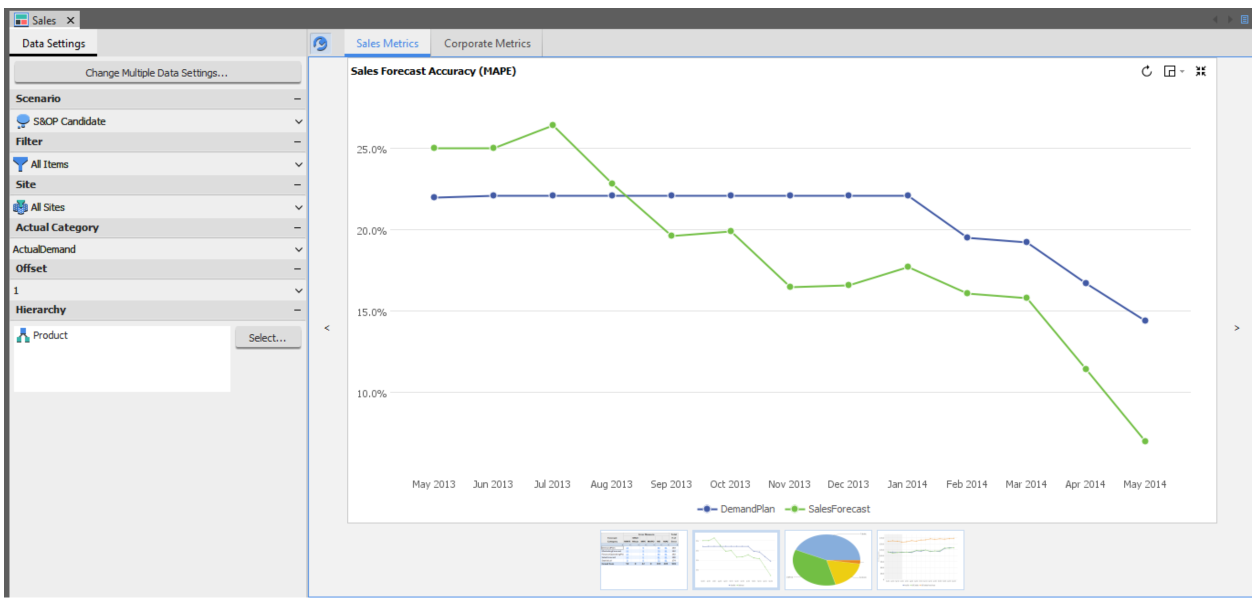
Sales | Sales Metrics | Corporate Metrics

Forecast Accuracy Exceptions

Forecast Category	Error Measure						Total # of Error
	MAPE	MAE/ Mean	MPE	MAPD	ME	MAE	
DemandPlan	13		9		80	81	183
MarketingForecast	13		9		79	81	182
FinanceOperatingPla	12		9		80	81	182
SalesForecast	12		6		81	81	180
Statistical	9		9		75	81	174
Grand Sum	**59**	**0**	**42**	**0**	**395**	**405**	**901**

This is the same widget that is described in the Demand Planner dashboard.

4.1.2.1.2. Sales Forecast Accuracy (MAPE)

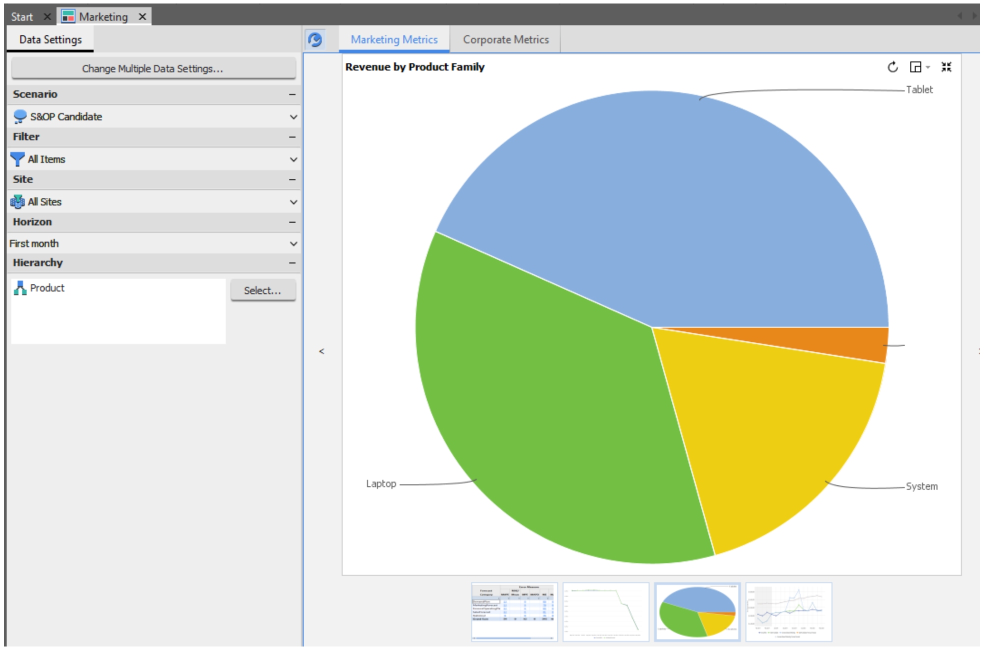


This reports the mean absolute percent error (MAPE) for the forecast by period and the actual demand for the period. The mean absolute percent error for each time period is the average of the absolute value of the percent difference between the forecast and actual demand for the given time period, and the eleven periods before it.

This widget is derived from the S&OP Forecast Accuracy Widgets workbook and Mean Absolute Percent Error Sales worksheet.

There is no drill-to-detail available for this widget.

4.1.2.1.3. Revenue by Product Family



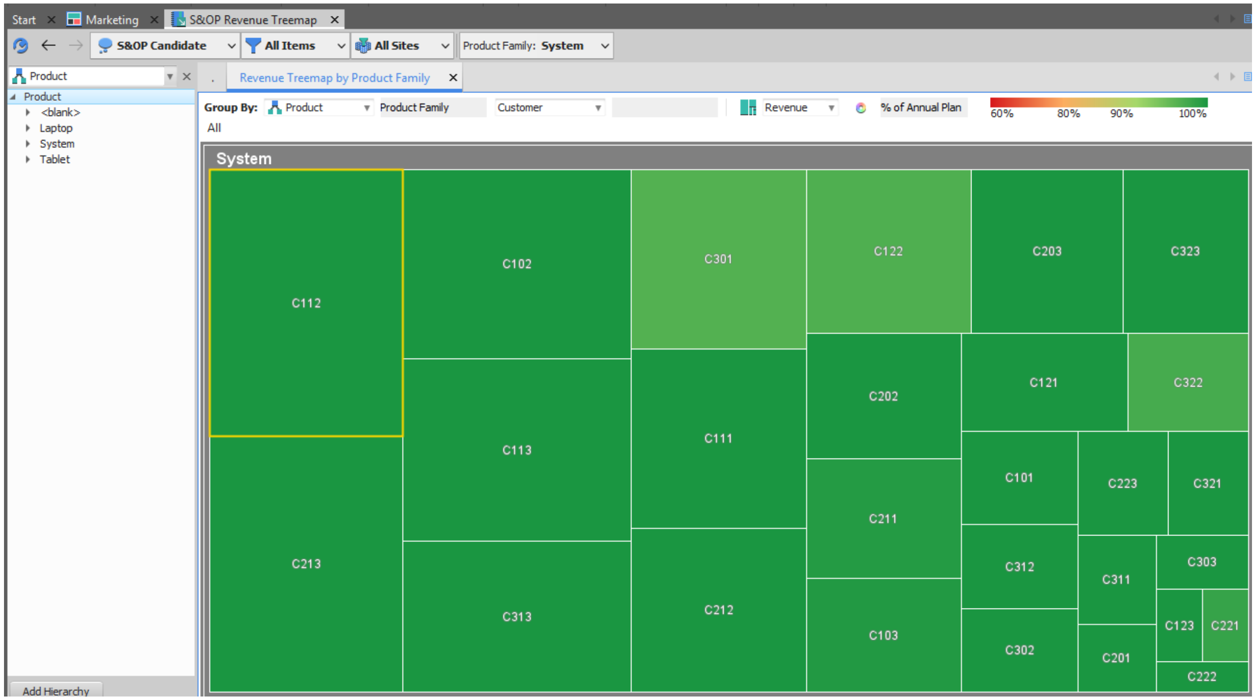
This reports the relative revenue for the forecast and actual orders for the selected horizon for individual product families.

The widget is derived from the **S&OP Widgets - Single Scenario** workbook and the **Revenue Pie Chart** worksheet.

The revenue portion is reported for only the top 5 items with the rest of the items added into the “Others” group. It also only reports revenue within a specified “horizon” as defined by the scorecard settings. The horizon is based on the later of due date or projected available date.

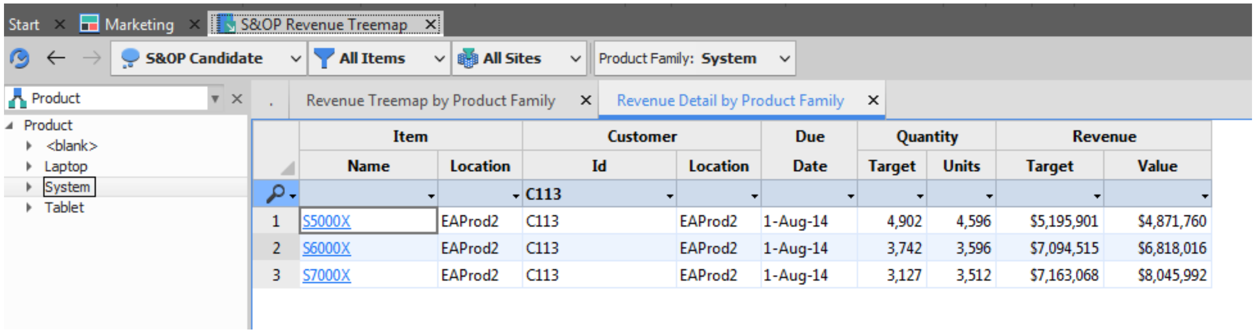
Product Family is based on Part.ProductGroup1 and may need to be configured.

Clicking on any of the drill-to-detail pie segments will open the **S&OP Revenue Treemap** workbook which shows the selected product revenue broken down by customer.



The size of the treemap segments correspond to relative revenue amounts while the color corresponds to a percentage of Annual Plan. The Annual Plan is a set of ForecastDetail records where the Header.Category is set to the $DP_DemandPlanValueTarget variable. On a default installation, this is set to DemandPlanValueTarget.

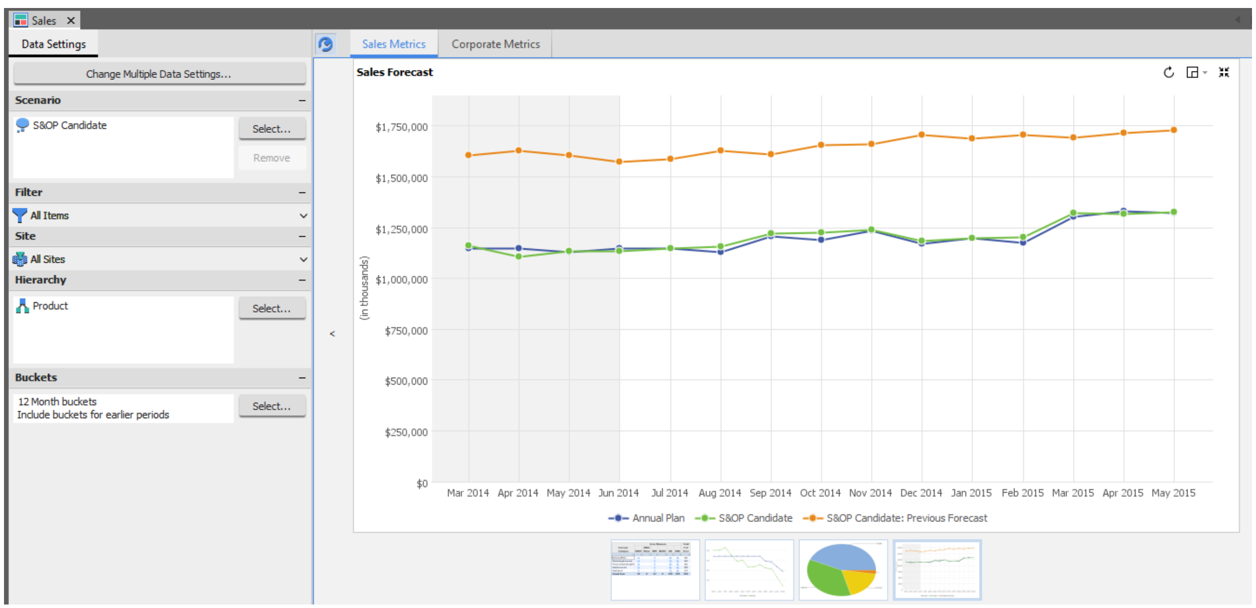
Double-clicking on any rectangle will expose the Revenue Detail by Product Family worksheet in the same workbook showing the revenue details of the particular segment.



This reports the target and actual quantity and revenue for the specific items selected in the selected horizon.

The DueDate is rounded to the CycleCalendar and the sheet is grouped. The DueDate for future orders is the max of DueDate or AvailableDate, consistent with the pie chart.

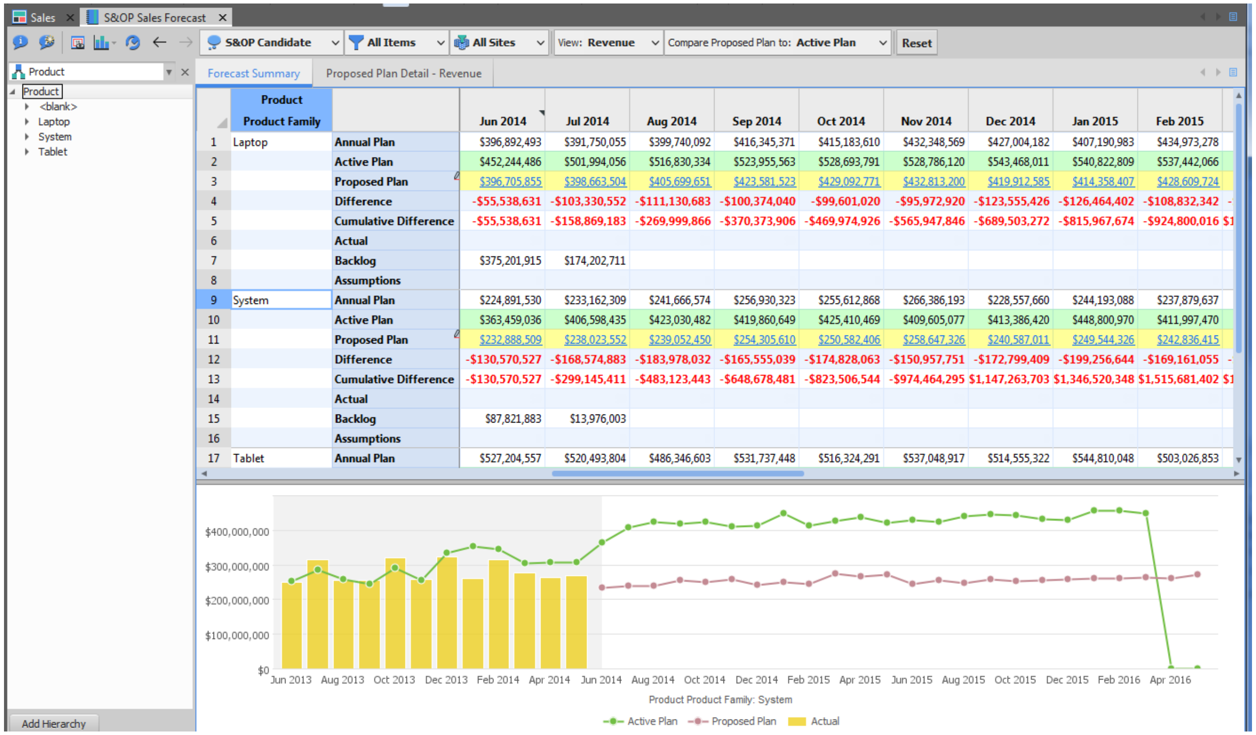
4.1.2.1.4. Sales Forecast



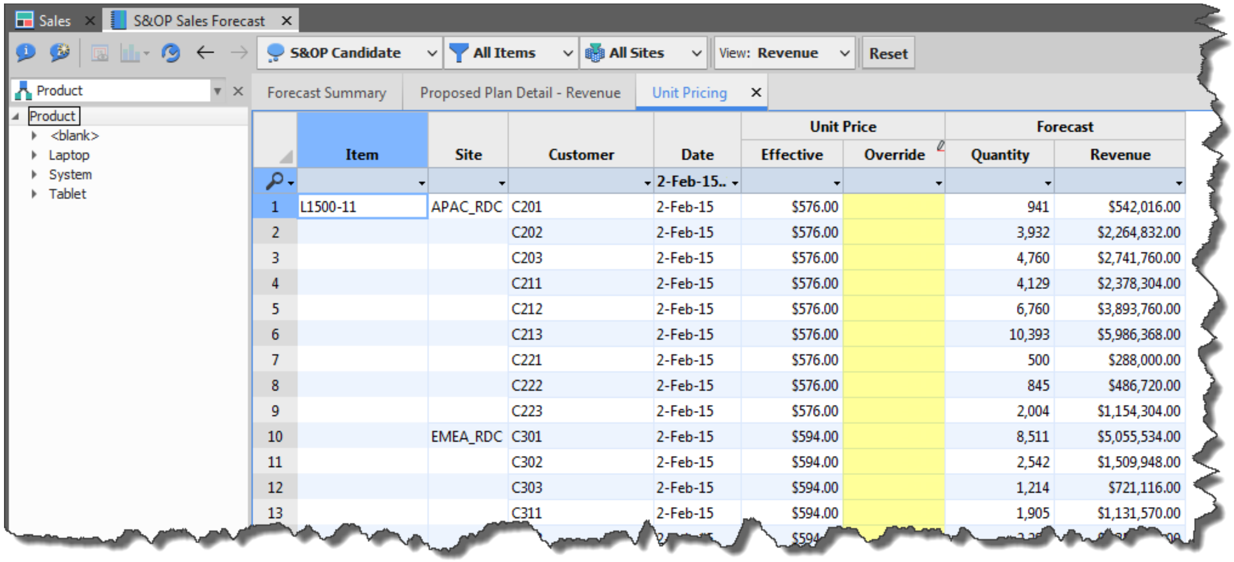
This reports the revenue in the sales forecast by period. History is displayed with a gray background.

This widget is derived from the **S&OP Widgets Workbook** and **Sales Value Chart** worksheet. The Annual Plan is derived from ForecastDetail records with a category set to $DP_SalesValueTarget (or 'SalesValueTarget' by default).

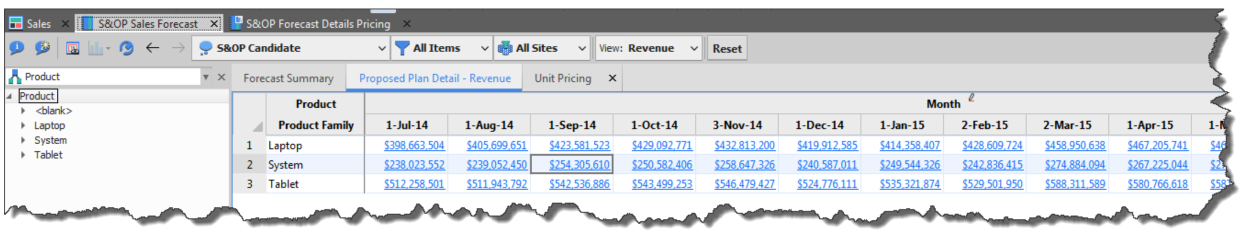
Clicking on the drill-to-detail values of the selected scenario forecast will open the **S&OP Sales Forecast** workbook using that scenario.



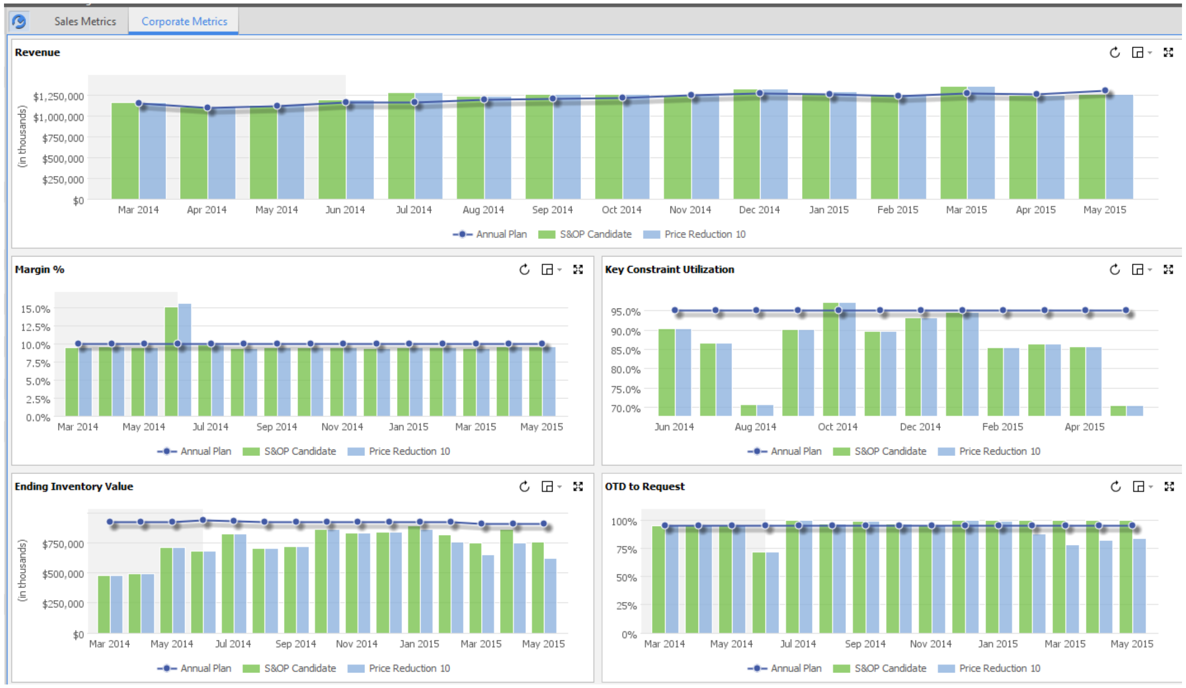
Clicking on the Proposed plan line in the graph takes you to the Unit Pricing sheet where prices can be edited. The Override column is the UnitPrice field on ForecastDetail for MPS parts in the $DP_SalesForecastCategory.



The Proposed Plan Detail – Revenue (or Units) tab allows you to import forecasts from Excel.



4.1.2.2. Corporate Metrics Tab

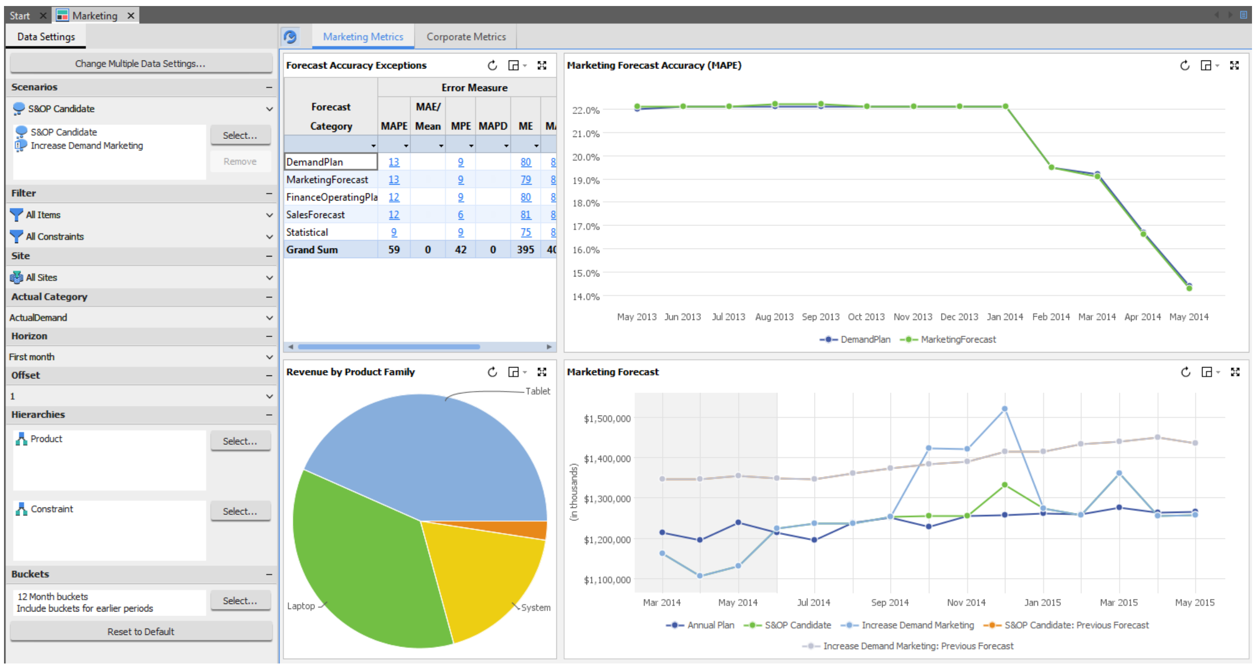


See the shared resources section earlier in this document for a description of Corporate Metrics.

4.1.3. Marketing Dashboard

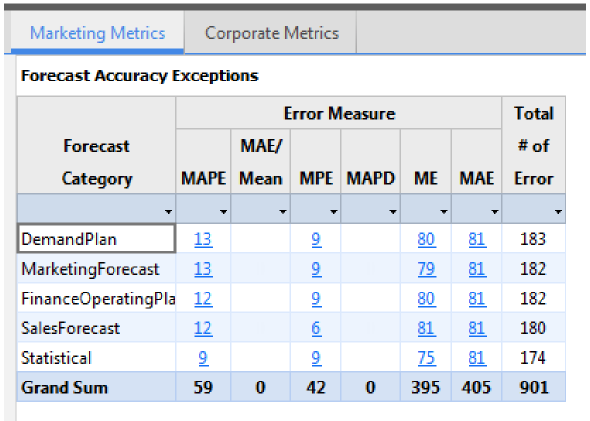
Dashboard Name	Marketing
Purpose	Contains a variety of graphs and links that are of interest to Marketing during the S&OP Demand Planning process. Provides the “jumping off” point for the Marketing role in the overall process and is opened at the start of the **S&OP Process – Marketing** task flow.

4.1.3.1. Marketing Metrics Tab



This is the first tab available on the Marketing dashboard. It is specific to the Marketing role.

4.1.3.1.1. Forecast Accuracy Exceptions



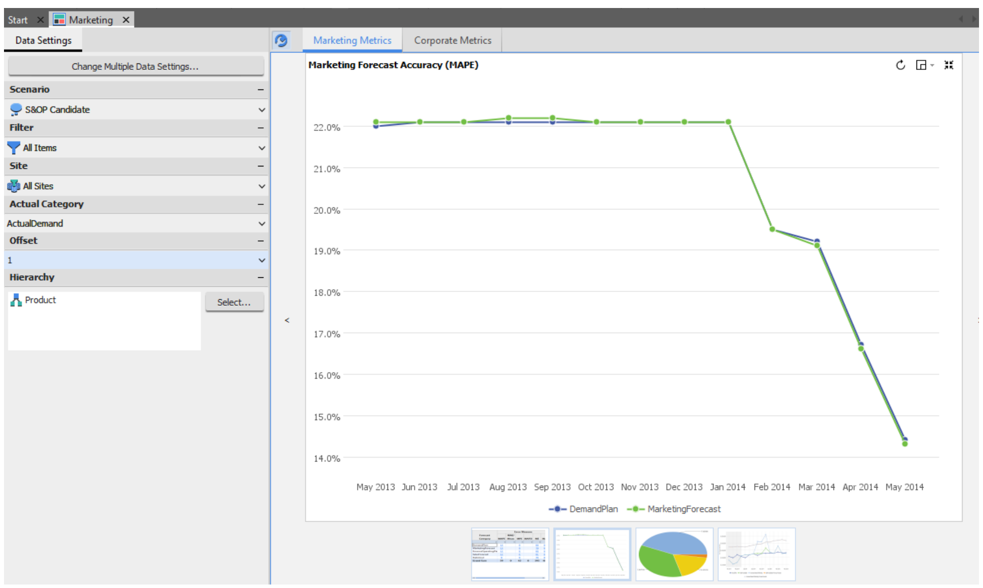
Marketing Metrics | Corporate Metrics

Forecast Accuracy Exceptions

Forecast Category	Error Measure						Total
	MAPE	MAE/ Mean	MPE	MAPD	ME	MAE	# of Error
DemandPlan	13		9		80	81	183
MarketingForecast	13		9		79	81	182
FinanceOperatingPla	12		9		80	81	182
SalesForecast	12		6		81	81	180
Statistical	9		9		75	81	174
Grand Sum	**59**	**0**	**42**	**0**	**395**	**405**	**901**

This is the same widget that is described in the Demand Planner dashboard.

4.1.3.1.2. Marketing Forecast Accuracy (MAPE)

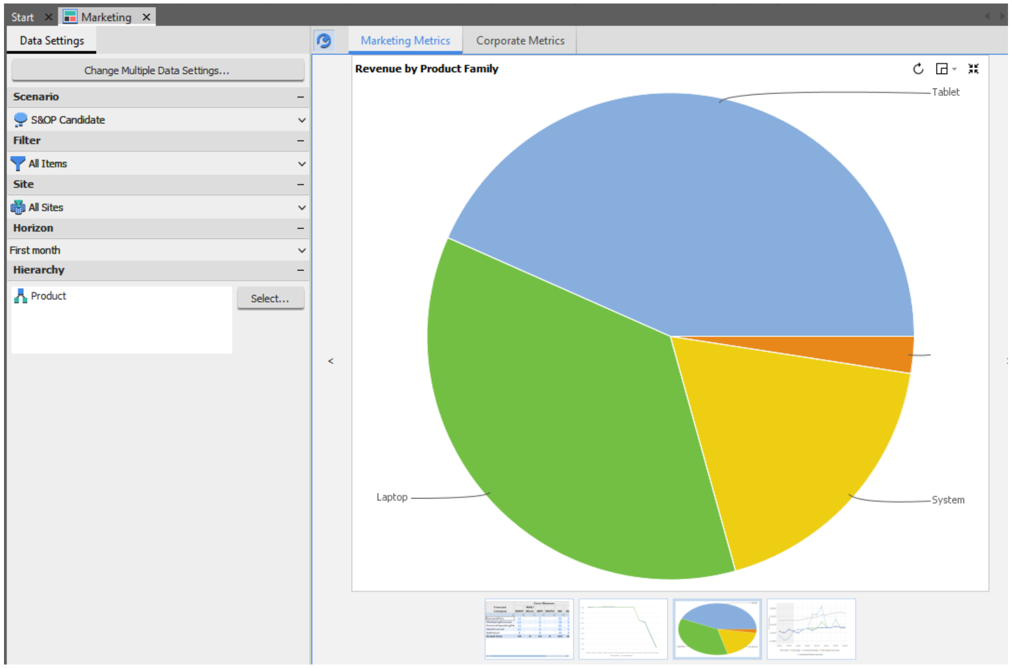


This reports the mean absolute percent error (MAPE) for the forecast by period and the actual demand for the period. The mean absolute percent error for each time period is the average of the absolute value of the percent difference between the forecast and actual demand for the given time period, and the eleven CycleCalendar periods before it.

This widget is derived from the **S&OP Forecast Accuracy Widgets** workbook and **Mean Absolute Percent Error Marketing** worksheet.

There is no drill-to-detail available for this widget.

4.1.3.1.3. Revenue by Product Family



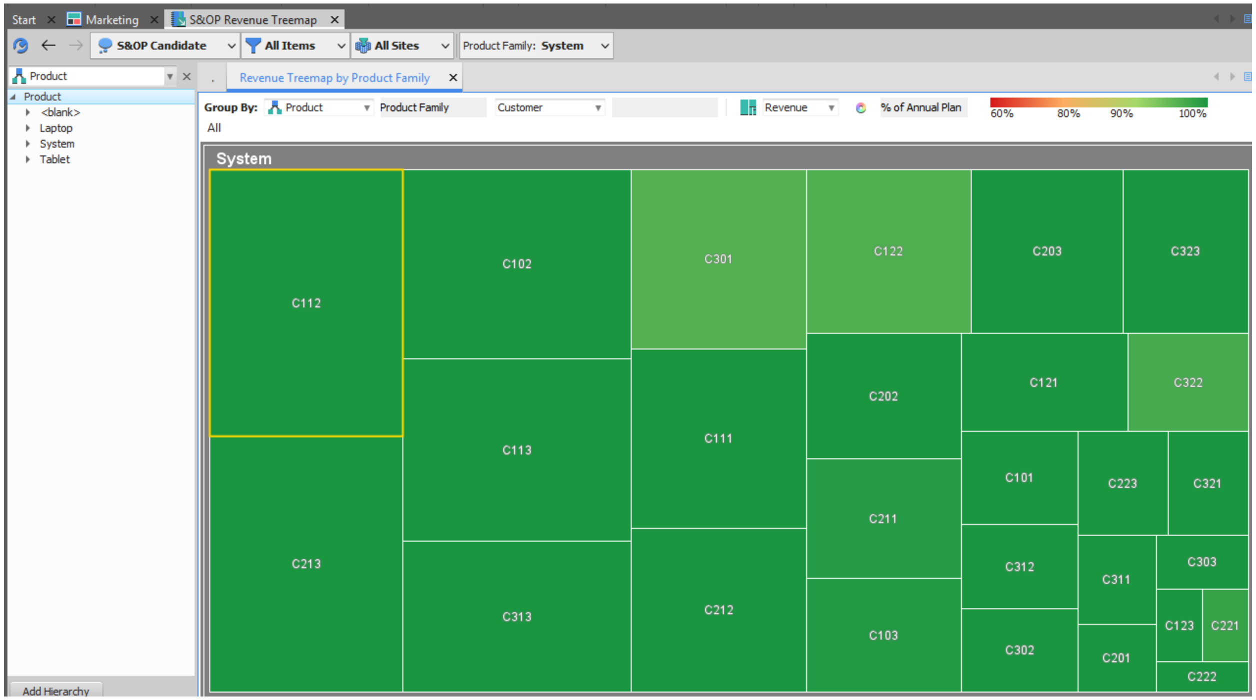
This reports the relative revenue for the forecast and actual orders for the selected horizon for individual product families.

The widget is derived from the **S&OP Widgets - Single Scenario** workbook and the **Revenue Pie Chart** worksheet.

The revenue portion is reported for only the top 5 items with the rest of the items added into the “Others” group. It also only reports revenue within a specified “horizon” as defined by the scorecard settings. The horizon is based on the later of due date or projected available date.

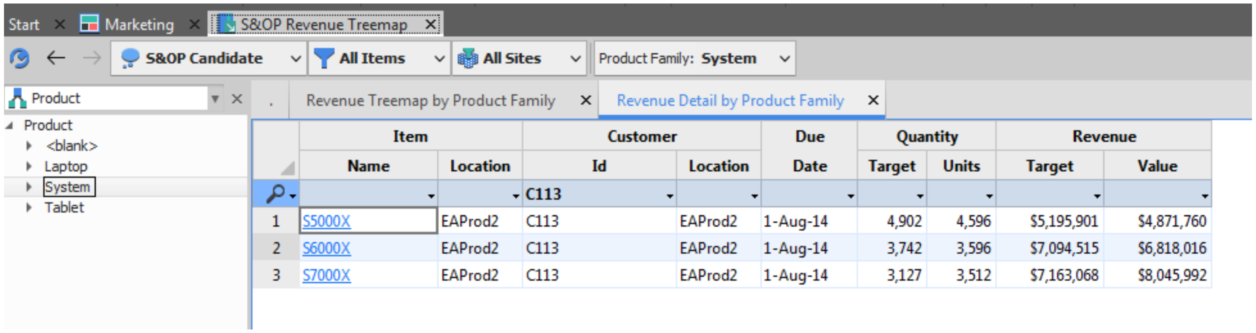
Product Family is based on Part.ProductGroup1 and may need to be configured.

Clicking on any of the drill-to-detail pie segments will open the **S&OP Revenue Treemap** workbook which shows the selected product revenue broken down by customer.



The size of the treemap segments correspond to relative revenue amounts while the color corresponds to a percentage of Annual Plan. The Annual Plan is a set of ForecastDetail records where the Header.Category is set to the $DP_DemandPlanValueTarget variable. On a default installation, this is set to DemandPlanValueTarget.

Double-clicking on any rectangle will expose the Revenue Detail by Product Family worksheet in the same workbook showing the revenue details of the particular segment.

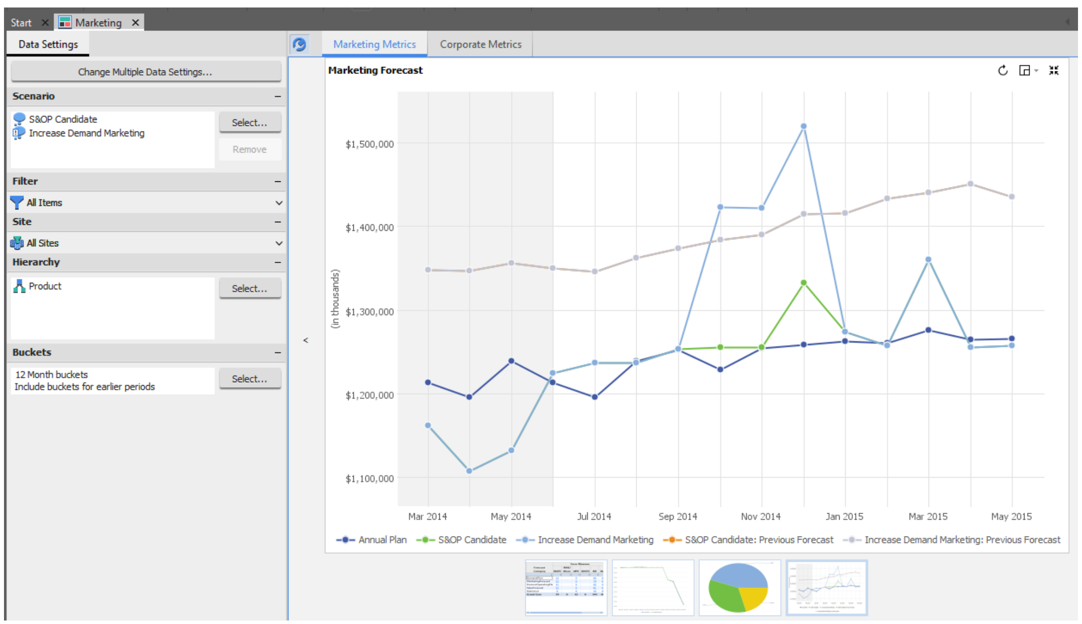


	Item		Customer		Due	Quantity		Revenue	
	Name	Location	Id	Location	Date	Target	Units	Target	Value
			C113						
1	S5000X	EAProd2	C113	EAProd2	1-Aug-14	4,902	4,596	$5,195,901	$4,871,760
2	S6000X	EAProd2	C113	EAProd2	1-Aug-14	3,742	3,596	$7,094,515	$6,818,016
3	S7000X	EAProd2	C113	EAProd2	1-Aug-14	3,127	3,512	$7,163,068	$8,045,992

This reports the target and actual quantity and revenue for the specific items selected in the selected horizon.

The DueDate is rounded to the CycleCalendar and the sheet is grouped. The DueDate for future orders is the max of DueDate or AvailableDate, consistent with the pie chart.

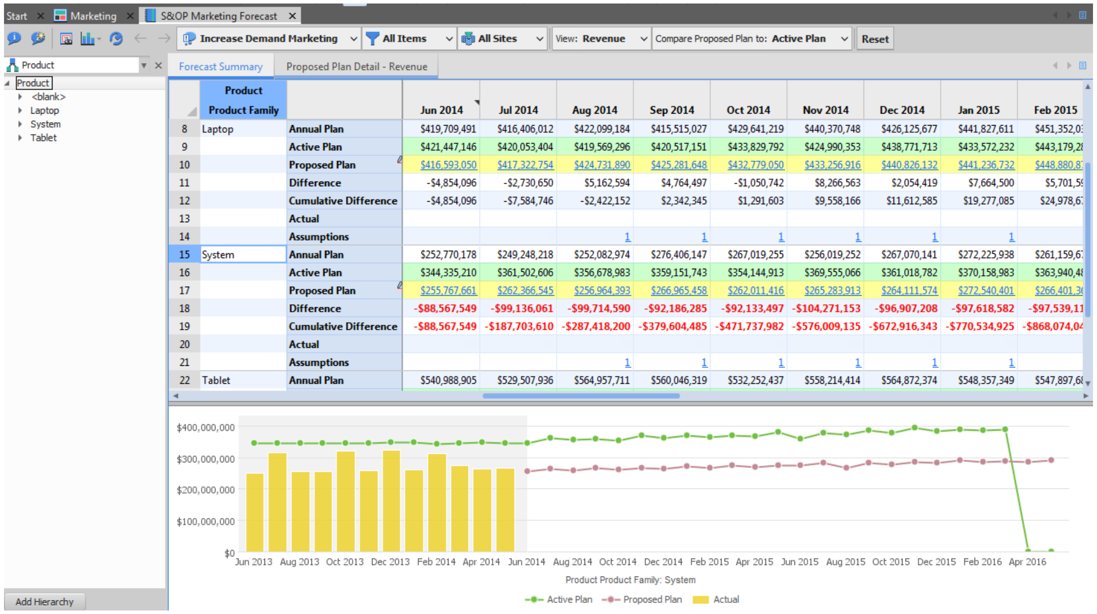
4.1.3.1.4. Marketing Forecast



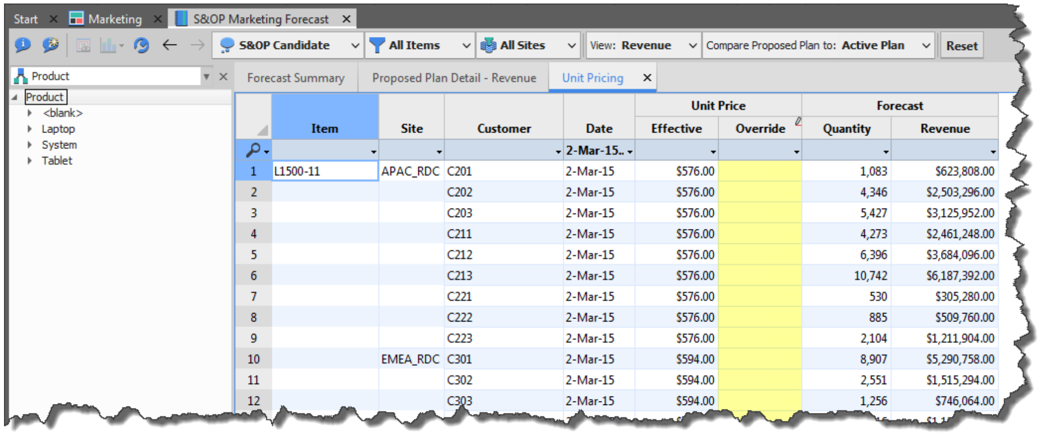
This reports the revenue in the marketing forecast by period. History is displayed with a gray background.

This widget is derived from the **S&OP Widgets Workbook** and **Marketing Value Chart** worksheet. The Annual Plan is derived from ForecastDetail records with a category set to $DP_MarketingValueTarget (or ‘MarketingValueTarget’ by default).

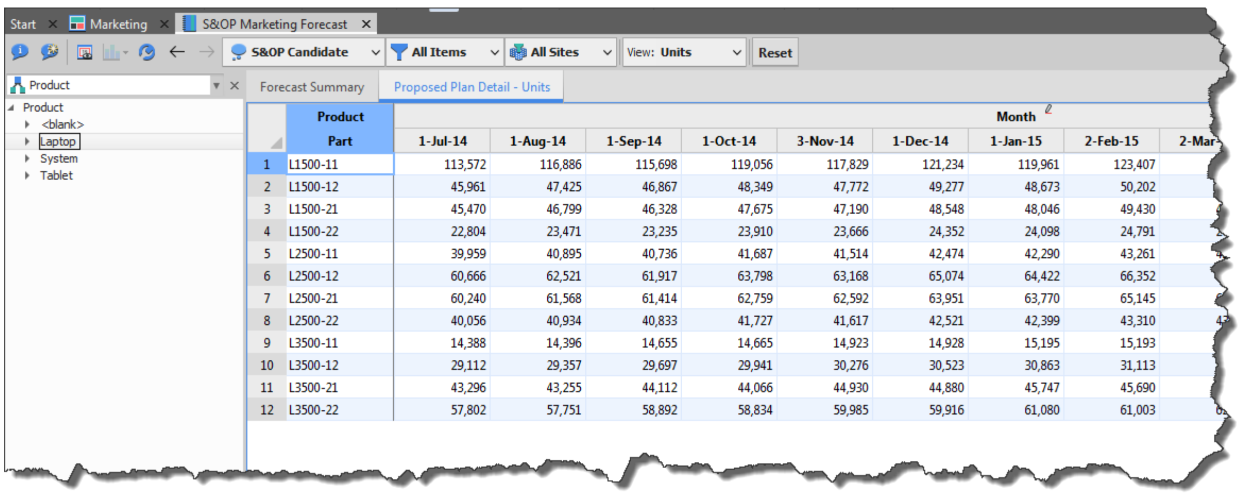
Clicking on the drill-to-detail values of the selected scenario forecast will open the **S&OP Marketing Forecast** workbook using that scenario.



Clicking on the Proposed plan line in the graph takes you to the Unit Pricing sheet where prices can be edited. The Override column is the UnitPrice field on ForecastDetail for MPS parts in the $DP_SalesForecastCategory.



The Proposed Plan Detail – Revenue (or Units) tab allows you to import forecasts from Excel. Units version is shown below.



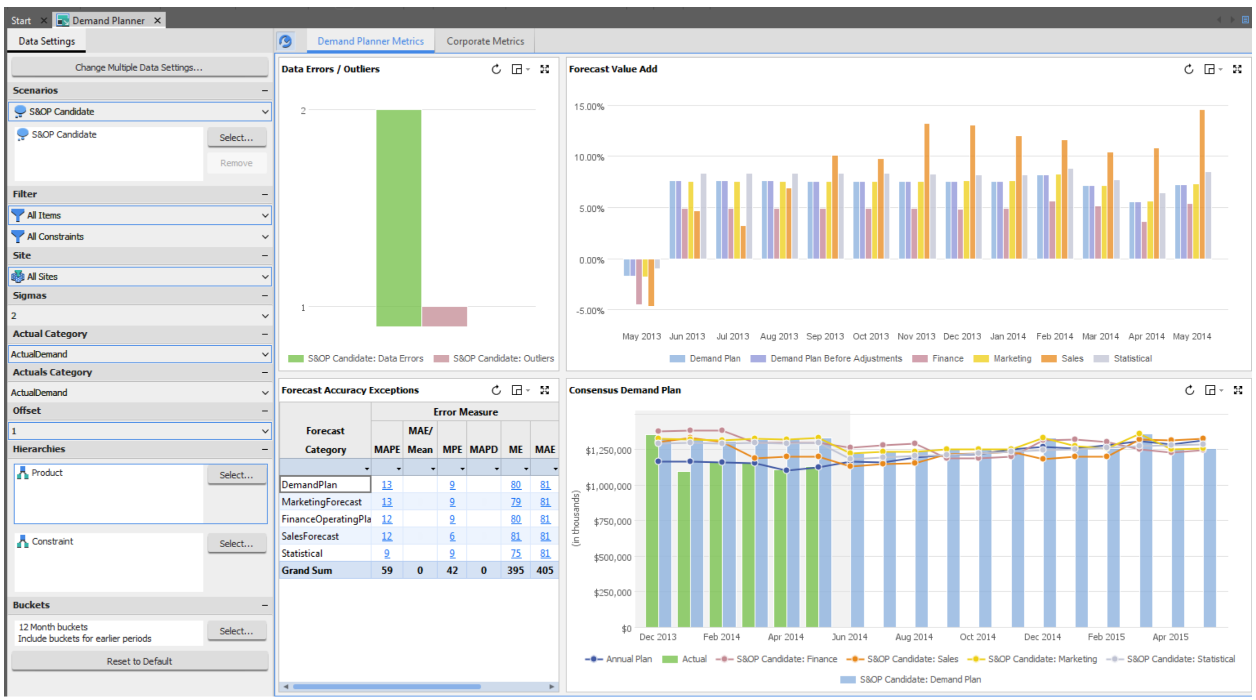
4.1.3.2. Corporate Metrics Tab

Corporate Metrics are described in the earlier shared resources section.

4.1.4. Demand Planner Dashboard

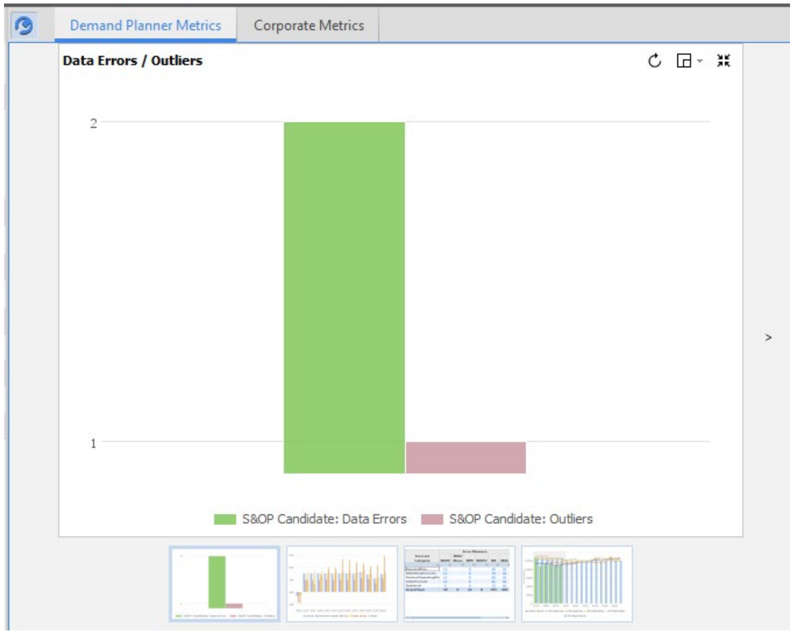
Dashboard Name	Demand Planner
Purpose	Contains a variety of graphs and links that are of interest to the Demand Planner during the S&OP Demand Planning process. Provides the “jumping off” point for the Demand Planner role in the overall process and is opened at the start of the **S&OP Process – Demand Planner** task flow.

4.1.4.1. Demand Planner Metrics Tab



This is the first tab available on the Demand Planner dashboard. It is specific to the Demand Planner role.

4.1.4.1.1. Data Errors / Outliers



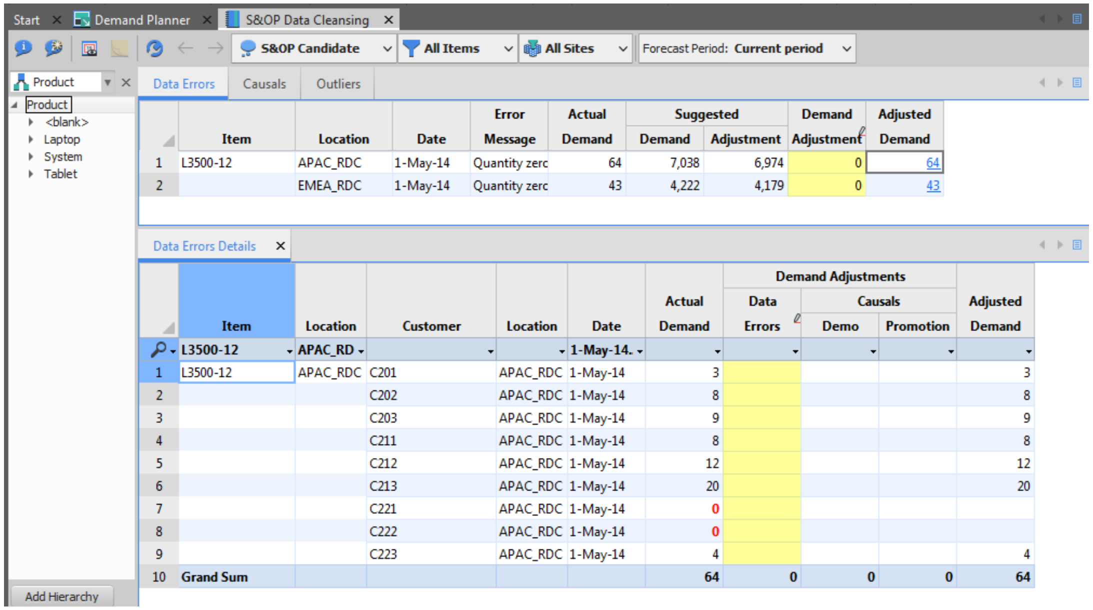
Counts the number of forecast items with data errors or outliers in each of the selected scenarios.

A data error is when at least one historical demand actual has a zero value or does not exist in a period for a forecast item.

An outlier is identified when a forecast item has actual demand quantities that fall outside of a calculated upper and lower limit based on standard statistical measures. For outliers, an average demand is calculated as a three month moving average. The upper and lower limits are then calculated as this average demand plus/minus two standard deviations.

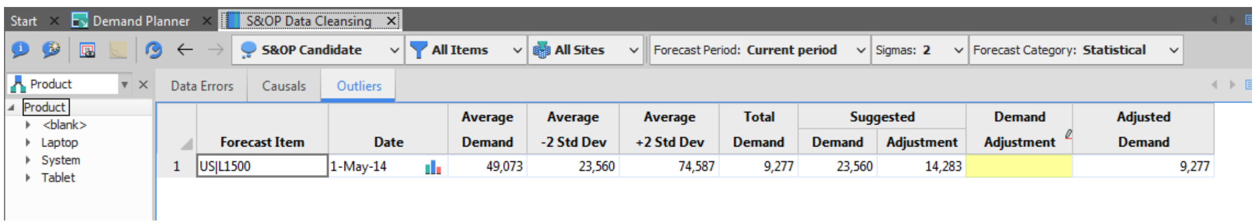
This widget is derived from the **S&OP Widgets Workbook** and **Data Errors / Outliers Chart** worksheet.

To see detailed data click the bar for a specific error/scenario and the **S&OP Data Cleansing** workbook will be opened at the appropriate category and scenario.



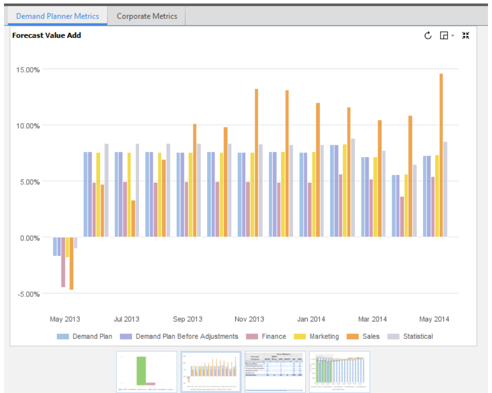
	Item	Location	Date	Error Message	Actual Demand	Suggested Demand	Suggested Adjustment	Demand Adjustment	Adjusted Demand
1	L3500-12	APAC_RDC	1-May-14	Quantity zerc	64	7,038	6,974	0	64
2		EMEA_RDC	1-May-14	Quantity zerc	43	4,222	4,179	0	43

	Item	Location	Customer	Location	Date	Actual Demand	Demand Adjustments: Data Errors	Demand Adjustments: Causals: Demo	Demand Adjustments: Causals: Promotion	Adjusted Demand
	L3500-12	APAC_RD			1-May-14.					
1	L3500-12	APAC_RDC	C201	APAC_RDC	1-May-14	3				3
2			C202	APAC_RDC	1-May-14	8				8
3			C203	APAC_RDC	1-May-14	9				9
4			C211	APAC_RDC	1-May-14	8				8
5			C212	APAC_RDC	1-May-14	12				12
6			C213	APAC_RDC	1-May-14	20				20
7			C221	APAC_RDC	1-May-14	0				
8			C222	APAC_RDC	1-May-14	0				
9			C223	APAC_RDC	1-May-14	4				4
10	Grand Sum					64	0	0	0	64



	Forecast Item	Date	Average Demand	Average -2 Std Dev	Average +2 Std Dev	Total Demand	Suggested Demand	Suggested Adjustment	Demand Adjustment	Adjusted Demand
1	US|L1500	1-May-14	49,073	23,560	74,587	9,277	23,560	14,283		9,277

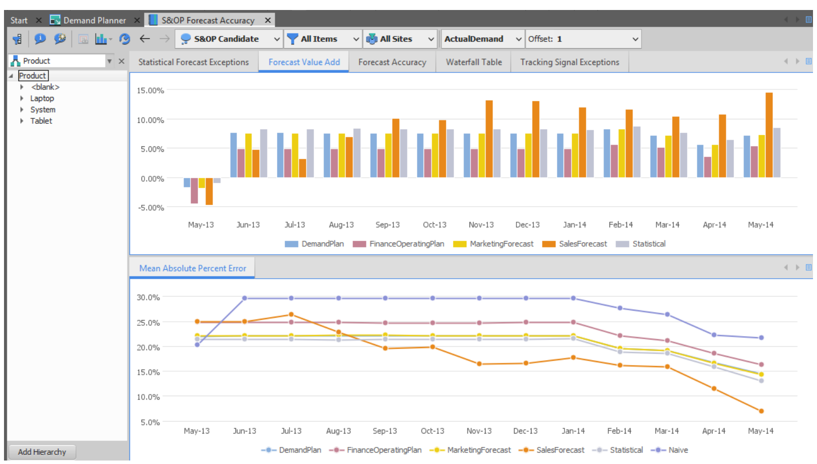
4.1.4.1.2. Forecast Value Add



Summarizes the forecast value add for each forecast stream included in the demand planning process. To evaluate if each forecast stream is more or less accurate than the naïve forecast, the value add is calculated. This is defined as the difference in mean absolute percent error of each forecast stream and the mean absolute percent error of the naive forecast. The naïve forecast for a period is the actual value from the previous period for the category selected. Positive forecast value add numbers indicate that the forecast stream is more accurate than the naïve forecast, negative forecast value add numbers indicate that the forecast stream is less accurate than the naïve forecast.

This widget is derived from the **S&OP Forecast Accuracy Widgets** workbook and **Forecast Value Add** worksheet.

To see detailed data click a bar for any period and RapidResponse will open the **S&OP Forecast Accuracy** workbook to the **Forecast Value Add** worksheet. This shows a graph of the MAPE's of the various forecasts. Note the Offset workbook control.



4.1.4.1.3. Forecast Accuracy Exceptions

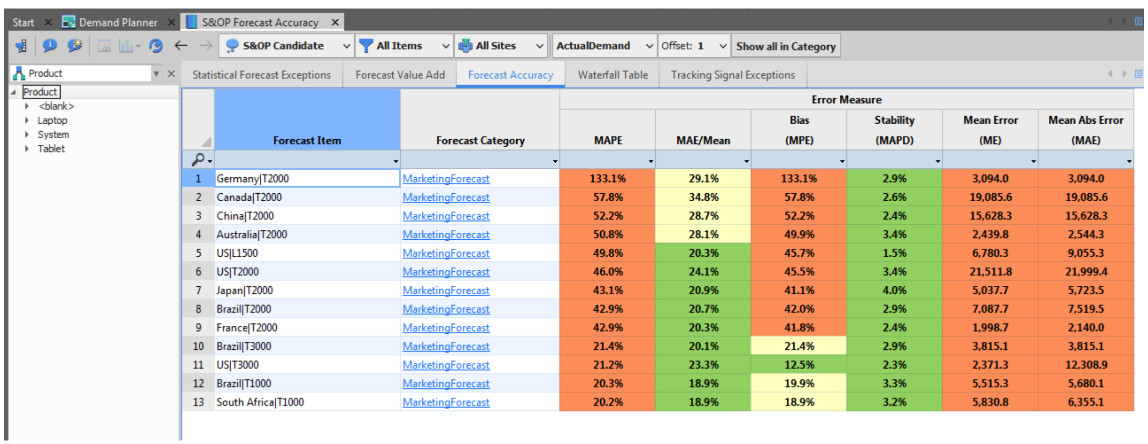
Demand Planner Metrics | Corporate Metrics

Forecast Accuracy Exceptions

Forecast Category	Error Measure MAPE	MAE/ Mean	MPE	MAPD	ME	MAE	Total # of Error
DemandPlan	13		9		80	81	183
MarketingForecast	13		9		79	81	182
FinanceOperatingPla	12		9		80	81	182
SalesForecast	12		6		81	81	180
Statistical	9		9		75	81	174
Grand Sum	**59**	**0**	**42**	**0**	**395**	**405**	**901**

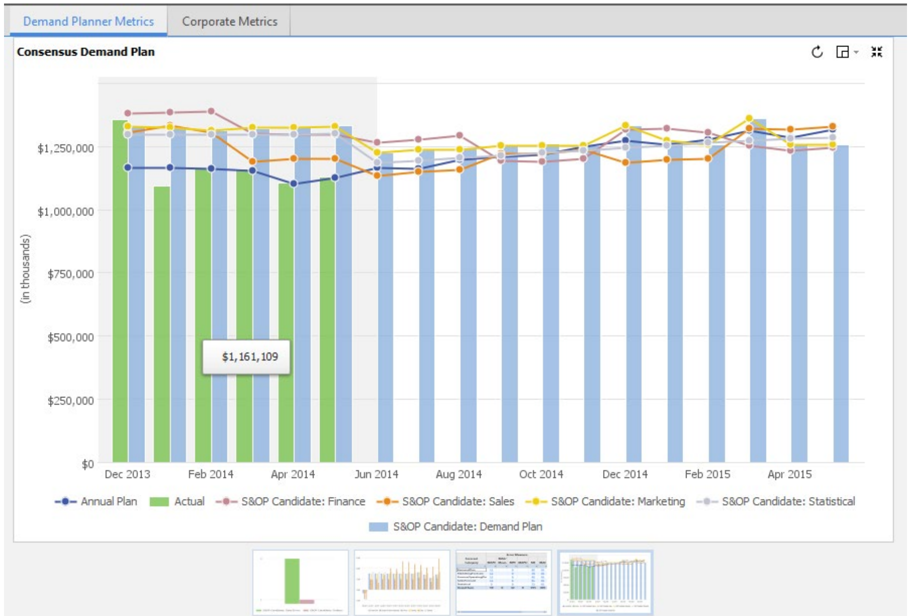
This shows the **Forecast Accuracy Exceptions** for a selection of categories as defined in the **S&OP Forecast Accuracy Widgets** workbook and the **Forecast Accuracy Exceptions** worksheet. The numbers that are reported are a count of Forecast Items where the column error measure exceeds a maximum defined in the **S&OP Forecast Accuracy Widgets** workbook. It summarizes where the error measure is above the critical limit for the forecast category. The accuracy is calculated by comparing the previous 12 months of forecast data to the selected actual demand category.

Clicking on any of the drill-to-detail values will open the **S&OP Forecast Accuracy** workbook to the **Forecast Accuracy** worksheet.



	Forecast Item	Forecast Category	Error Measure MAPE	MAE/Mean	Bias (MPE)	Stability (MAPD)	Mean Error (ME)	Mean Abs Error (MAE)
1	Germany|T2000	MarketingForecast	133.1%	29.1%	133.1%	2.9%	3,094.0	3,094.0
2	Canada|T2000	MarketingForecast	57.8%	34.8%	57.8%	2.6%	19,085.6	19,085.6
3	China|T2000	MarketingForecast	52.2%	28.7%	52.2%	2.4%	15,628.3	15,628.3
4	Australia|T2000	MarketingForecast	50.8%	28.1%	49.9%	3.4%	2,439.8	2,544.3
5	US|L1500	MarketingForecast	49.8%	20.3%	45.7%	1.5%	6,780.3	9,055.3
6	US|T2000	MarketingForecast	46.0%	24.1%	45.5%	3.4%	21,511.8	21,999.4
7	Japan|T2000	MarketingForecast	43.1%	20.9%	41.1%	4.0%	5,037.7	5,723.5
8	Brazil|T2000	MarketingForecast	42.9%	20.7%	42.0%	2.9%	7,087.7	7,519.5
9	France|T2000	MarketingForecast	42.9%	20.3%	41.8%	2.4%	1,998.7	2,140.0
10	Brazil|T3000	MarketingForecast	21.4%	20.1%	21.4%	2.9%	3,815.1	3,815.1
11	US|T3000	MarketingForecast	21.2%	23.3%	12.5%	2.3%	2,371.3	12,308.9
12	Brazil|T1000	MarketingForecast	20.3%	18.9%	19.9%	3.3%	5,515.3	5,680.1
13	South Africa|T1000	MarketingForecast	20.2%	18.9%	18.9%	3.2%	5,830.8	6,355.1

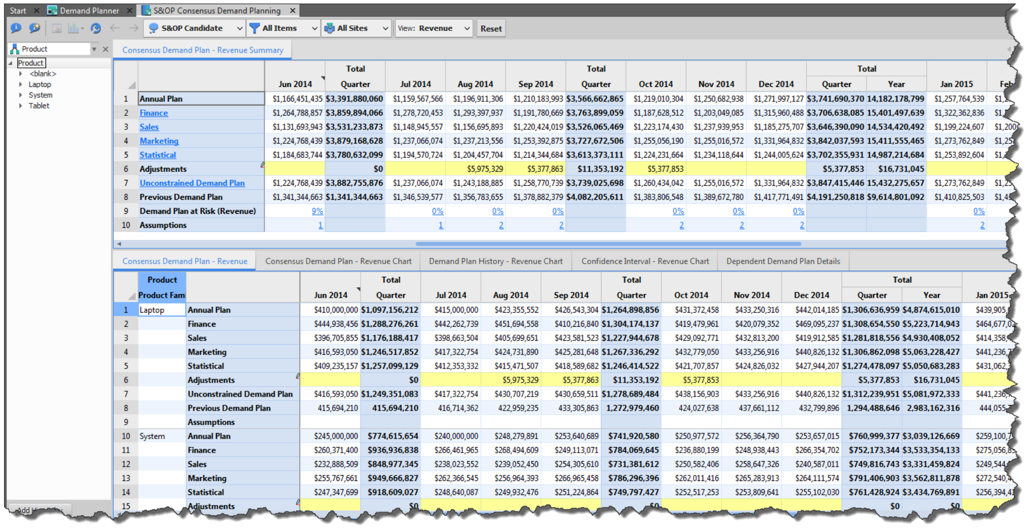
4.1.4.1.4. Consensus Demand Plan



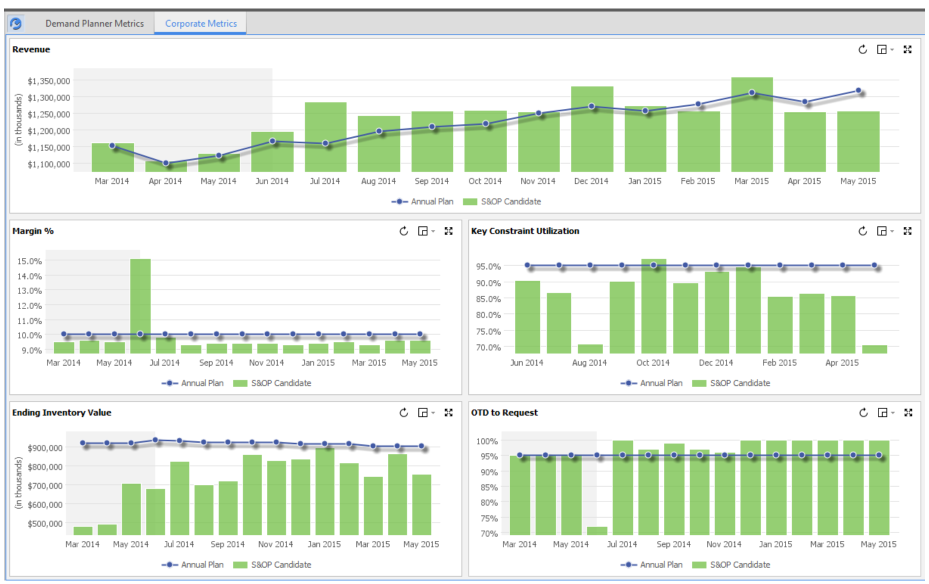
Compares the demand plan to the functional forecast input streams and the statistical forecast. You can also compare the demand plan to the annual plan. Historical data is displayed with a gray background.

This widget is derived from the **S&OP Consensus Demand Planning Widgets** workbook and **Consensus Demand Plan - Revenue Chart** worksheet. The Annual Plan is derived from ForecastDetail records with a category set to $ DP_DemandPlanValueTarget (or 'DemandPlanValueTarget' by default).

To see detailed data click one of the bars summarizing the demand plan for a specific future period and RapidResponse the **S&OP Consensus Demand Planning** workbook will open.



4.1.4.2. Corporate Metrics Tab



The Corporate Metrics Dashboard tab is described earlier in the shared resources section.

4.2. Scorecards and Metrics

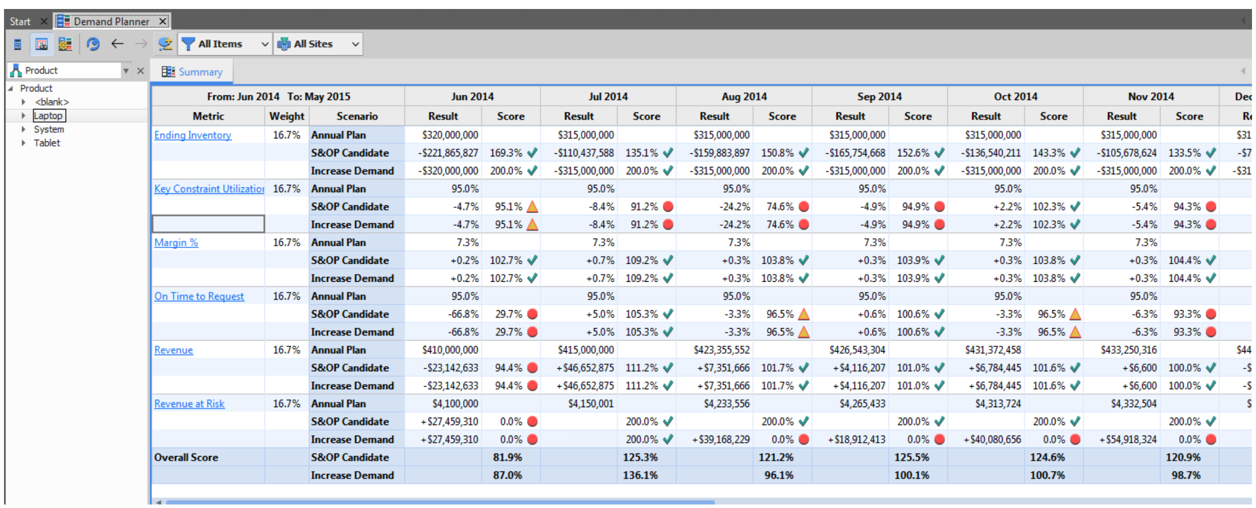
Outline the applicable Scorecard and its construction for the application. Insert the standard Scorecard snippets(s) in this section. If additional configuration details are required, include those as regular text.

4.2.1. Demand Planner Scorecard

Scorecard Name	Demand Planner
Purpose	Allows the demand planner to compare and evaluate key corporate performance metrics from alternate scenarios against the annual plan and each other. The user can see the impact of different scenarios on overall high level corporate metrics and make a selection about which scenario to choose.

Definitions:

- Annual Plan – a set of ForecastDetail records where the HistoricalDemandCategory has a Type.ProcessingRule of Target. Each metric in the Scorecard defines its own Annual Plan category in the scorecard properties. The annual plan is defined and maintained in the "Annual plan target scenario" maintained in Global Settings. This defaults to the Baseline scenario.



From: Jun 2014 To: May 2015			Jun 2014		Jul 2014		Aug 2014		Sep 2014		Oct 2014		Nov 2014	
Metric	Weight	Scenario	Result	Score	Result	Score	Result	Score	Result	Score	Result	Score	Result	Score
Ending Inventory	16.7%	Annual Plan	$320,000,000		$315,000,000		$315,000,000		$315,000,000		$315,000,000		$315,000,000	
		S&OP Candidate	-$221,865,827	169.3%	-$110,437,588	135.1%	-$159,883,897	150.8%	-$165,754,668	152.6%	-$136,540,211	143.3%	-$105,678,624	133.5%
		Increase Demand	-$320,000,000	200.0%	-$315,000,000	200.0%	-$315,000,000	200.0%	-$315,000,000	200.0%	-$315,000,000	200.0%	-$315,000,000	200.0%
Key Constraint Utilizatior	16.7%	Annual Plan	95.0%		95.0%		95.0%		95.0%		95.0%		95.0%	
		S&OP Candidate	-4.7%	95.1%	-8.4%	91.2%	-24.2%	74.6%	-4.9%	94.9%	+2.2%	102.3%	-5.4%	94.3%
		Increase Demand	-4.7%	95.1%	-8.4%	91.2%	-24.2%	74.6%	-4.9%	94.9%	+2.2%	102.3%	-5.4%	94.3%
Margin %	16.7%	Annual Plan	7.3%		7.3%		7.3%		7.3%		7.3%		7.3%	
		S&OP Candidate	+0.2%	102.7%	+0.7%	109.2%	+0.3%	103.8%	+0.3%	103.9%	+0.3%	103.8%	+0.3%	104.4%
		Increase Demand	+0.2%	102.7%	+0.7%	109.2%	+0.3%	103.8%	+0.3%	103.9%	+0.3%	103.8%	+0.3%	104.4%
On Time to Request	16.7%	Annual Plan	95.0%		95.0%		95.0%		95.0%		95.0%		95.0%	
		S&OP Candidate	-66.8%	29.7%	+5.0%	105.3%	-3.3%	96.5%	+0.6%	100.6%	-3.3%	96.5%	-6.3%	93.3%
		Increase Demand	-66.8%	29.7%	+5.0%	105.3%	-3.3%	96.5%	+0.6%	100.6%	-3.3%	96.5%	-6.3%	93.3%
Revenue	16.7%	Annual Plan	$410,000,000		$415,000,000		$423,355,552		$426,543,304		$431,372,458		$433,250,316	
		S&OP Candidate	-$23,142,633	94.4%	+$46,652,875	111.2%	+$7,351,666	101.7%	+$4,116,207	101.0%	+$6,784,445	101.6%	+$6,600	100.0%
		Increase Demand	-$23,142,633	94.4%	+$46,652,875	111.2%	+$7,351,666	101.7%	+$4,116,207	101.0%	+$6,784,445	101.6%	+$6,600	100.0%
Revenue at Risk	16.7%	Annual Plan	$4,100,000		$4,150,001		$4,233,556		$4,265,433		$4,313,724		$4,332,504	
		S&OP Candidate	+$27,459,310	0.0%		200.0%		200.0%		200.0%		200.0%		200.0%
		Increase Demand	+$27,459,310	0.0%		200.0%	+$39,168,229	0.0%	+$18,912,413	0.0%	+$40,080,656	0.0%	+$54,918,324	0.0%
Overall Score		S&OP Candidate		81.9%		125.3%		121.2%		125.5%		124.6%		120.9%
		Increase Demand		87.0%		136.1%		96.1%		100.1%		100.7%		98.7%

The above screen shot illustrates the option to bucket by time and show multiple scenarios.

All of the metrics for this scorecard are derived from the **S&OP Corporate Metrics** workbook with the exception of the **Revenue at Risk** metric which is defined in the **S&OP Demand Metrics** workbook.

See the earlier Corporate Metrics Scorecard sub-section of the Shared Resources section for an explanation of all but the Revenue at Risk metric.

1.1.1.1 Revenue at Risk metric

The Revenue at Risk metric displays the value of the revenue for late customer orders. Customer orders are considered late if their available date is later than the disaggregation calendar period of the demand date (configurable).

This does not report all orders where AvailableDate > DueDate. It looks for orders where the AvailableDate is later than the bucket of the DueDate based on SOPAnalyticsConfiguration.DisaggregationCalendar. $DP_RevenueAtRiskIntervalCount is normally set to 1 but you can increase that if you want to flag lateness as more than one DisaggregationCalendar period. The formula is: *Period + $DP_RevenueAtRiskIntervalCount SOPConfiguration!DisaggregationCalendar <= AvailableDate*

The Annual Plan in this case is based on the Revenue At Risk Target. This means the HistoricalDemandCategory.Value='RevenueAtRiskTarget' and the associated profile variable is $DP_RevenueAtRiskTarget.

All of the “revenue” calculations are based on the EffectiveUnitPrice calculated on each of the demands.

There are three detailed worksheets reported with this metric.

Revenue: by Period by Demand

Displays the total and late value for each independent demand, sorted by period, in which one of the following values has changed from scenario to scenario: Available Date, Quantity, Late Quantity, Revenue, or Revenue at Risk.

Column	Description
Period	The reporting period of the demand due date.
Order - Number:Line	The demand order number and independent demand line identification for the order separated by a colon.
Order - Type	The demand type for the demand order.
Order - Item	The part name for the independent demand.
Order - Location	The part site associated with the independent demand.
Available Date	The date at which the demand is expected to be available for delivery.
Quantity	The quantity of the demand order.

Late Quantity	The quantity of the demand that is expected to available after the demand date.
Revenue	The revenue associated with the demand order.
Revenue at Risk	The revenue associated with the demand order that is expected to be late.

Revenue: by Manufacturing Region

Displays the total and late value for each independent demand, sorted by part region, in which one of the following values has changed from scenario to scenario: Quantity, Late Quantity, Revenue, or Late Revenue.

Column	Description
Group	The regional group that the order site is associated with. This is a new field path off of the Site table. It is the Part.Site.Mfg::Region.RegionGroup.Id
Region	The region that the order site is associated with. This is a new field path off of the Site table. It is the Part.Site.Mfg::Region.Id
Location	The site associated with the order part. (Part.Site)
Period	The reporting period of the demand due date.
Quantity	The quantity of the demand order.
Late Quantity	The quantity of the demand that is expected to available after the demand date.
Revenue	The revenue associated with the demand order.
Revenue at Risk	The revenue associated with the demand order that is expected to be late.

Revenue: by Customer Region

Displays the total and late value for each independent demand, sorted by region, in which one of the following values has changed from scenario to scenario: Quantity, Late Quantity, Revenue, or Late Revenue.

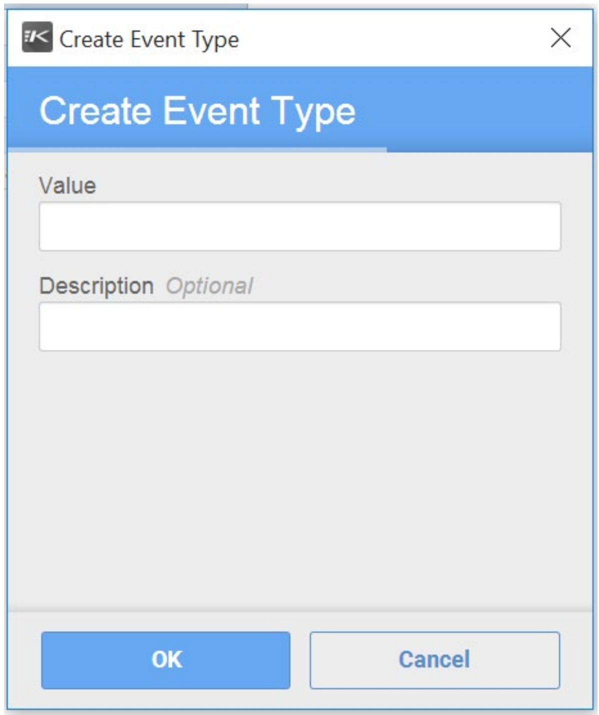
Column	Description
Group	The regional group that the order site is associated with. This is a new field path off of the Site table. It is the Part.Site.Mfg::Region.RegionGroup.Id
Region	The region that the order site is associated with. This is a new field path off of the Site table. It is the Part.Site.Mfg::Region.Id
Customer	The customer associated with the order. (Order.Customer.Id or PartCustomer.Customer.Id)
Period	The reporting period of the demand due date.
Quantity	The quantity of the demand order.

Late Quantity	The quantity of the demand that is expected to available after the demand date.
Revenue	The revenue associated with the demand order.
Revenue at Risk	The revenue associated with the demand order that is expected to be late.

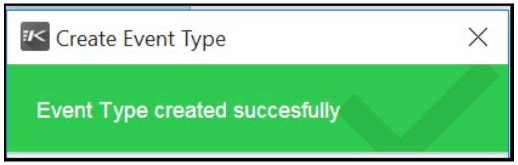
4.3. Forms

The following Forms are available for the Demand Planning Process (specifically Event Planning):

4.3.1. Create Event Type



The Create Event Type form is called from within **Manage Events** workbook when Insert Records button is pressed on the **Event Type** worksheet.



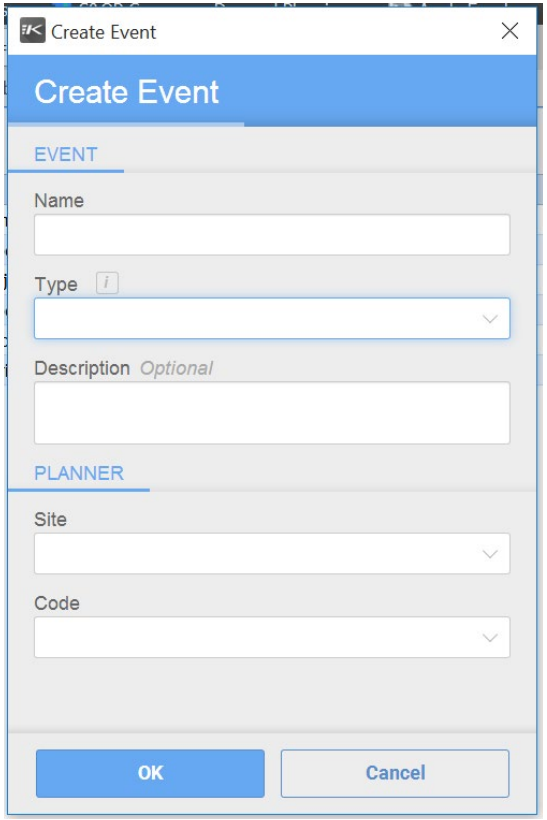
The following are the parameters to this new form, which will create new records in the **EventType** table.

- Value - Set the unique Value for Event Type.
- Description - Set the Description for Event Type.

The Event Types created by the above Form will then be used in the Create Event Form to create multiple Events based on the Event Types.

4.3.2. Create Event

The Create Event form is called from within **Manage Events** workbook when Insert Records button is pressed on the **Event** worksheet.



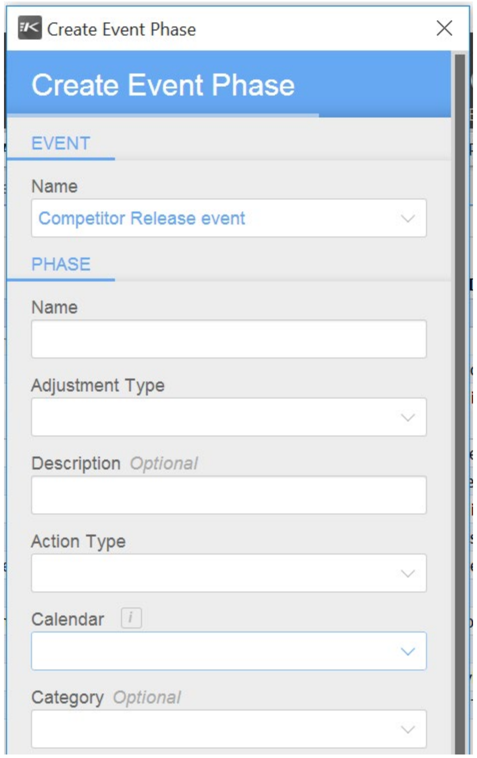
The following are the parameters to this new form, which will create new records in the **Event** table.

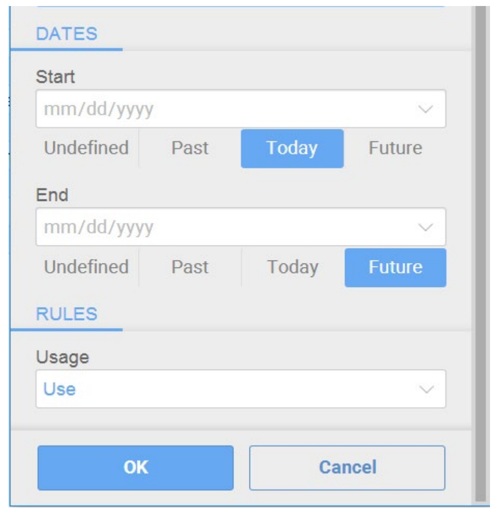
- Name - The name of the event.
- Type - The type of the event.
- Description - A description of the event.
- Site - The site of the planner code that is associated with the event.
- Code - The planner code that is associated with the event.

The Event created by the above Form will then be used in the Event Phase Form to create multiple Event Phases based on the Event created using above.

4.3.3. Create Event Phase

The Create Event Phase form is called from within **Manage Events** workbook when Insert Records button is pressed on the **Event Phase** worksheet.





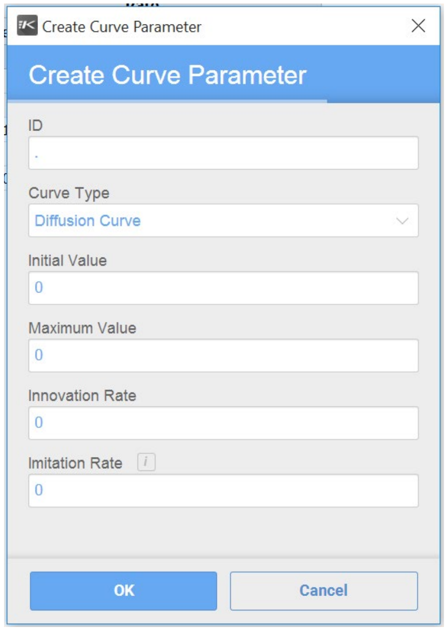
The following are the parameters to this new form, which will create new records in the **EventPhase** table.

- Event Name - Select the event that this phase is associated with.
- Phase Name - A name of this event phase.
- Adjustment Type - Describes what component of the forecast this phase will affect (Quantity or Price)
- Description - A description of this event phase.
- Action Type - Describes what technique the event phase will use to adjust the forecast.
- Calendar - Calendar to be used when applying the adjustment specified in this event phase.
- Category - If defined, the category that this event phase can be applied to. If left blank, the event phase can be applied to all categories.
- Start Date - The start date of the phase (inclusive).
- End Date - The end date of the phase (exclusive).
- Usage - The usage rule determines if this event phase will be used or ignored.

The Event Phase created by the above Form will then be used in the Apply Event Form to create multiple Events using this Event Phase.

4.3.4. Create Curve Parameter

The Create Curve Parameter form is called from within **Manage Events** workbook when Insert Records button is pressed on the **Curve Parameters** worksheet.

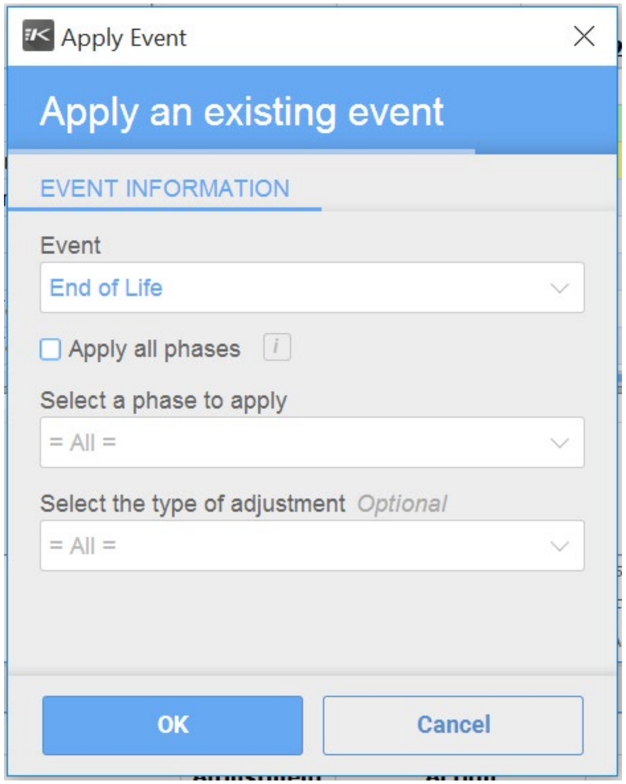


The following are the parameters to this new form, which will create new records in the **CurveParameters** table.

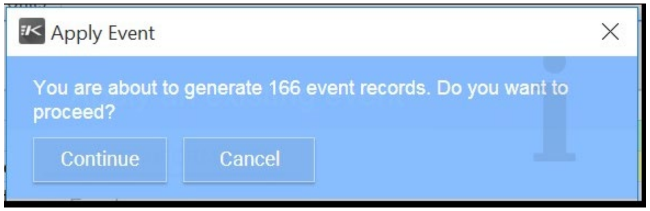
- ID - The unique identifier for Curve Parameter.
- Curve Type - Select the Curve Type from available Curve Types.
- Initial Value - Set the Initial Value for the Curve.
- Maximum Value - Set the Maximum Value for Diffusion Curve Type.
- Innovation Rate - Set the Innovation Rate for Diffusion Curve Type.
- Imitation Rate - Set the Innovation Rate for Diffusion Curve Type.

The Curve Parameters created by the above Form will then be used in the Event Phase Form to create multiple Event Phases.

4.3.5. Apply Event



The Apply Event form is called from Sales, Marketing and Finance workbooks when 'Apply an existing event' button from toolbar is pressed while on the **Event or Event Phase** worksheets.



The following are the parameters to this new form.

- Event - The name of event that will be applied to the forecast.
- Apply all phases (Checkbox) - All phases of an event will be applied to the forecast if this checkbox is checked.
- Select a phase to apply - You can apply only a single phase of the event by selecting a phase here.
- Select the type of Adjustment - Determines if the event phase will apply a price or quantity adjustment.

4.4. Scripts

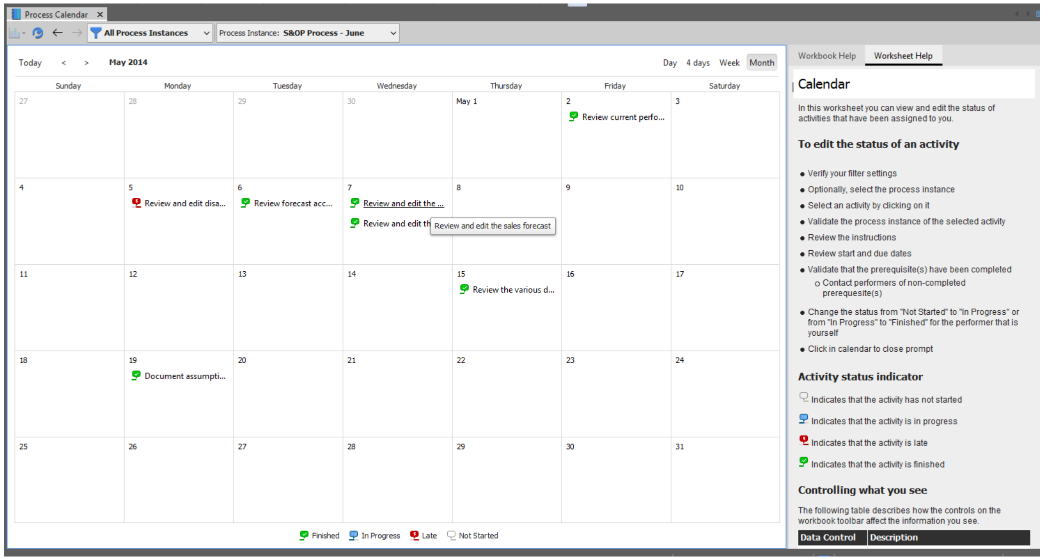
The following Scripts are available for the Demand Planning Process (specifically Event Planning):

Script Name	Description	Associated Form
Create Event	Creates a new Event in Event Table.	Create Event
Create Event Phase	Creates a new Event Phase in EventPhase Table.	Create Event Phase
Create Event Curve	Creates a new Event Curve Parameters in CurveParameters Table.	Create Curve Parameter
Create Event Type	Creates a new Event Type in EventType table.	Create Event Type
Apply Event	Applies the Event.	Apply Event
Copy Event	Copies the existing Event.	Copy Event

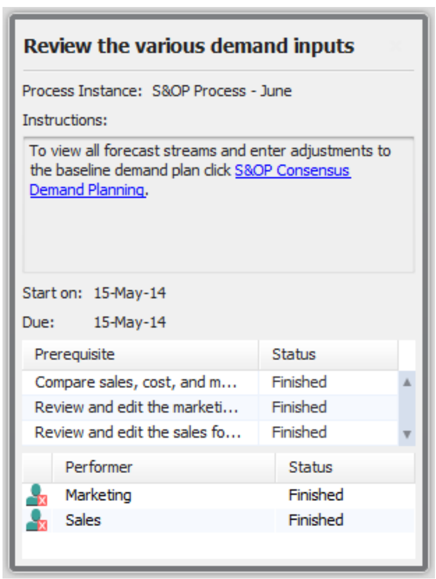
4.5. Workbooks

4.5.1. Process Calendar

Workbook Name	Process Calendar
Purpose	This workbook is intended for process activities performers to update the status of the scheduled activities assigned to them. Use this workbook to: • For all process instances that have activities assigned to you: • Display all process activities on the day that they are scheduled to start • Provide an activity visual status indicator • Edit the status of activities
Is Standard PDR?	Yes
Key Data Elements	Activity performers, description, prerequisites, status and start and due dates by due date.
Key Calculations	Determination of whether or not an activity is late.



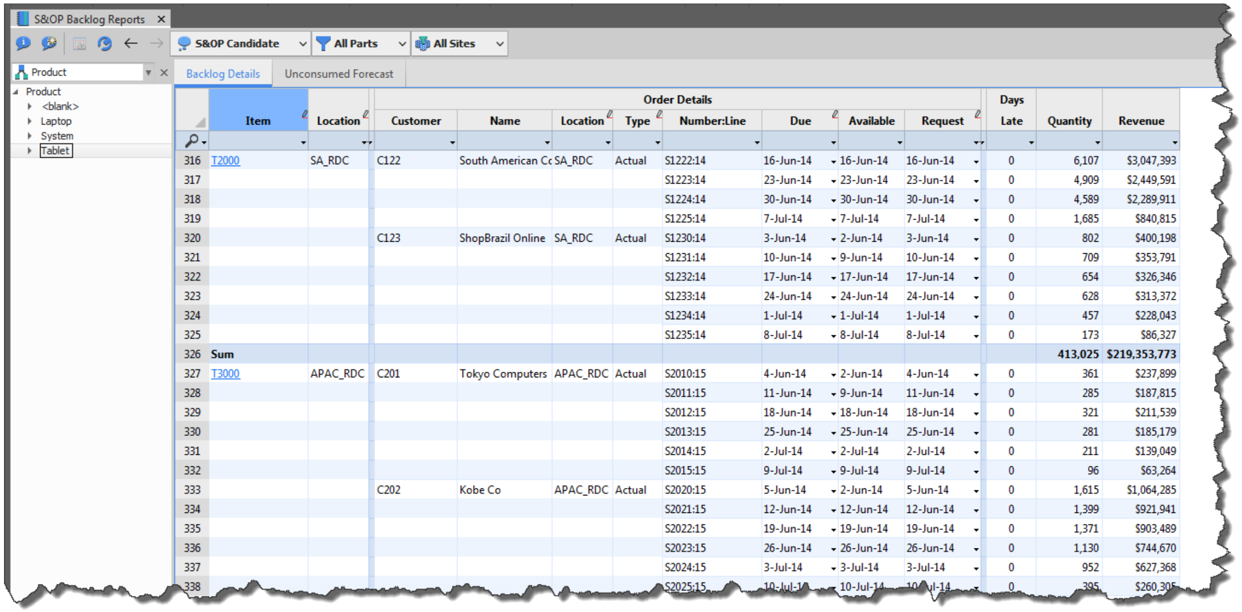
Clicking on any activity, the activity status and detailed information is made available. Note that this data is maintained in a specialized hidden scenario.



The status of each of the user-related tasks can be edited in this dialog.

4.5.2. S&OP Backlog Reports

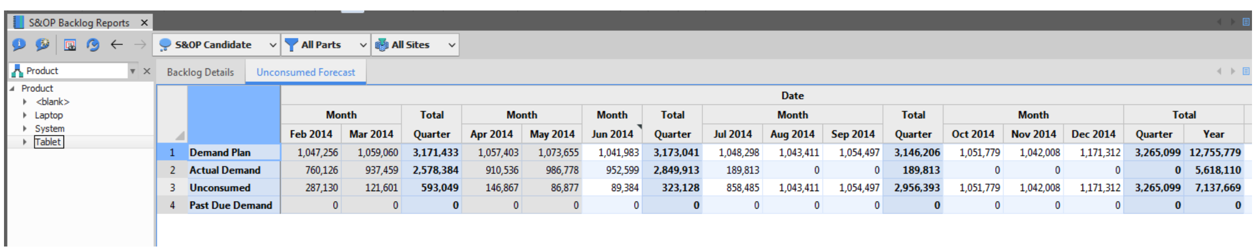
Workbook Name	S&OP Backlog Reports
Purpose	This workbook is normally opened from the **S&OP Backlog Treemap** report but may be opened individually if shared. The workbook reports specific backlog details and a summary of unconsumed forecast.
Is Standard PDR?	Yes
Key Data Elements	For backlog details, by specific lines, this reports: Due Date, Request Date, Available Date Days Late (from Due Date) Quantity Revenue For unconsumed forecast, this reports Demand plan (past and current consensus forecast) Actual demand (backlog) Unconsumed forecast Past due demand (if recorded as $DP_PastDueBacklogCategory in HistoricalDemandActuals)
Key Calculations	Availability of backlog (days late and available date) Revenue of backlog Consensus forecast (unconstrained demand plan) reported as demand plan Forecast consumption



	Item	Location	Order Details: Customer	Name	Location	Type	Number:Line	Due	Available	Request	Days Late	Quantity	Revenue
316	T2000	SA_RDC	C122	South American Cc	SA_RDC	Actual	S1222:14	16-Jun-14	16-Jun-14	16-Jun-14	0	6,107	$3,047,393
317							S1223:14	23-Jun-14	23-Jun-14	23-Jun-14	0	4,909	$2,449,591
318							S1224:14	30-Jun-14	30-Jun-14	30-Jun-14	0	4,589	$2,289,911
319							S1225:14	7-Jul-14	7-Jul-14	7-Jul-14	0	1,685	$840,815
320			C123	ShopBrazil Online	SA_RDC	Actual	S1230:14	3-Jun-14	2-Jun-14	3-Jun-14	0	802	$400,198
321							S1231:14	10-Jun-14	9-Jun-14	10-Jun-14	0	709	$353,791
322							S1232:14	17-Jun-14	17-Jun-14	17-Jun-14	0	654	$326,346
323							S1233:14	24-Jun-14	24-Jun-14	24-Jun-14	0	628	$313,372
324							S1234:14	1-Jul-14	1-Jul-14	1-Jul-14	0	457	$228,043
325							S1235:14	8-Jul-14	8-Jul-14	8-Jul-14	0	173	$86,327
326	**Sum**											**413,025**	**$219,353,773**
327	T3000	APAC_RDC	C201	Tokyo Computers	APAC_RDC	Actual	S2010:15	4-Jun-14	2-Jun-14	4-Jun-14	0	361	$237,899
328							S2011:15	11-Jun-14	9-Jun-14	11-Jun-14	0	285	$187,815
329							S2012:15	18-Jun-14	18-Jun-14	18-Jun-14	0	321	$211,539
330							S2013:15	25-Jun-14	25-Jun-14	25-Jun-14	0	281	$185,179
331							S2014:15	2-Jul-14	2-Jul-14	2-Jul-14	0	211	$139,049
332							S2015:15	9-Jul-14	9-Jul-14	9-Jul-14	0	96	$63,264
333			C202	Kobe Co	APAC_RDC	Actual	S2020:15	5-Jun-14	2-Jun-14	5-Jun-14	0	1,615	$1,064,285
334							S2021:15	12-Jun-14	12-Jun-14	12-Jun-14	0	1,399	$921,941
335							S2022:15	19-Jun-14	19-Jun-14	19-Jun-14	0	1,371	$903,489
336							S2023:15	26-Jun-14	26-Jun-14	26-Jun-14	0	1,130	$744,670
337							S2024:15	3-Jul-14	3-Jul-14	3-Jul-14	0	952	$627,368
338							S2025:15	10-Jul-14	10-Jul-14	10-Jul-14	0	395	$260,305

The **Backlog Details** worksheet displays all of the IndependentDemand records where Order.Type.ProcessingRule in ('SalesActual','Regular') and EffectiveDemand > 0.

The revenue reported here is the effective demand quantity (Quantity) * the effective unit price. This would not include any portion of the line that had already been shipped (ShippedQty).



		Date: Month Feb 2014	Month Mar 2014	Total Quarter	Month Apr 2014	Month May 2014	Month Jun 2014	Total Quarter	Month Jul 2014	Month Aug 2014	Month Sep 2014	Total Quarter	Month Oct 2014	Month Nov 2014	Month Dec 2014	Total Quarter	Total Year
1	**Demand Plan**	1,047,256	1,059,060	**3,171,433**	1,057,403	1,073,655	1,041,983	**3,173,041**	1,048,298	1,043,411	1,054,497	**3,146,206**	1,051,779	1,042,008	1,171,312	**3,265,099**	**12,755,779**
2	**Actual Demand**	760,126	937,459	**2,578,384**	910,536	986,778	952,599	**2,849,913**	189,813	0	0	**189,813**	0	0	0	**0**	**5,618,110**
3	**Unconsumed**	287,130	121,601	**593,049**	146,867	86,877	89,384	**323,128**	858,485	1,043,411	1,054,497	**2,956,393**	1,051,779	1,042,008	1,171,312	**3,265,099**	**7,137,669**
4	**Past Due Demand**	0	0	**0**	0	0	0	**0**	0	0	0	**0**	0	0	0	**0**	**0**

The **Unconsumed Forecast** worksheet displays bucketed quantities for the Demand Plan, Actual Demand, Unconsumed Forecast and Past Due Demand summarized for all items in the selection.

The Demand Plan is derived from both history and projected consensus forecast. Historically, it is taken from the last recorded series of HistoricalDemandSeriesDetail records where the Category is set to $DP_DemandTypeForConsensus (DemandPlan). For future values, it is taken from the calculated ConsensusForecast.

The Actual Demand is also derived from derived from both history and projected actual demands. Historically, it is taken from the HistoricalDemandActual records where the Category is set to $DP_ActualsCategory (Shipment). For future values, it is taken from the IndependentDemand records where Order.Type.ProcessingRule in ('SalesActual','Regular') and EffectiveDemand > 0. This is the same set of data as the Backlog Details.

The Unconsumed row is the difference between Demand Plan and Actual Demand.

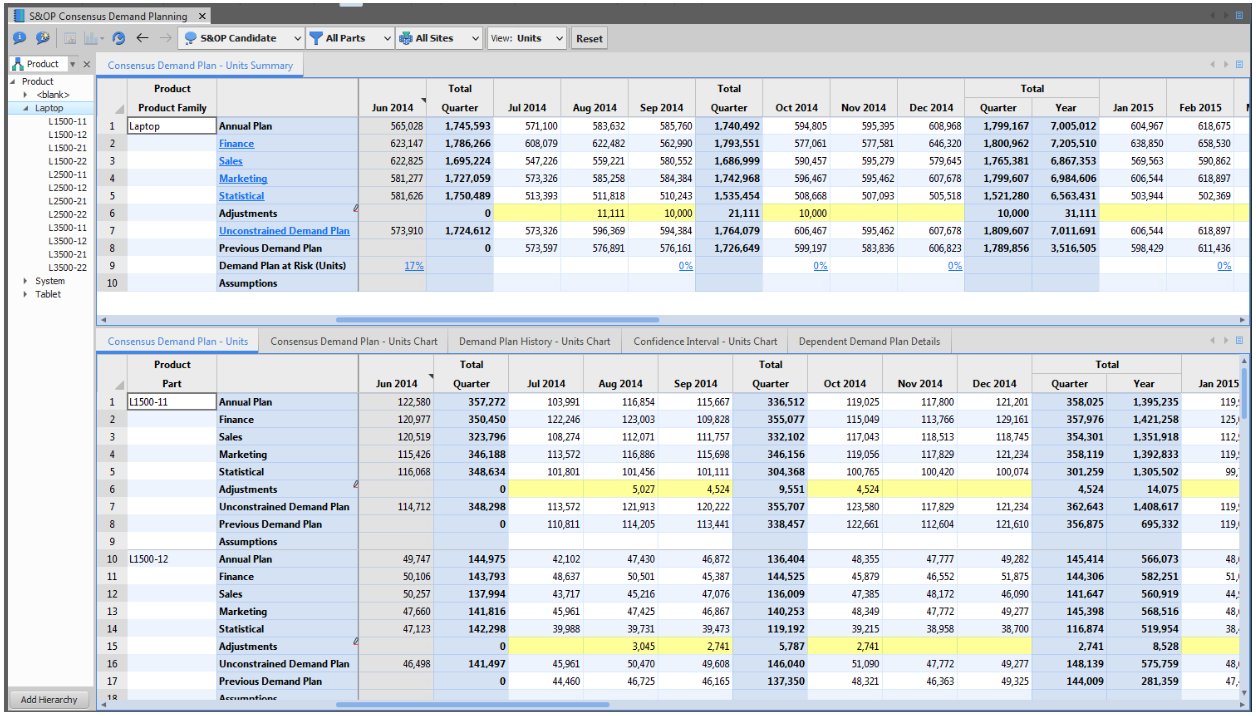
The Past Due row is talking about records stored in the HistoricalDemandActual table where the Header.Category is set to $DP_PastDueBacklogCategory (PastDue) and the Date + 1 S&OP cycle forecast calendar is before the forecast start date. This usually means anything from the previous months (or S&OP planning cycle) and earlier that was not shipped on time. This is purely a historical measure and does not include any projected lateness or misses. This data is recorded automatically by the **S&OP Insert Backlog into History** scheduled task which calls the **Write Backlog** data change in the **S&OP Write History Records** workbook and includes all IndependentDemand records where (DueDate < Part.PlanningCalendars.RunDate.FirstDate + 0 $DP_CycleForecastCalendar) and EffectiveDemand > 0 and Order.Type.ProcessingRule = 'SalesActual'.

4.5.3. S&OP Consensus Demand Planning

Workbook Name	S&OP Consensus Demand Planning
Purpose	Use this workbook to generate a consensus demand forecast that combines the statistical demand forecast with the functional forecasts from sales, marketing and finance. You can adjust forecasts to more accurately identify, interpret, and react to changes in demand. In addition, you can use this workbook to compare the statistical forecast generated for a product with the dependent demand calculated as a result of exploding through a planning BOM.
Is Standard PDR?	Yes
Key Data Elements	• Annual Plan • Cleansed Finance, Sales and Marketing forecasts • Statistical forecast • Editable row of adjustments to the consensus forecast • Previous consensus forecast • Assumptions
Key Calculations	Consensus forecast (unconstrained demand plan) Demand Plan at risk

Definitions:

Annual Plan – a set of ForecastDetail records where the HistoricalDemandCategory has a Type.ProcessingRule of Target. If you are reporting Units, then the reported Annual Plan has a category of $DP_DemandPlanValueTarget (DemandPlanValueTarget) if you are reporting by Revenue or $DP_DemandPlanVolumeTarget (DemandPlanVolumeTarget) if you are reporting by Units.



	Product Product Family		Jun 2014	Total Quarter	Jul 2014	Aug 2014	Sep 2014	Total Quarter	Oct 2014	Nov 2014	Dec 2014	Total Quarter	Total Year	Jan 2015	Feb 2015
1	Laptop	Annual Plan	565,028	1,745,593	571,100	583,632	585,760	1,740,492	594,805	595,395	608,968	1,799,167	7,005,012	604,967	618,675
2		Finance	623,147	1,786,266	608,079	622,482	562,990	1,793,551	577,061	577,581	646,320	1,800,962	7,205,510	638,850	658,530
3		Sales	622,825	1,695,224	547,226	559,221	580,552	1,686,999	590,457	595,279	579,645	1,765,381	6,867,353	569,563	590,862
4		Marketing	581,277	1,727,059	573,326	585,258	584,384	1,742,968	596,467	595,462	607,678	1,799,607	6,984,606	606,544	618,897
5		Statistical	581,626	1,750,489	513,393	511,818	510,243	1,535,454	508,668	507,093	505,518	1,521,280	6,563,431	503,944	502,369
6		Adjustments		0		11,111	10,000	21,111	10,000			10,000	31,111		
7		Unconstrained Demand Plan	573,910	1,724,612	573,326	596,369	594,384	1,764,079	606,467	595,462	607,678	1,809,607	7,011,691	606,544	618,897
8		Previous Demand Plan		0	573,597	576,891	576,161	1,726,649	599,197	583,836	606,823	1,789,856	3,516,505	598,429	611,436
9		Demand Plan at Risk (Units)	17%				0%		0%		0%				0%
10		Assumptions													

	Product Part		Jun 2014	Total Quarter	Jul 2014	Aug 2014	Sep 2014	Total Quarter	Oct 2014	Nov 2014	Dec 2014	Total Quarter	Total Year	Jan 2015
1	L1500-11	Annual Plan	122,580	357,272	103,991	116,854	115,667	336,512	119,025	117,800	121,201	358,025	1,395,235	119,
2		Finance	120,977	350,450	122,246	123,003	109,828	355,077	115,049	113,766	129,161	357,976	1,421,258	125,
3		Sales	120,519	323,796	108,274	112,071	111,757	332,102	117,043	118,513	118,745	354,301	1,351,918	112,
4		Marketing	115,426	346,188	113,572	116,886	115,698	346,156	119,056	117,829	121,234	358,119	1,392,833	119,
5		Statistical	116,068	348,634	101,801	101,456	101,111	304,368	100,765	100,420	100,074	301,259	1,305,502	99,
6		Adjustments		0		5,027	4,524	9,551	4,524			4,524	14,075	
7		Unconstrained Demand Plan	114,712	348,298	113,572	121,913	120,222	355,707	123,580	117,829	121,234	362,643	1,408,617	119,
8		Previous Demand Plan		0	110,811	114,205	113,441	338,457	122,661	112,604	121,610	356,875	695,332	119,
9		Assumptions												
10	L1500-12	Annual Plan	49,747	144,975	42,102	47,430	46,872	136,404	48,355	47,777	49,282	145,414	566,073	48,
11		Finance	50,106	143,793	48,637	50,501	45,387	144,525	45,879	46,552	51,875	144,306	582,251	51,
12		Sales	50,257	137,994	43,717	45,216	47,076	136,009	47,385	48,172	46,090	141,647	560,919	44,
13		Marketing	47,660	141,816	45,961	47,425	46,867	140,253	48,349	47,772	49,277	145,398	568,516	48,
14		Statistical	47,123	142,298	39,988	39,731	39,473	119,192	39,215	38,958	38,700	116,874	519,954	38,
15		Adjustments		0		3,045	2,741	5,787	2,741			2,741	8,528	
16		Unconstrained Demand Plan	46,498	141,497	45,961	50,470	49,608	146,040	51,090	47,772	49,277	148,139	575,759	48,
17		Previous Demand Plan		0	44,460	46,725	46,165	137,350	48,321	46,363	49,325	144,009	281,359	47,

This workbook is the resource used by the Demand Planner in the development of the unconstrained demand plan. It is used to "adjust" the demand plan to the from the default calculated consensus forecast. The workbook also contains the command to delete existing adjustments and overrides.

The only part of this worksheet that is editable is the "Adjustments" row (by default). However, an author can un-hide the "Promotional Adjustments" and "Forecast Override" rows which are also editable. Disaggregation for these rows is typically driven by the calculated consensus forecast in order to only make adjustments to those Part/Customer/Categories that are actually contributing to the demand plan. Adjustment categories have a HistoricalDemandCategoryType.ProcessingRule of Forecast and a ConsensusForecastWeight of 100%. However, the Override category will have a HistoricalDemandCategoryType.ProcessingRule of ForecastOverride and the weight is, therefore, irrelevant.

There is another HistoricalDemandCategoryType.ProcessingRule of ReBalancingForecastOverride that is available and not used in the PDRs. This type of adjustment is a final adjustment AFTER the override has been applied. For example, a forecast that has been explicitly overridden with a quantity of 100 can be further "adjusted" by this category by adding or subtracting some quantity to the override. The **S&OP Demand Supply Balancing** workbook allows you to edit this.

There are a variety of drill-to-details capabilities exposed here as well. The four row headers "Finance", "Sales", "Marketing" and "Statistical" will each open a workbook corresponding specifically to those forecast categories ("**S&OP Finance Operating Plan**", "**S&OP Sales Forecast**", "**S&OP Marketing Forecast**" and "**S&OP Statistical Forecasting**"). These are described below.

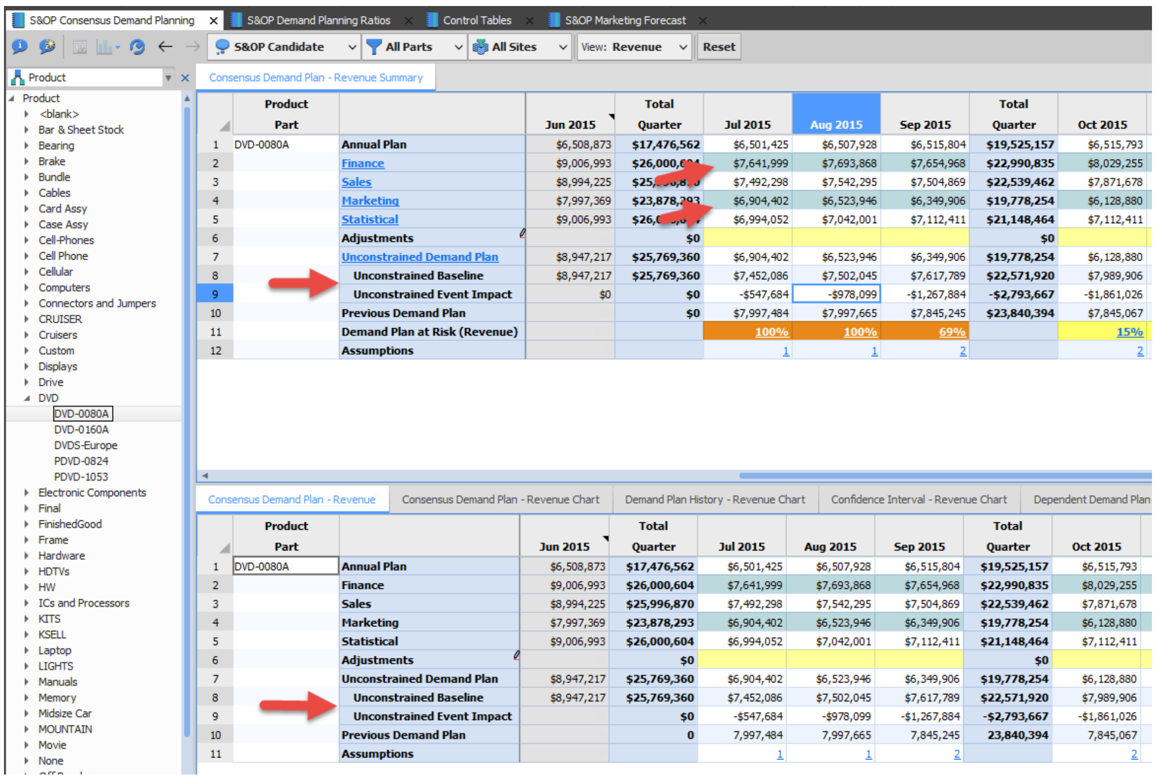
The other row header drill-to-details is on the “Unconstrained Demand Plan” header and will open the “**S&OP Demand Planning Ratios**” workbook to give the planner a chance to adjust the weights or ratios from each forecast demand stream that will drive the calculated consensus forecast.

Within the data, there are two additional drill-to-details links. The data in the “Demand at Risk” row will open the “**S&OP Demand Plan at Risk Treemap**” workbook to get more details about the consensus demand plan that is at risk of not being fulfilled by the supply plan.

Finally, the number displayed in the “Assumptions” row indicate the total number of assumptions effective for the period. Clicking on this number will open a worksheet in the bottom pane listing those specific assumptions.

Also available in the bottom pane, are a variety of charts showing the consensus demand plan, demand plan history, confidence intervals and dependent demand plan. As with other places in our reports, the confidence interval charts only show one side of the interval (either the upper bound or the lower bound). The other side is in the underlying report but would need to be un-hidden by an author.

If Event Planning is enabled, two new rows are added Unconstrained Baseline and Unconstrained Event Adjustment showing the impact any event phases active in that period has on the Uncostrained Demand Plan. There will be conditional formatting in place to show the user which forecast categories have active events in place. The user can drill into those categories in order to review the applied events. Charts will also include this two new rows.

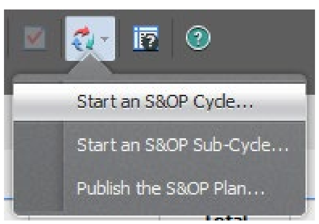


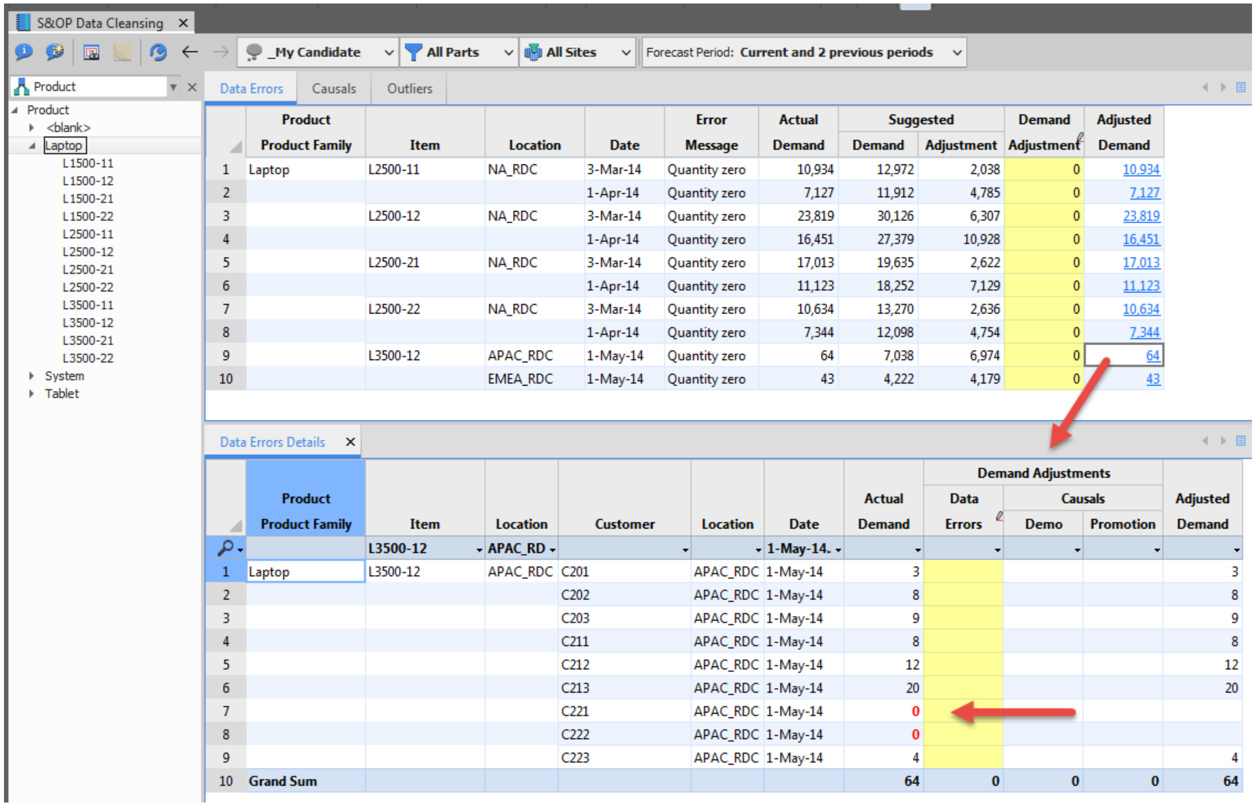
4.5.4. S&OP Data Cleansing

Workbook Name	S&OP Data Cleansing
Purpose	Use this workbook to ensure that the historical data used to generate the statistical forecast is free of errors and anomalies. Cleanse historical data to remove data errors, causal factors, and outliers. It is (HIGHLY) recommended that you cleanse the data in the following order: Data errors Causal factors Outliers
Is Standard PDR?	Yes
Key Data Elements	Bucketed historical demand actuals with and without adjustments. Various causal factors that affect those actuals.
Key Calculations	Moving average of "adjusted" historical demand actuals, standard deviations of adjusted actuals from that moving average.

This is the primary workbook used by the Demand Planner role to perform the Sales Data Conditioning function. This is a prerequisite activity for both Statistical Forecast Generation and for Demand Input Collection. The result of this "cleansing" will have an effect on disaggregation of the Finance, Sales and Marketing forecasts as well as affecting the calculated results of the statistical forecast (and its associated disaggregation).

This is the first step in the S&OP Cycle (not the S&OP Process). That is, once the S&OP Process Owner starts a new S&OP Cycle, then this will be the first task that must be completed using the S&OP Data Cleansing workbook. It is either opened directly by the Demand Planner or accessed indirectly through the **Condition Sales Data** task flow or the **Demand Planner** dashboard.



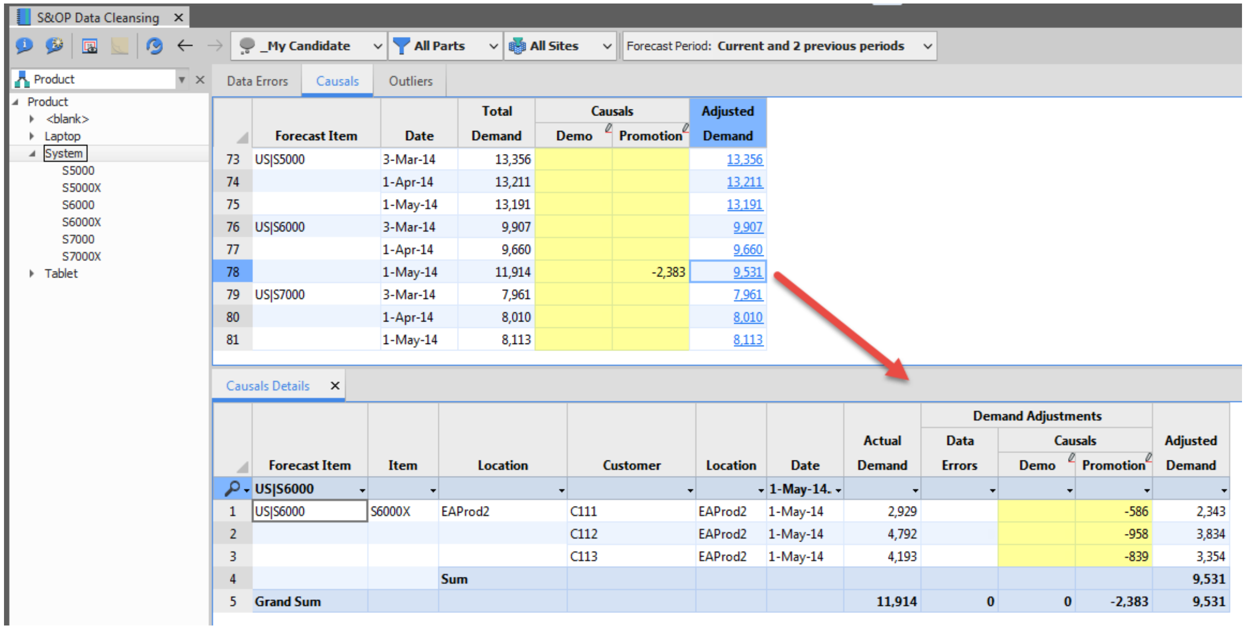


In the Data Errors worksheet, you will see instances where there is either no HistoricalDemandActual data for a part (Error Message = “No record”) or, if there is HistoricalDemandActual data but the quantity is zero (Error Message = “Quantity zero”). It is, of course, possible that this data condition is legitimate and can be ignored. However, this check will highlight the other cases as well where there really was some actual shipments but they have not been recorded. The idea here is to either get the extract fixed when there is missing data or to make adjustments in the worksheet for those cases where there is known missing historical data.

If it is clear that there is missing data but the correct values are not known, the user may elect to simply copy the “Suggested Adjustment” quantity into the “Demand Adjustment” in order to (at least temporarily) set the effective historical quantity to a moving average quantity. The number of intervals used in the moving average calculation is defined as a workbook variable and is, by default, set to 3. The calendar for these intervals is defined by the ForecastItemParameters Type.IntervalsCalendar.

When the Demand Planner enters an adjustment here, it is inserted into the CausalFactorDetail table (rather than directly editing historical demand actuals) with a CausalFactor.Category of ‘DemandAdjustment’.

An author can configure the top sheet to summarize by Item - Location (Site) – Customer instead of or in addition to Item – Location.



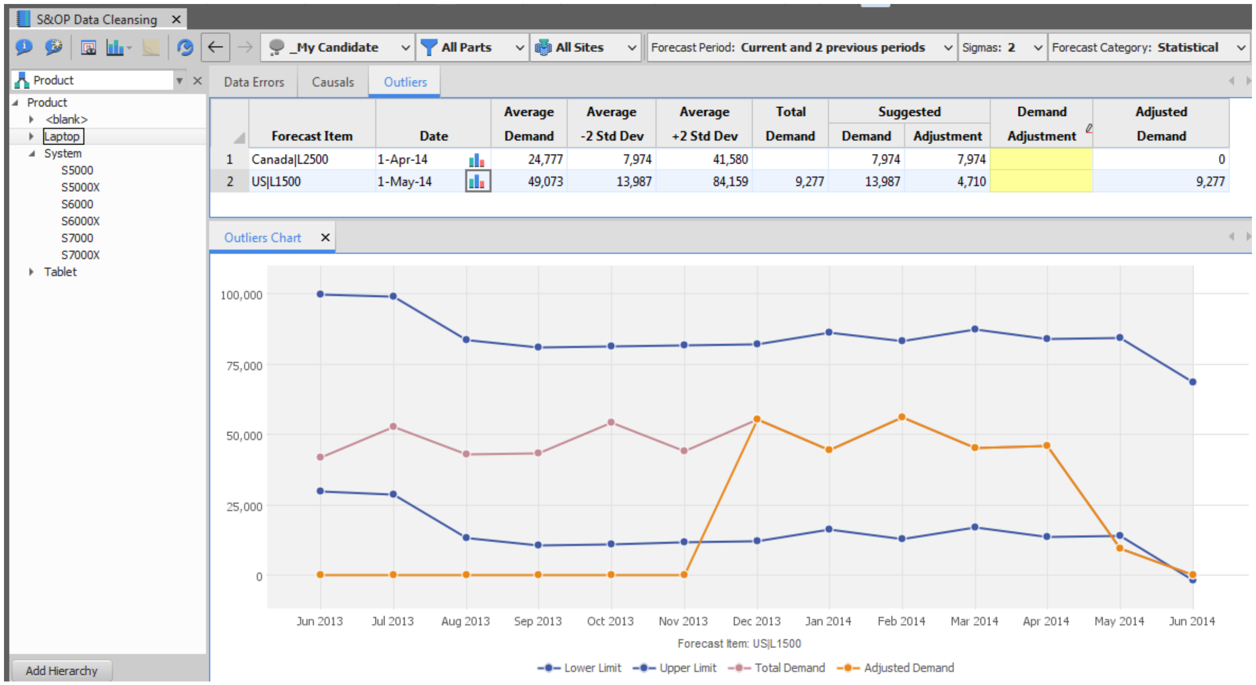
	Forecast Item	Date	Total Demand	Causals Demo	Causals Promotion	Adjusted Demand
73	US|S5000	3-Mar-14	13,356			13,356
74		1-Apr-14	13,211			13,211
75		1-May-14	13,191			13,191
76	US|S6000	3-Mar-14	9,907			9,907
77		1-Apr-14	9,660			9,660
78		1-May-14	11,914		-2,383	9,531
79	US|S7000	3-Mar-14	7,961			7,961
80		1-Apr-14	8,010			8,010
81		1-May-14	8,113			8,113

	Forecast Item	Item	Location	Customer	Location	Date	Actual Demand	Demand Adjustments: Data Errors	Demand Adjustments: Causals Demo	Demand Adjustments: Causals Promotion	Adjusted Demand
	US|S6000					1-May-14.					
1	US|S6000	S6000X	EAProd2	C111	EAProd2	1-May-14	2,929			-586	2,343
2				C112	EAProd2	1-May-14	4,792			-958	3,834
3				C113	EAProd2	1-May-14	4,193			-839	3,354
4			Sum								9,531
5	Grand Sum						11,914	0	0	-2,383	9,531

Once the missing historical data has been dealt with in the **Data Errors** worksheet, the user can then enter information about special extra data in the history. Events such as demos and special promotions and fire sales may need to be “removed” from the history in order to make the resulting “adjusted history” more reflective of what would be normally expected. If the event that created the actual history is not something that would have been forecasted normally or would be an event that should not be included in statistical forecasting, then it should be effectively removed from the history before we would use the history for either statistical forecasting or for disaggregation. This is done using the **Causals** worksheet.

When the Demand Planner enters an adjustment here, it is inserted into the CausalFactorDetail table (rather than directly editing historical demand actuals) with a CausalFactor.Category of either ‘Promotion’ or ‘Demo/Test’ depending on which column the adjustment is entered on.

There is an additional type of causal that can be entered by simply un-hiding the ExtraCausal column in the worksheet. This give a 3rd category value of ‘ExtraCausal’ but the column header can be edited to make it represent anything you might need. The adjustments are stored in the CausalFactorDetail table which references HistoricalDemandHeader. If you are using SEIP or MEIO you can set the CausalFactorCategory.ProcessingRule to decide whether adjustments impact SafetyStock calculations.



Finally, once the data errors and special event causals have been dealt with using the previous two worksheets, we should have a more statistically valid set of history to work with. It is entirely possible at this point that there is still some form of event that was not forecasted or even foreseeable that is affecting the history. Perhaps events such as natural disasters or economic down-turns might cause an unexpected surge or dip in the historical demands. We can use the third **Outliers** worksheet to detect and compensate for these additional historical data anomalies.

Data that is displayed in this worksheet is showing ForecastItems where the bucketed historical data is outside the range of some number (sigma) of standard deviations from the moving average (same number of intervals as the **Data Errors** worksheet uses to get the Suggested Adjustment). So, for example, a 3 month moving average is calculated based on the adjusted (so far) historical demand and a standard deviation calculated from this moving average. The upper bound line is drawn at the moving average + 2 standard deviations (sigma = 2) while the lower bound is drawn at the moving average - 2 standard deviations. Sigma can be set in the data settings for the workbook. If any of the bucketed adjusted historical demands are outside that range, it is flagged as a possible outlier and reported. Clicking on the little graph icon in the top pane will show the associated graph.

The test for outliers respects the Forecast Period worksheet control setting at the top but the graph shows all buckets. The Adjusted Demand graph line shows zero if it is prior to the Forecast Period setting plus the number of periods in the moving average. These are the relevant points for the outlier test.

Demand Adjustment

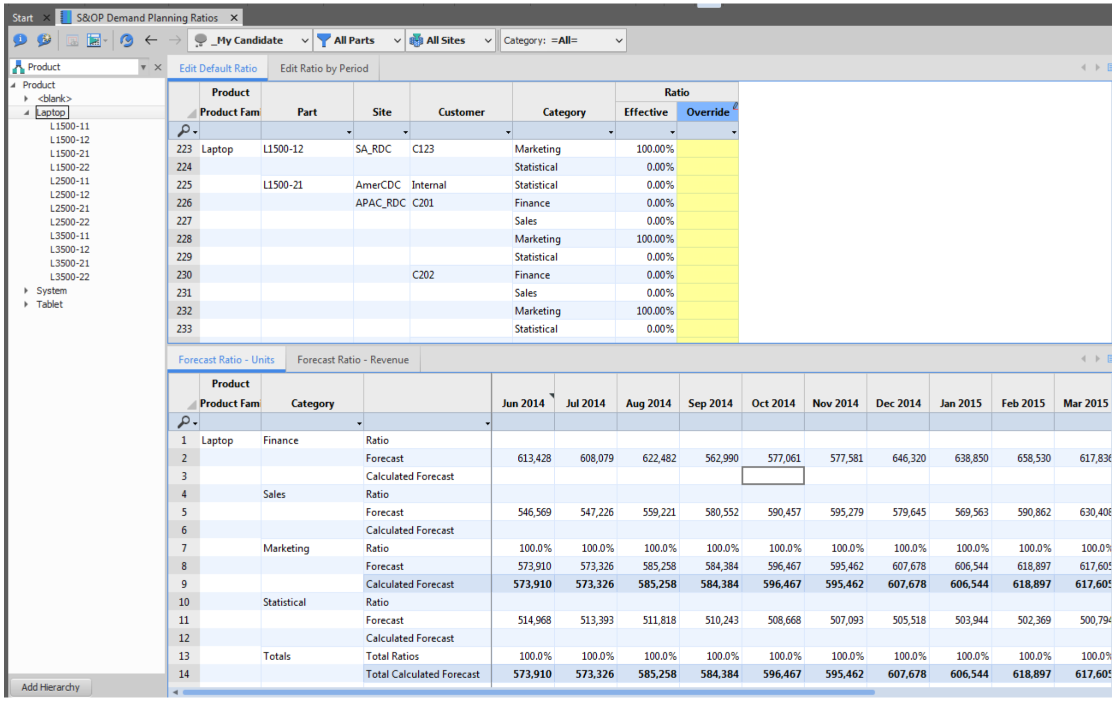
This adjustment for outliers is separate from the Causal adjustments. Causal adjustments are defined at the part, customer, and category level but demand outlier adjustments are associated with ForecastItemParameters as are the data for the graph. This means that forecast items that belong to more than one forecast category in ForecastItemParameters can use a different set of outlier adjustments for each category. You set the Forecast Category with the worksheet control.

The suggested adjustment is whatever it takes to bring the adjusted history to the nearest bound. That is, if sigma = 2 and the actual is higher than the moving average plus 2 standard deviations, then the suggestion will be the amount it take to bring it back down to the moving average plus 2 standard deviations. It will be up to the best judgment of the demand planner to decide if more or less adjustment is required.

4.5.5. S&OP Demand Planning Ratios

Workbook Name	S&OP Demand Planning Ratios
Purpose	Use this workbook to evaluate and make adjustments to the forecast ratios that are assigned to individual forecast categories. These adjustments allow you to refine the calculated consensus demand plan. You can also: Adjust forecast ratios for specific items across all time periods. Adjust individual forecast ratios for a specific time period. Review the forecast using the adjusted ratios.
Is Standard PDR?	Yes
Key Data Elements	Effective ratio (weight) per PartCustomer and Category with override and time phased override, the forecast by category and the calculated consensus forecast from each category (if any) over time.
Key Calculations	Calculated forecast and effective ratio per category over time.

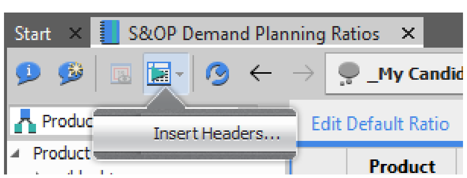
In the Create a Consensus Demand Plan task flow, the first active step (step 6) is to adjust the consensus demand weights based on the reviews that the Demand Planner has done in the preceding steps. This comes down to a judgment call on the part of the Demand Planner as to which forecasting category or categories should be used to drive the calculated unconstrained consensus demand plan.



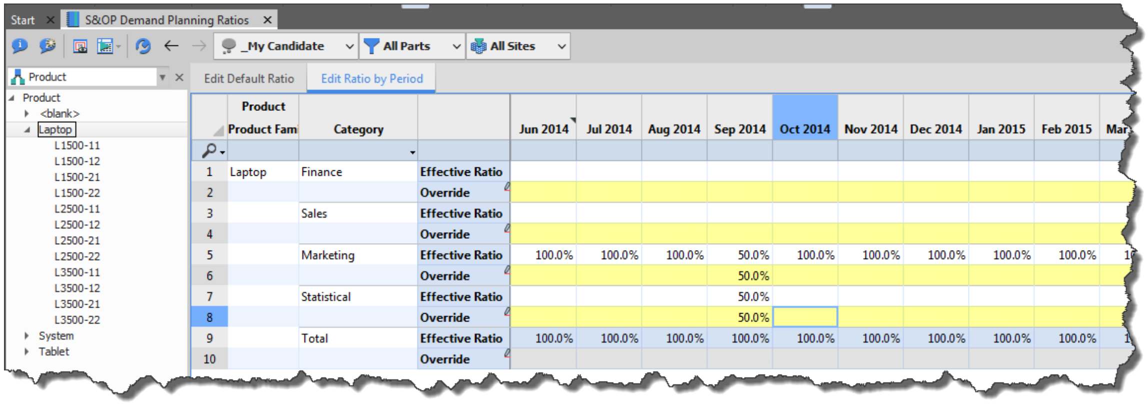
There are actually three levels at which RapidResponse can define the consensus forecast weights. Only the last two levels are maintained by this workbook. The highest level default weight is purely by Category and is defined in the **Control Tables** workbook at integration time using the Historical Demand Category worksheet. This is not covered in this process flow but determines the reported effective ratio above when there are no overrides specified.

In the Edit Default Ratio worksheet, you have the opportunity to override those category defaults for specific Part/Customer/Categories (HistoricalDemandHeaders). The **Control Table** setting might indicate that the default weighting will assume that all the consensus forecast is coming from the Sales forecast. However, in the Edit Default Ratio worksheet you can indicate that for specific Part/Customers, the consensus forecast will come 50% from Sales and 50% from Statistical.

An interesting thing to note about this worksheet is that it only shows values for already existing Part/Customer/Category headers. For example, in the above screen shot, you will not see a Marketing category for Part L-1500-21 in Site AmerCDC although you do see it for the same Part in Site APAC_RDC. There is no "header" record for Marketing in Site AmerCDC. In order to edit those default values, a header record needs to be created. A data command is provided to do this.



Running this command will add in any missing header records for whatever is selected in the associated filters and hierarchy.



The second worksheet in the top pane allows the Demand Planner to override the weights at final level; Part/Customer/Category/Date. In this case, the Demand Planner can indicate that the consensus forecast will come 50% from Marketing and 50% from Statistical in September and then revert back to the defaults.

An interesting thing to watch for in this worksheet is the "Total" effective ratio. If, for example, you were to set the ratios to 50% and 25% and 35% the total becomes highlighted and 110%. This is flagged as unexpected but it will still work. It's just that the calculated consensus forecast ends up being inflated beyond 100%. Similarly, the total could end up adding up to something less 100% in which case the forecast is smaller than the inputs. This might be useful if, for example, Sales is always the best forecast source but they are always 10% high. In this case, maybe you really do want to drive the consensus forecast entirely from Sales but only something like 90% of it. It will be highlighted but it will work.

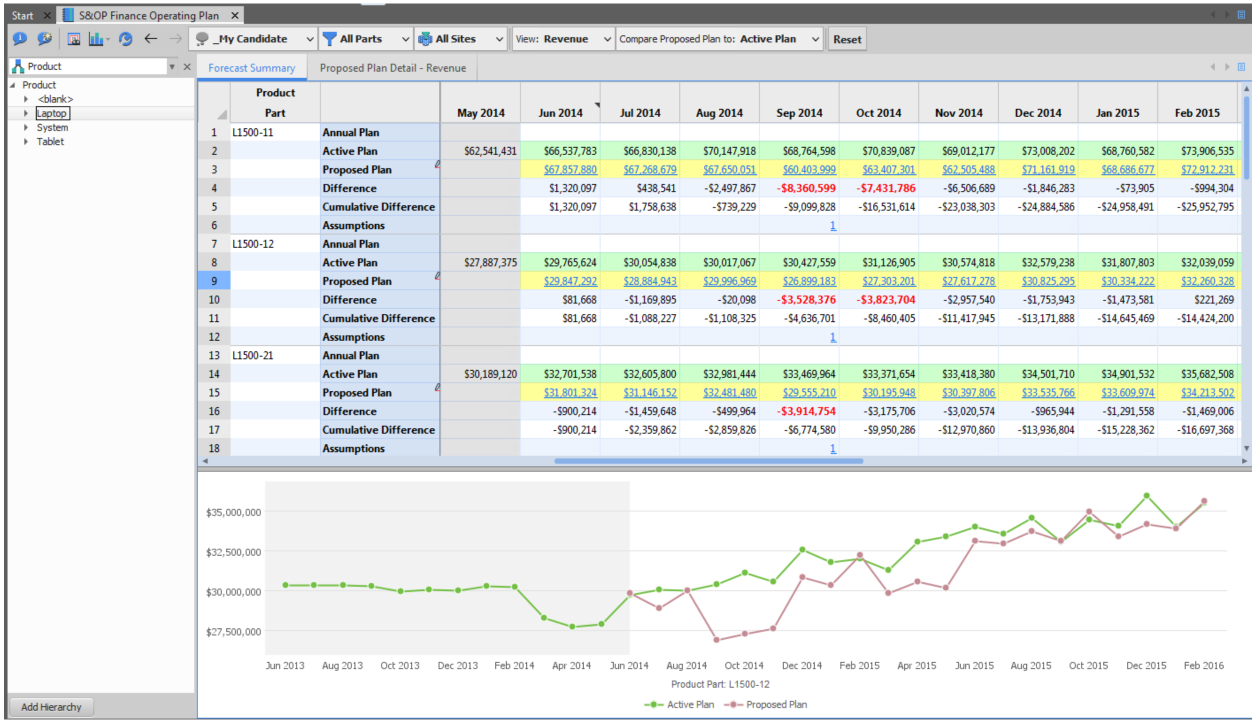
The bottom pane shows worksheets displaying the effective ratio, input forecasts and resulting calculated consensus forecast by period for each of the categories along with the total. One worksheet shows this in units while the other displays in revenue. These are the views that show the forecasts per category and resulting consensus demand based on the edited (or default) ratios in the top pane.

4.5.6. S&OP Finance Operating Plan

Workbook Name	S&OP Finance Operating Plan
Purpose	Use this workbook to: • Review or modify the finance forecast (by units or revenue). • Compare the optimistic finance plans to the pessimistic finance plans. This option is available depending on your company's processes. • Evaluate the differences between the annual or active plans to the proposed plan. • View the number of assumptions for a given period. • Create, modify and apply events to the forecast
Is Standard PDR?	Yes
Key Data Elements	Annual Plan, Active Plan, Proposed Plan (editable if user is in Finance group), Difference and Cumulative Difference, Assumptions.
Key Calculations	The differences are a comparison between the Proposed Plan and either the Annual Plan or the Active Plan.

This workbook is used by the Finance planner to enter maintain the Finance Forecast (not the Annual Plan). If the user belongs to the Finance group, then the Proposed Plan row in this report is editable and a couple of other worksheets are visible. This is the primary resource used in the Task Flow called “Input Forecast – Finance”. The planner can either use this workbook with units or with their chosen currency.

The reported Annual Plan consists of ForecastDetail records where the category is either $DP_BudgetVolumeTarget (BudgetVolumeTarget if editing in Units) or $DP_BudgetValueTarget (BudgetValueTarget if editing in Revenue).



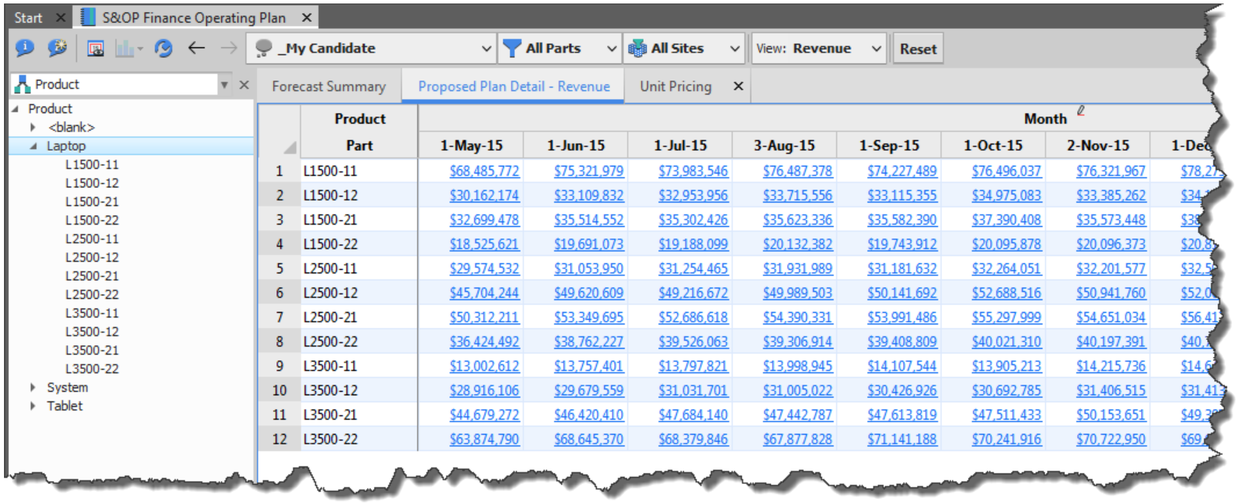
The Active Plan consists of the set of HistoricalDemandSeriesDetail records with a category of $DP_BudgetForecastCategory (FinanceOperatingPlan) and the last Series on the last AsOfDate.

The Proposed Plan consists of ForecastDetail records with the same category of $DP_BudgetForecastCategory (FinanceOperatingPlan). This is the editable row (either units or revenue) that the Finance planner will be maintaining.

There is a control on the toolbar allowing the planner to compare the “Proposed Plan” to either the “Active Plan” or the “Annual Plan” (as defined above). This comparison row has a special background color. Values in the “Difference” row are highlighted in red if they deviate from the selected plan by more than the specified tolerance threshold. This “tolerance threshold” is specified in the Data Settings dialog.

Finally, the number displayed in the “Assumptions” row indicate the total number of assumptions effective for the period. Clicking on this number will open a worksheet listing those specific assumptions.

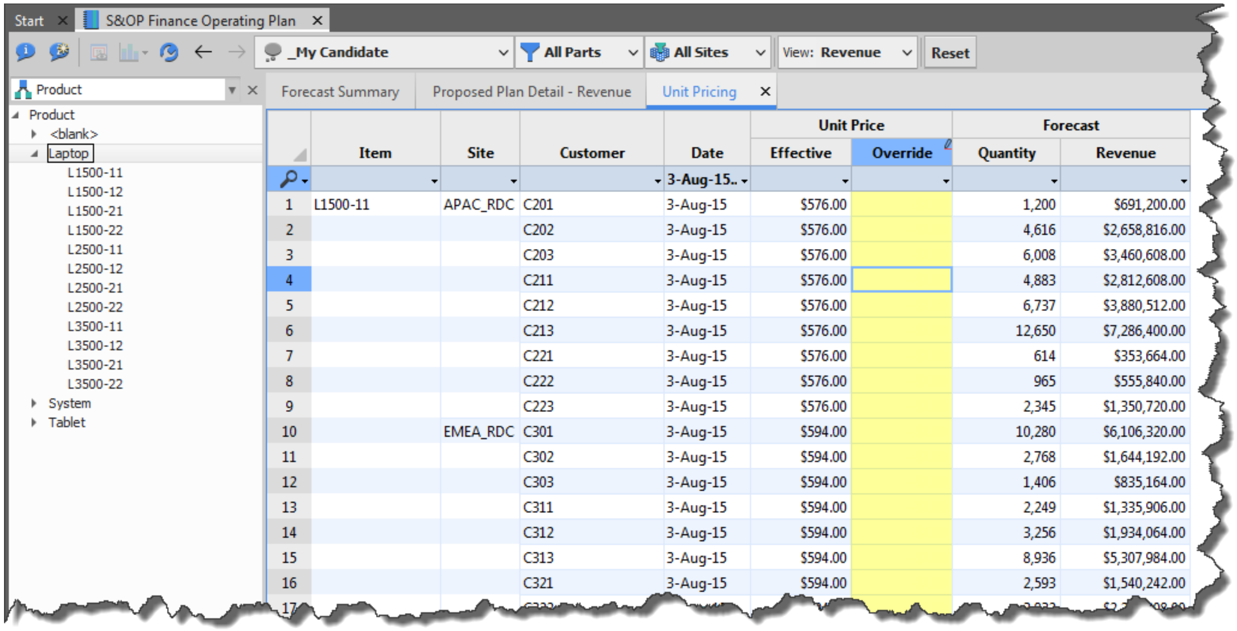
If the user is in the ‘Finance’ group there will be a second editable worksheet available to them in the top pane that reports a bucketed detailed view of the Finance Operating Plan broken out by one level below the selected hierarchy level.



	Product	Month							
	Part	1-May-15	1-Jun-15	1-Jul-15	3-Aug-15	1-Sep-15	1-Oct-15	2-Nov-15	1-Dec
1	L1500-11	$68,485,772	$75,321,979	$73,983,546	$76,487,378	$74,227,489	$76,496,037	$76,321,967	$78,27
2	L1500-12	$30,162,174	$33,109,832	$32,953,956	$33,715,556	$33,115,355	$34,975,083	$33,385,262	$34,
3	L1500-21	$32,699,478	$35,514,552	$35,302,426	$35,623,336	$35,582,390	$37,390,408	$35,573,448	$38
4	L1500-22	$18,525,621	$19,691,073	$19,188,099	$20,132,382	$19,743,912	$20,095,878	$20,096,373	$20,8
5	L2500-11	$29,574,532	$31,053,950	$31,254,465	$31,931,989	$31,181,632	$32,264,051	$32,201,577	$32,5
6	L2500-12	$45,704,244	$49,620,609	$49,216,672	$49,989,503	$50,141,692	$52,688,516	$50,941,760	$52,0
7	L2500-21	$50,312,211	$53,349,695	$52,686,618	$54,390,331	$53,991,486	$55,297,999	$54,651,034	$56,41
8	L2500-22	$36,424,492	$38,762,227	$39,526,063	$39,306,914	$39,408,809	$40,021,310	$40,197,391	$40,
9	L3500-11	$13,002,612	$13,757,401	$13,797,821	$13,998,945	$14,107,544	$13,905,213	$14,215,736	$14,6
10	L3500-12	$28,916,106	$29,679,559	$31,031,701	$31,005,022	$30,426,926	$30,692,785	$31,406,515	$31,41
11	L3500-21	$44,679,272	$46,420,410	$47,684,140	$47,442,787	$47,613,819	$47,511,433	$50,153,651	$49,3
12	L3500-22	$63,874,790	$68,645,370	$68,379,846	$67,877,828	$71,141,188	$70,241,916	$70,722,950	$69,

The intent of this worksheet is to provide an easy way to edit the details more directly. This has been designed to be able to import the Finance Operating Plan from a similarly formatted Excel worksheet. There is a version of this for units and for revenue.

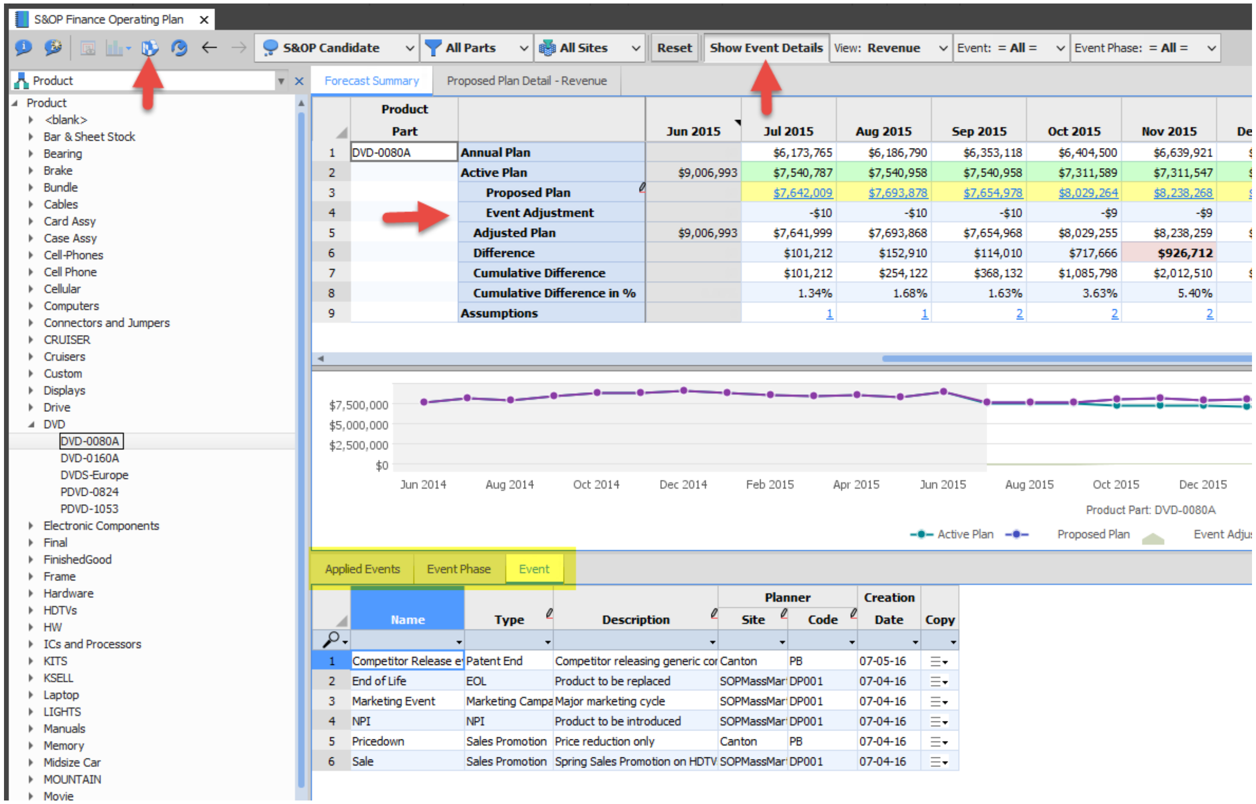
If the Finance planner is working with revenue rather than units, there is also the opportunity to set the effective price for Part/Customers for specific date ranges. Clicking on any of the Proposed Plan revenue values will open the Unit Pricing worksheet.



The current pricing is reported in this worksheet along with the ability to set new unit prices for the Part/Customer (ForecastDetail).

If Event Planning is enabled, new options become available in order to allow creating, modifying and applying events. Two new rows are added **Event Adjustment** and **Adjusted Plan.** There will be an option to Show Event Details, which will make three new worksheets available in a lower pane

Events Phases can be applied from either the **Event Phase** or **Event worksheet**. The Apply Event icon in the toolbar will become enabled when the cursos is in any cell of these worksheets. Only members of the Finance group will be able to apply events. The **Applied Events** worksheet will list those events currently applied at the selected hierarchy level.



4.5.7. S&OP Forecast Accuracy

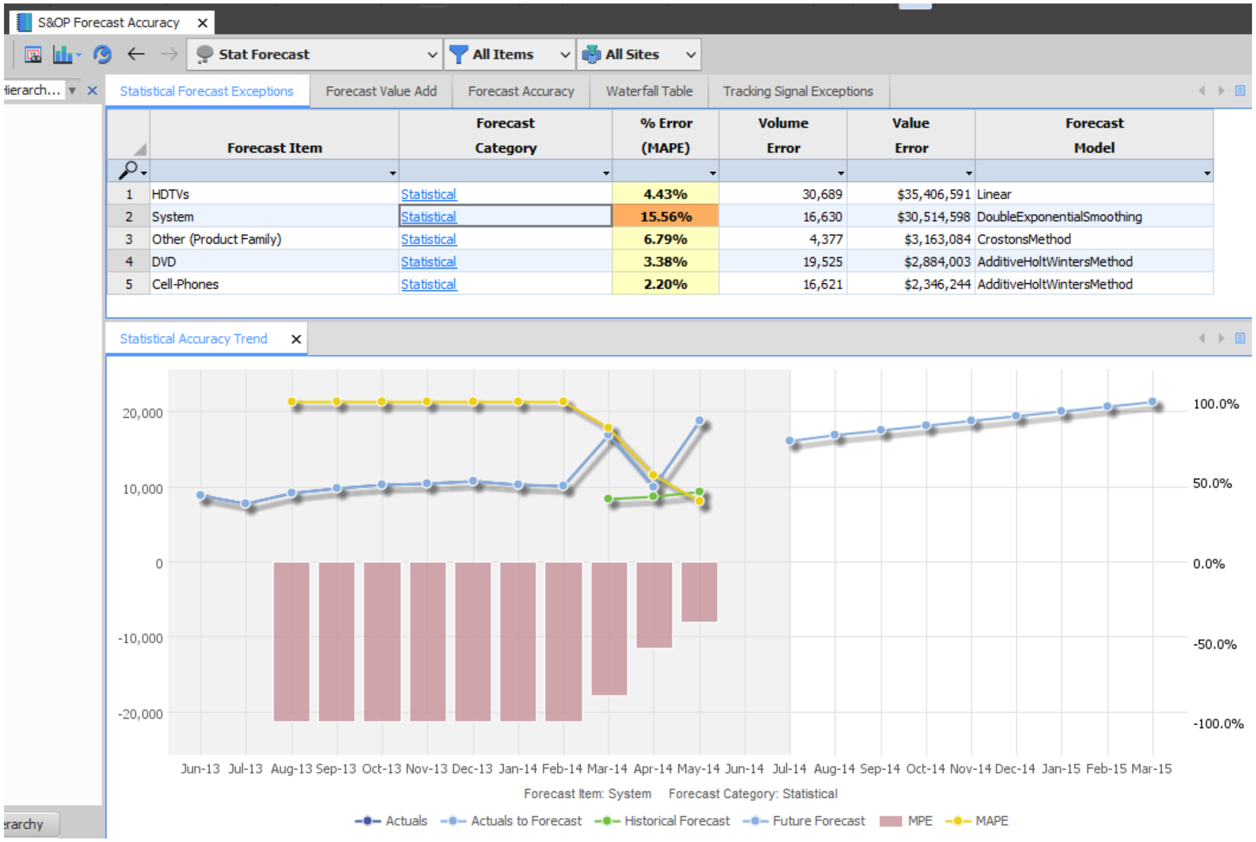
Workbook Name	S&OP Forecast Accuracy
Purpose	Use this workbook to review: • Forecast accuracy measures • Forecast value add • Waterfall worksheets
Is Standard PDR?	Yes
Key Data Elements	By ForecastItem or by hierarchy node selection, various measures are reported against selected forecast categories.
Key Calculations	There is quite a list of measures reported in different ways in this workbook. All provide different information about the usefulness and variability of the selected forecast categories. These are best described in the RapidResponse Solutions Guide.

Definitions:

- **AdjustedRSquared**: calculates the ratio between the regression sum of squares and total sum of squares measures similar to the RSquared measure, but is influenced by the number of values in the input data set.
  The closer to 1.0, the better the forecast.
- **MeanAbsoluteError (MAE)**: calculates the absolute value of the variance between the historical actual values and the predicted forecast values.
  The smaller the value, the better the forecast.
- **MeanAbsoluteErrorByMean**: calculates the absolute value of the variance between the historical actual values and the predicted forecast values, and then divides it by the mean quantity of the input data.
  The smaller the value, the better the forecast.
- **MeanAbsolutePercentDeviation (MAPD)**: This is not actually an error measure at all. It is. Rather, a measure of stability of a forecast. It does not compare to any actuals. Instead, it is calculating the percent deviation from the mean of the forecast itself.
  The smaller the value, the more stable the forecast. A horizontal straight line has an MAPD of zero.
- **MeanAbsolutePercentageError (MAPE)**: This is the absolute value of the variance between the historical actual values and the predicted forecast values with respect to the actual values, expressed as a percentage. Zero values in the input data are ignored, and are not included in the total count of points. This tends to be a very popular measure and is referred to as the MAPE. *The smaller the value, the better the forecast.*

- **MeanError (ME)**: calculates the average variance between the historical actual values and the predicted forecast values.
  The closer to zero, the better the forecast.
- **MeanSquareError**: calculates the average of the residual sum of squares for the forecast.
  The smaller the value, the better the forecast.
- **MeanPercentageError (MPE)**: calculates the average variance between the historical actual values and the predicted forecast values, expressed as a percentage.
  The closer to zero, the better the forecast.
- **RegressionSumSquares**: calculates the difference between the total sum of squares of the input data and the residual sum of squares. .
  The larger the value, the better the forecast.
- **ResidualSumSquares**: calculates the sum of the square of differences between the historical actual values and the values predicted by the statistical forecast.
  The smaller the value, the better the forecast.
- **RootMeanSquareError**: determines the square root of the value produced by the mean square errors measure.
  The smaller the value, the better the forecast.
- **RSquared**: calculates the ratio between the regression sum of squares and total sum of squares measures.
  The closer to 1.0, the better the forecast.

The first worksheet in the top pane is the Statistical Forecast Exceptions report. This reports some forecast accuracy indicators for the generated statistical forecast for ForecastItems belonging to any of the Part/Customers selected in the hierarchy. Clicking on the Forecast Category will raise a Statistical Forecast Accuracy Trend graph in the bottom pane.


S&OP Forecast Accuracy
Stat Forecast
All Items
All Sites
Statistical Forecast Exceptions
Forecast Value Add
Forecast Accuracy
Waterfall Table
Tracking Signal Exceptions
Forecast Item
Forecast Category
% Error (MAPE)
Volume Error
Value Error
Forecast Model
1 HDTVs Statistical 4.43% 30,689 $35,406,591 Linear
2 System Statistical 15.56% 16,630 $30,514,598 DoubleExponentialSmoothing
3 Other (Product Family) Statistical 6.79% 4,377 $3,163,084 CrostonsMethod
4 DVD Statistical 3.38% 19,525 $2,884,003 AdditiveHoltWintersMethod
5 Cell-Phones Statistical 2.20% 16,621 $2,346,244 AdditiveHoltWintersMethod
Statistical Accuracy Trend
20,000
10,000
0
-10,000
-20,000
100.0%
50.0%
0.0%
-50.0%
-100.0%
Jun-13 Jul-13 Aug-13 Sep-13 Oct-13 Nov-13 Dec-13 Jan-14 Feb-14 Mar-14 Apr-14 May-14 Jun-14 Jul-14 Aug-14 Sep-14 Oct-14 Nov-14 Dec-14 Jan-15 Feb-15 Mar-15
Forecast Item: System Forecast Category: Statistical
Actuals
Actuals to Forecast
Historical Forecast
Future Forecast
MPE
MAPE

The data shows:

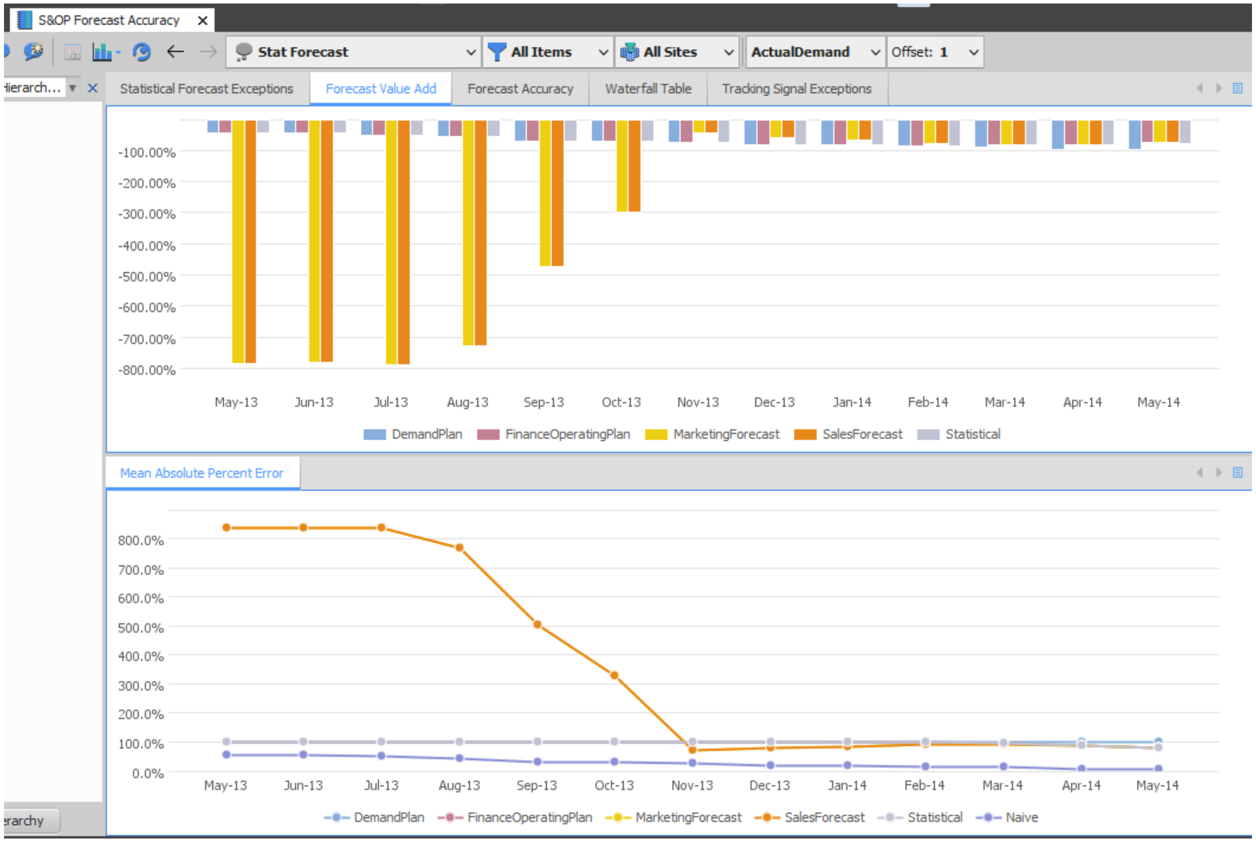
- **% Error (MAPE)**: The average of the absolute value of the percentage difference between the forecast and actual demand. The background color of the error measure value depends on the limits entered in the Data Settings dialog:
  - Green—Less than or equal to the % Error (MAPE) Warning Limit (set in the Data Settings dialog)
  - Yellow—Greater than the % Error (MAPE) Warning Limit and less than or equal to the % Error (MAPE) Critical Limit
  - Orange—Greater than the % Error (MAPE) Critical Limit
- **Volume Error**: Total forecast quantity for the next 6 months multiplied by the MAPE.
- **Value Error**: The total forecast value for the next 6 months multiplied by the MAPE.
- **Forecast Model**: The forecast model used to calculate the statistical forecast.

In this case, the reported MAPE (Mean Absolute Percent Error) is the one that was calculated and stored in the ForecastItemFitOutput table when the statistical forecast was last generated. It is not being recalculated at this time.

The Graph shows:

- **Actuals** – a set of actuals that could be passed to stat forecasting without regard to horizon but with all adjustments included
- **Actuals to Forecast** – same as Actuals but only reporting the actuals in the specified past horizon (limited by EffectiveHistoryStartDate).
- **Historical Forecast** – the stat forecast saved at the end of the last S&OP planning cycle in the HistoricalDemandSeriesDetails prior to the Actuals (lag = 1 S&OP cycle).
- **Future Forecast** – the ForecastDetails records currently generated for the statistical forecast.
- **MPE and MAPE** – as described above but comparing Historical Forecast to Actuals (above) in rolling blocks of 3 graph buckets at a time. (The 3 is set in a hidden workbook variable.) Because the Historical Forecast is so recent, only the latest point is likely to be valid.

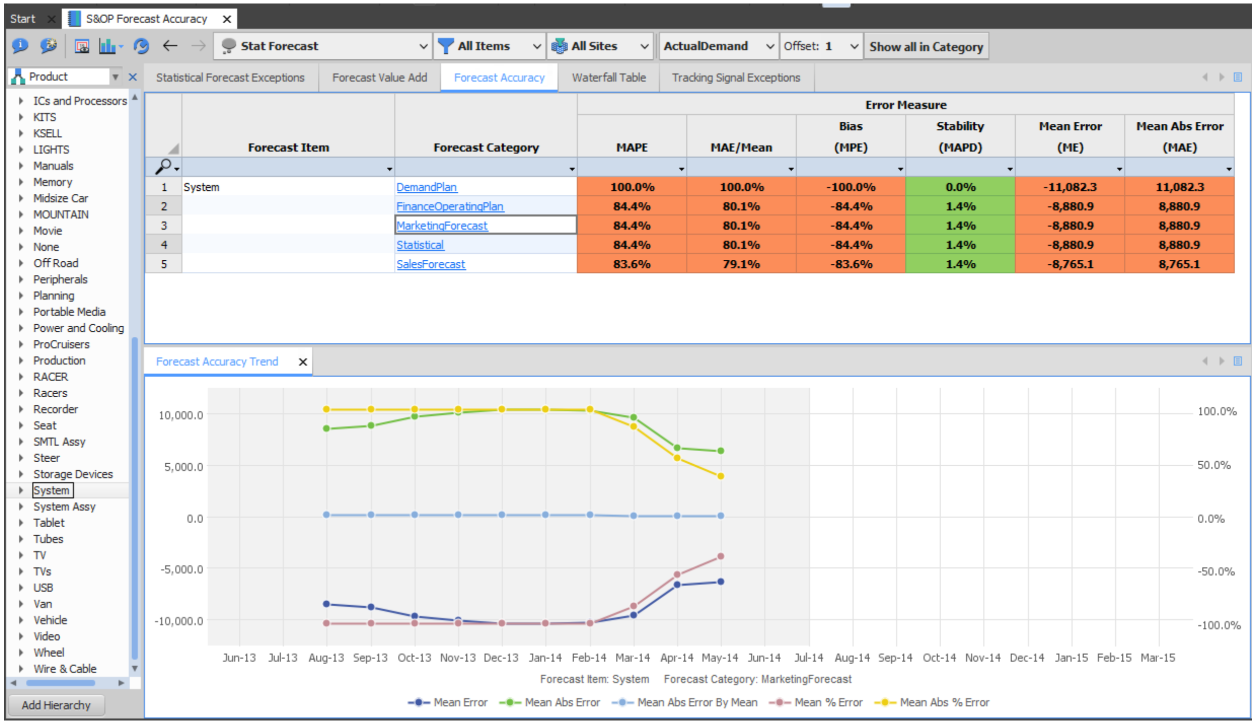
The second worksheet in the top pane is the Forecast Value Add report. Use this worksheet to view the forecast value add for each forecast stream included in the demand planning process. To evaluate if each forecast stream is more or less accurate than the naïve forecast, the value add is calculated. This is defined as the difference in mean absolute percent error of each forecast stream and the mean absolute percent error of the naive forecast.



The naïve forecast for a period is the actual value from a previous period for the category selected. This time, we are comparing a particular forecast stream from some number of intervals prior to the occurrence of the "actual" to determine the MAPE. That is, if we are looking for the forecast error related to last April's "actual" shipment, we could look at the forecast made in the previous March (Offset=1), or the previous February (offset=2) or even the forecast made 1 lead-time before the "actual" and rounded back to the previous $DP_CycleForecastCalendar (Month). It will be this forecast that is compared with the actual to get the errors and return the MAPE calculation.

You can add and remove categories to the charts to compare them.

The third worksheet in the top pane is the Forecast Accuracy report. Use this worksheet to review forecast accuracy for the selected category of forecast. The accuracy is calculated by comparing the forecast to the selected category of actual demand and providing a summary for the previous 12 months.



The data shows:

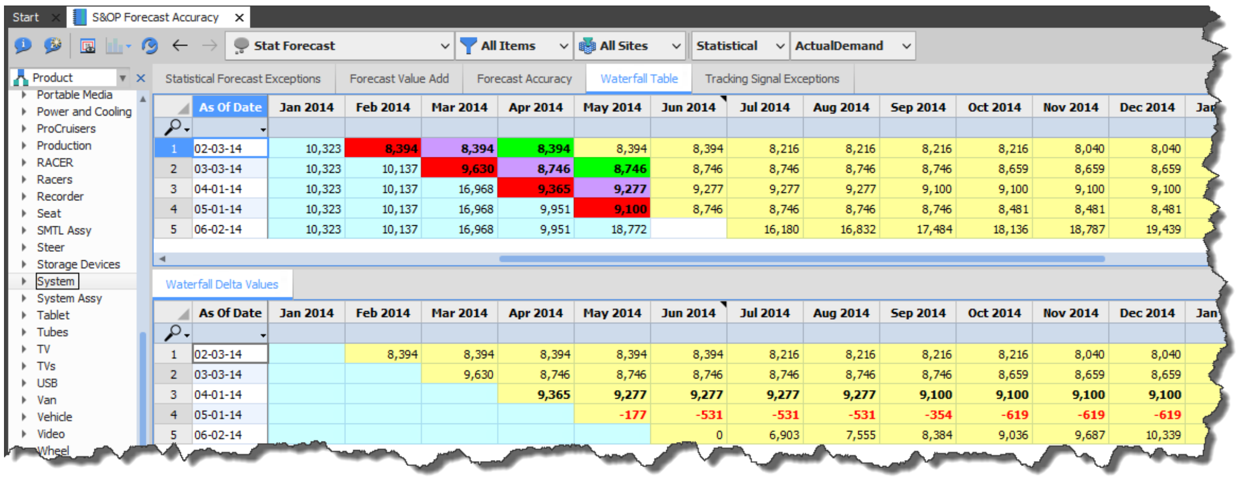
- **MAPE**: The average of the absolute value of the percentage difference between the forecast and actual demand.
- **MAE/Mean**: The average of absolute differences between the forecast and actual demand divided by the average actual demand.
- **Bias - (MPE)**: The average of the percentage difference between the forecast and actual demand. This can be positive or negative.
- **Stability - (MAPD)**: The average of the absolute percentage deviations between a forecast value and the median forecast value in the series. This is always a positive value. Note that this does not represent "error". A larger number here indicates that the forecast is expected to be more variable than a smaller number. If the forecast were to be the same number for the entire 12 months (naive) then this value would be zero.
- **Mean Error - (ME)**: The average of the difference between the forecast and actual demand. This can be positive or negative.
- **Mean Abs Error - (MAE)**: The average of the absolute value of the difference between the forecast and the actual demand.

Each measure in this report has a background color that is determined by the value of the measure and the corresponding warning and critical limits for the measure that are set up in the Data Setup dialog.

The fourth worksheet in the top pane is the Waterfall Table report. Use this worksheet to review historical demand and forecast quantities to view trends and see how closely the past forecasts have matched actual demands.

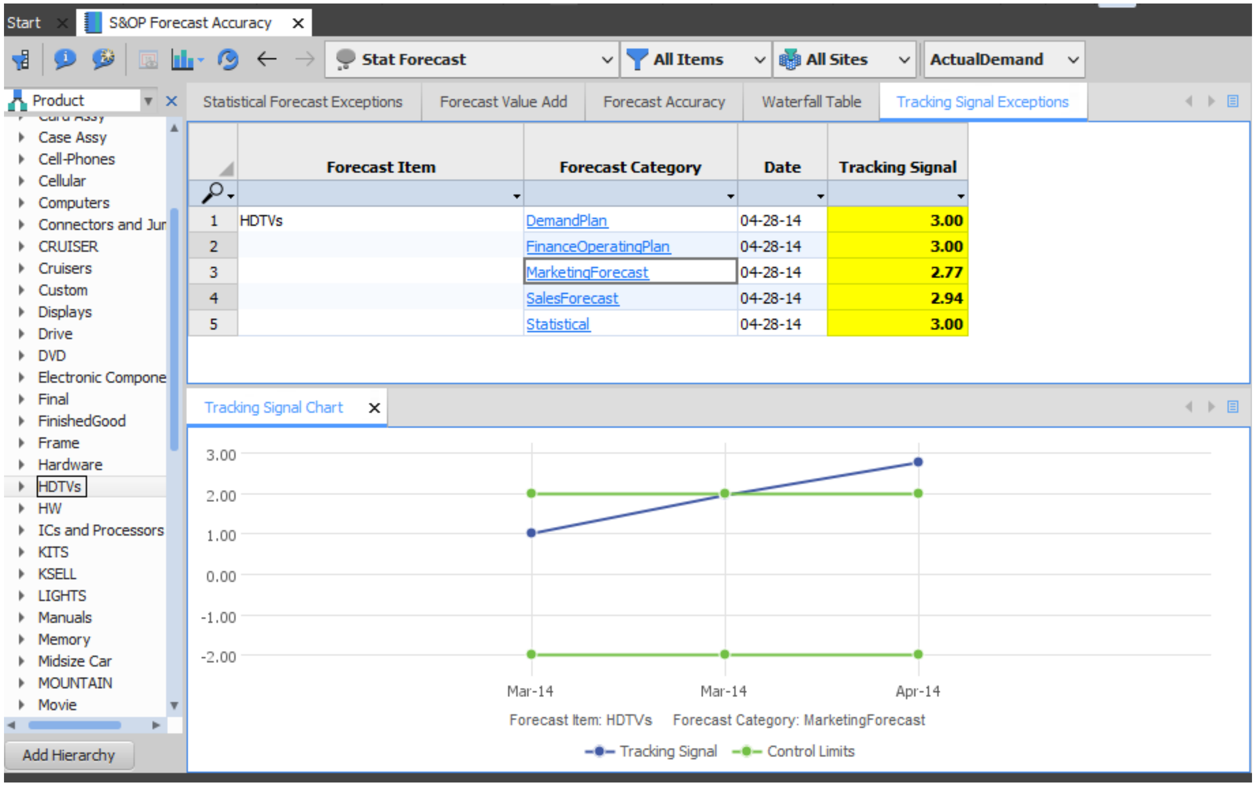
The background color of the values in the quantity by Date column signified the following:

- Yellow: forecast demand
- Blue: actual historical
- Red: forecast for the current period that was set in a previous cycle. Offset 1 establishes the number of periods in the past to use.
- Purple: forecast for the current period that was set in a previous cycle. Offset 2 establishes the number of periods in the past to use.
- Green: forecast for the current period that was set in a previous cycle. Offset 3 establishes the number of periods in the past to use.



This is a very familiar view for looking at a set of time-series data like forecasts. The top pane shows the actuals and forecasts. The bottom pane shows the difference between the forecast quantity and the last baseline forecast quantity. If the forecast is a baseline forecast, then the actual forecast quantity is shown, and all subsequent values are the difference between the last baseline and the forecast on the given as of date. Baseline are the bolded row.

The fifth (last) worksheet in the top pane is the Tracking Signal Exceptions report. Use this worksheet to compare actual data to forecast values in the selected forecast stream, in order to monitor and evaluate the effectiveness of forecasting methods used.



The data shows:

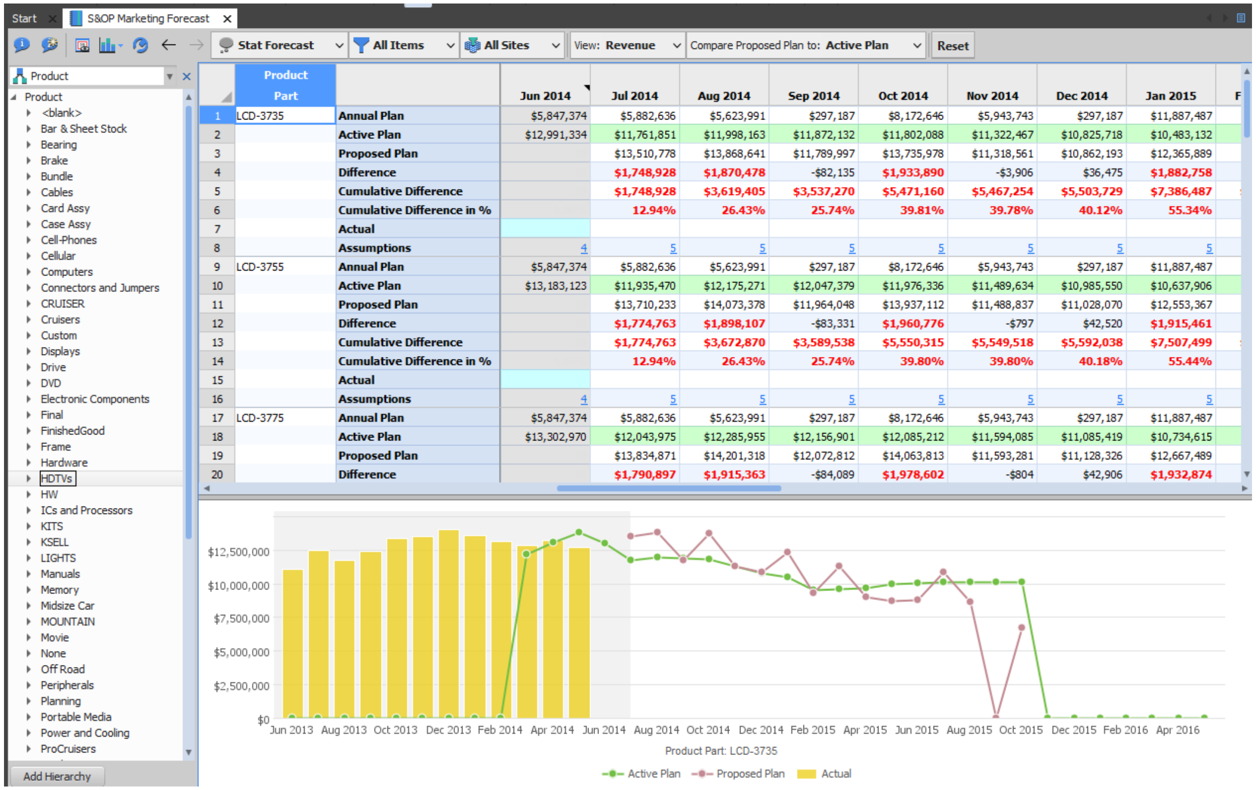
- **Date**: The date of the last period in the interval reported.
- **Tracking Signal**: The sum of the differences between the forecast and actual values divided by the mean absolute deviation (MAD). The movement of the tracking signal is compared to the control limits specified in the Data Settings. As long as the tracking signal is within these limits, the forecast is considered in control. Values outside this range indicate that the model used should be reevaluated.

4.5.8. S&OP Marketing Forecast

Workbook Name	S&OP Marketing Forecast
Purpose	Use this workbook to: • Review or modify the marketing forecast (by units or revenue). • Compare the optimistic marketing plans to the pessimistic marketing plans. This option is available depending on your company's processes. • Evaluate the differences between the annual or active plans to the proposed plan. • View the number of assumptions for a given period. • Create, modify and apply events to the forecast
Is Standard PDR?	Yes
Key Data Elements	Annual Plan, Active Plan, Proposed Plan (editable if user is in Marketing group), Difference and Cumulative Difference, Assumptions.
Key Calculations	The differences are a comparison between the Proposed Plan and either the Annual Plan or the Active Plan.

This workbook is used by the Marketing planner to enter maintain the Marketing Forecast. If the user belongs to the Marketing group, then the Proposed Plan row in this report is editable. This is the primary resource used in the Task Flow called “Input Forecast – Marketing”. The planner can either use this workbook with units or with their chosen currency.

The reported Annual Plan consists of ForecastDetail records where the category is either $DP_MarketingVolumeTarget (MarketingVolumeTarget if editing in units) or $DP_MarketingValueTarget (MarketingValueTarget if editing in units).



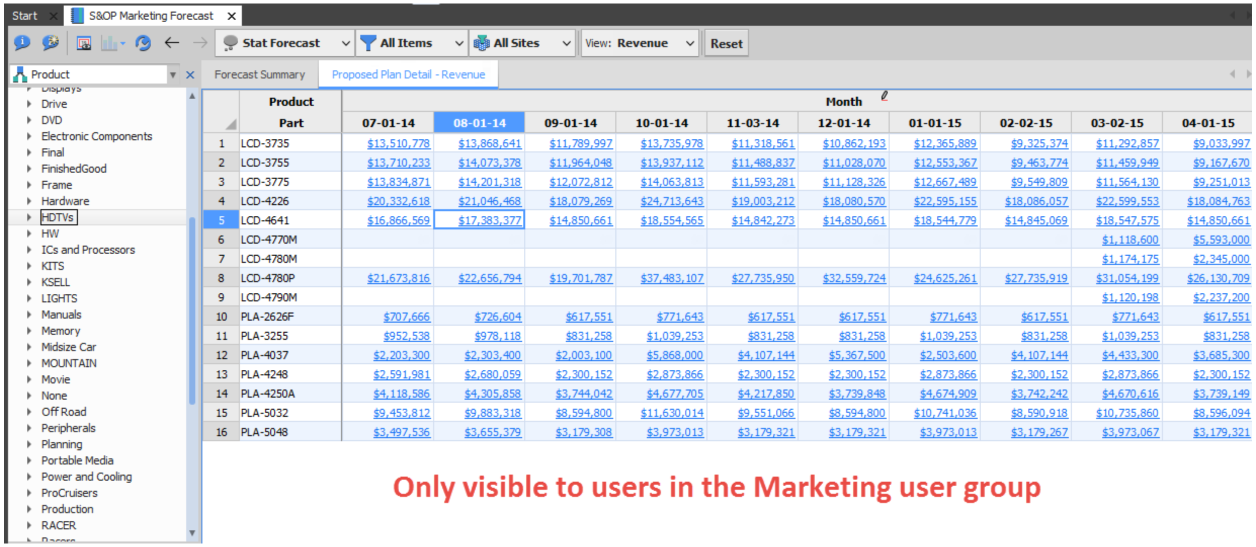
The Active Plan consists of the set of HistoricalDemandSeriesDetail records with a category of $DP_MarketingForecastCategory (MarketingForecast) and the last Series on the last AsOfDate.

The Proposed Plan consists of ForecastDetail records with the same category of $DP_MarketingForecastCategory (MarketingForecast). This is the editable row (either units or revenue) that the Marketing planner will be maintaining.

There is a control on the toolbar allowing the planner to compare the “Proposed Plan” to either the “Active Plan” or the “Annual Plan” (as defined above). This comparison row has a special background color. Values in the “Difference” row are highlighted in red if they deviate from the selected plan by more than the specified tolerance threshold. This “tolerance threshold” is specified in the Data Settings dialog.

Finally, the number displayed in the “Assumptions” row indicate the total number of assumptions effective for the period. Clicking on this number will open a worksheet listing those specific assumptions.

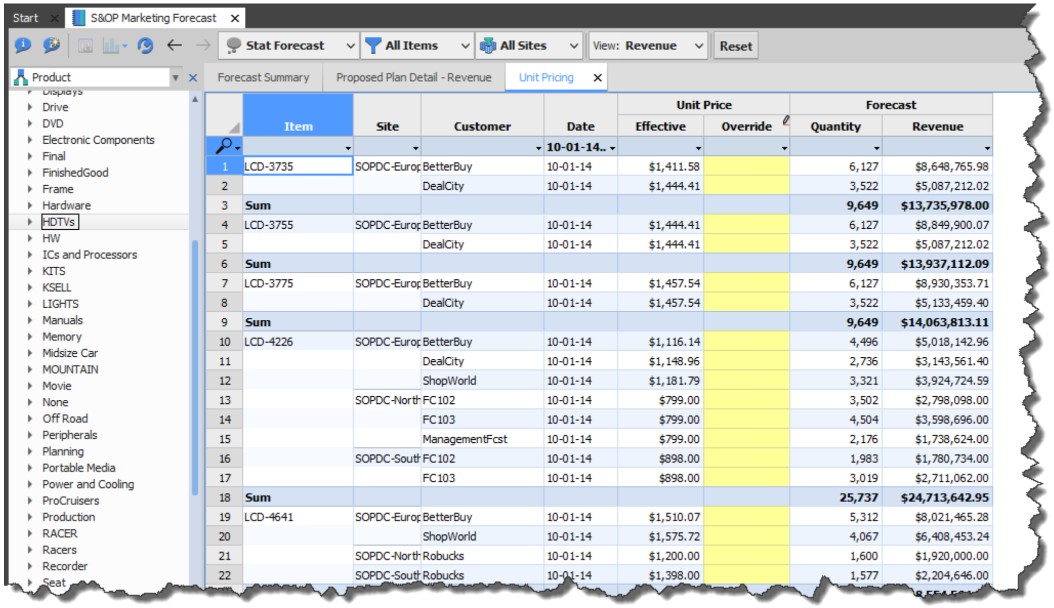
If the Marketing planner is editing in either revenue or units there will be a second worksheet available to them in the top pane that will report a bucketed detailed view of the Marketing Plan broken out by one level below the selected hierarchy level.



	Product	Month									
	Part	07-01-14	08-01-14	09-01-14	10-01-14	11-03-14	12-01-14	01-01-15	02-02-15	03-02-15	04-01-15
1	LCD-3735	$13,510,778	$13,868,641	$11,789,997	$13,735,978	$11,318,561	$10,862,193	$12,365,889	$9,325,374	$11,292,857	$9,033,997
2	LCD-3755	$13,710,233	$14,073,378	$11,964,048	$13,937,112	$11,488,837	$11,028,070	$12,553,367	$9,463,774	$11,459,949	$9,167,670
3	LCD-3775	$13,834,871	$14,201,318	$12,072,812	$14,063,813	$11,593,281	$11,128,326	$12,667,489	$9,549,809	$11,564,130	$9,251,013
4	LCD-4226	$20,332,618	$21,046,468	$18,079,269	$24,713,643	$19,003,212	$18,080,570	$22,595,155	$18,086,057	$22,599,553	$18,084,763
5	LCD-4641	$16,866,569	$17,383,377	$14,850,661	$18,554,565	$14,842,273	$14,850,661	$18,544,779	$14,845,069	$18,547,575	$14,850,661
6	LCD-4770M									$1,118,600	$5,593,000
7	LCD-4780M									$1,174,175	$2,345,000
8	LCD-4780P	$21,673,816	$22,656,794	$19,701,787	$37,483,107	$27,735,950	$32,559,724	$24,625,261	$27,735,919	$31,054,199	$26,130,709
9	LCD-4790M									$1,120,198	$2,237,200
10	PLA-2626F	$707,666	$726,604	$617,551	$771,643	$617,551	$617,551	$771,643	$617,551	$771,643	$617,551
11	PLA-3255	$952,538	$978,118	$831,258	$1,039,253	$831,258	$831,258	$1,039,253	$831,258	$1,039,253	$831,258
12	PLA-4037	$2,203,300	$2,303,400	$2,003,100	$5,868,000	$4,107,144	$5,367,500	$2,503,600	$4,107,144	$4,433,300	$3,685,300
13	PLA-4248	$2,591,981	$2,680,059	$2,300,152	$2,873,866	$2,300,152	$2,300,152	$2,873,866	$2,300,152	$2,873,866	$2,300,152
14	PLA-4250A	$4,118,586	$4,305,858	$3,744,042	$4,677,705	$4,217,850	$3,739,848	$4,674,909	$3,742,242	$4,670,616	$3,739,149
15	PLA-5032	$9,453,812	$9,883,318	$8,594,800	$11,630,014	$9,551,066	$8,594,800	$10,741,036	$8,590,918	$10,735,860	$8,596,094
16	PLA-5048	$3,497,536	$3,655,379	$3,179,308	$3,973,013	$3,179,321	$3,179,321	$3,973,013	$3,179,267	$3,973,067	$3,179,321

The intent of this worksheet is to provide an easy way to edit the details more directly. This has been designed to be able to import the Marketing Plan from a similarly formatted Excel worksheet. There is a version of this for units and for revenue.

If the Marketing planner is working with revenue rather than units, there is also the opportunity to set the effective price for Part/Customers for specific date ranges. Clicking on any of the Proposed Plan revenue values will open the Unit Pricing worksheet.

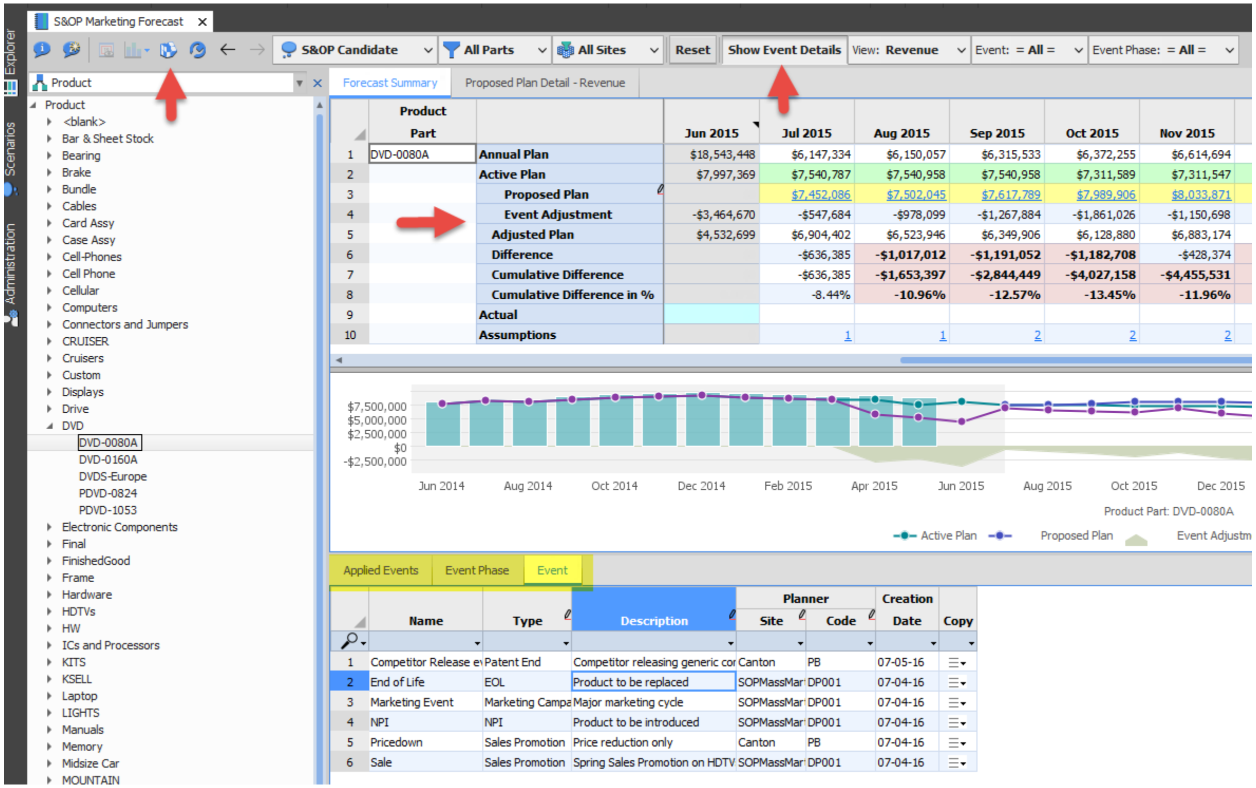


	Item	Site	Customer	Date	Unit Price: Effective	Unit Price: Override	Forecast: Quantity	Forecast: Revenue
				10-01-14..				
1	LCD-3735	SOPDC-Europ	BetterBuy	10-01-14	$1,411.58		6,127	$8,648,765.98
2			DealCity	10-01-14	$1,444.41		3,522	$5,087,212.02
3	**Sum**						**9,649**	**$13,735,978.00**
4	LCD-3755	SOPDC-Europ	BetterBuy	10-01-14	$1,444.41		6,127	$8,849,900.07
5			DealCity	10-01-14	$1,444.41		3,522	$5,087,212.02
6	**Sum**						**9,649**	**$13,937,112.09**
7	LCD-3775	SOPDC-Europ	BetterBuy	10-01-14	$1,457.54		6,127	$8,930,353.71
8			DealCity	10-01-14	$1,457.54		3,522	$5,133,459.40
9	**Sum**						**9,649**	**$14,063,813.11**
10	LCD-4226	SOPDC-Europ	BetterBuy	10-01-14	$1,116.14		4,496	$5,018,142.96
11			DealCity	10-01-14	$1,148.96		2,736	$3,143,561.40
12			ShopWorld	10-01-14	$1,181.79		3,321	$3,924,724.59
13		SOPDC-North	FC102	10-01-14	$799.00		3,502	$2,798,098.00
14			FC103	10-01-14	$799.00		4,504	$3,598,696.00
15			ManagementFcst	10-01-14	$799.00		2,176	$1,738,624.00
16		SOPDC-South	FC102	10-01-14	$898.00		1,983	$1,780,734.00
17			FC103	10-01-14	$898.00		3,019	$2,711,062.00
18	**Sum**						**25,737**	**$24,713,642.95**
19	LCD-4641	SOPDC-Europ	BetterBuy	10-01-14	$1,510.07		5,312	$8,021,465.28
20			ShopWorld	10-01-14	$1,575.72		4,067	$6,408,453.24
21		SOPDC-North	Robucks	10-01-14	$1,200.00		1,600	$1,920,000.00
22		SOPDC-South	Robucks	10-01-14	$1,398.00		1,577	$2,204,646.00

The current pricing is reported in this worksheet along with the ability to set new prices for the Part/Customer starting on an effective date.

If Event Planning is enabled, new options become available in order to allow creating, modifying and applying events. Two new rows are added **Event Adjustment** and **Adjusted Plan.** There will be an option to Show Event Details, which will make three new worksheets available in a lower pane

Events Phases can be applied from either the **Event Phase** or **Event worksheet**. The Apply Event icon in the toolbar will become enabled when the cursos is in any cell of these worksheets. Only members of the Marketing group will be able to apply events. The **Applied Events** worksheet will list those events currently applied at the selected hierarchy level.

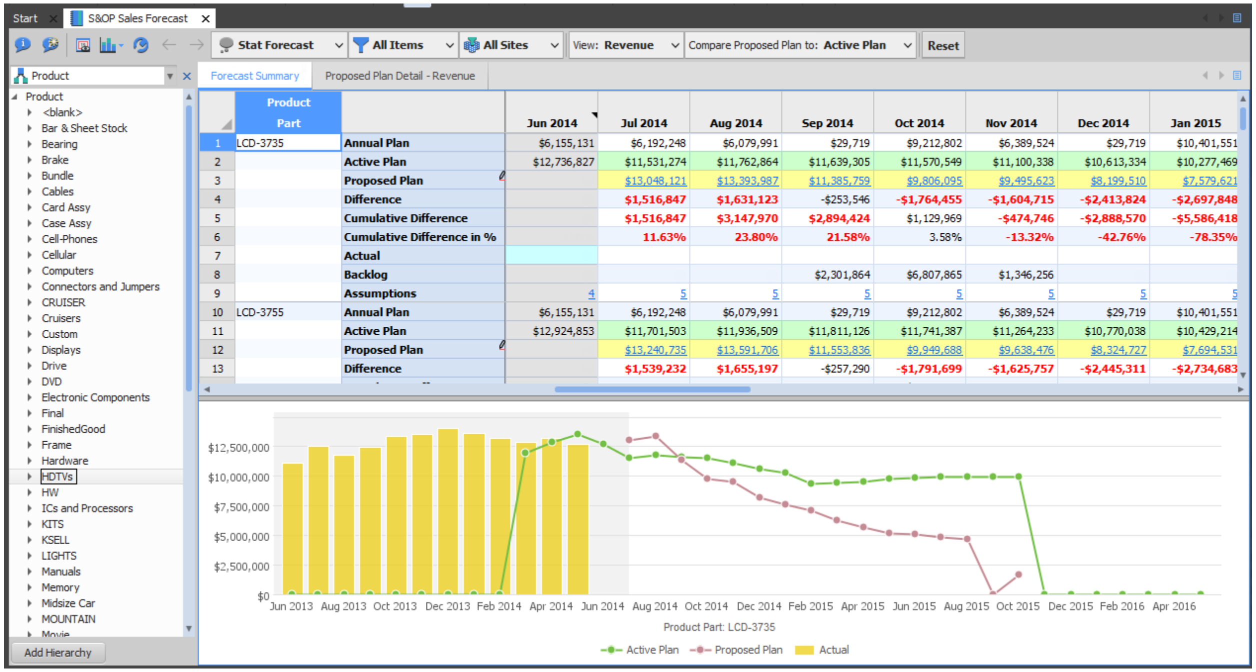

S&OP Marketing Forecast
S&OP Candidate
All Parts
All Sites
Reset
Show Event Details
View: Revenue
Event: = All =
Event Phase: = All =
Product
Forecast Summary
Proposed Plan Detail - Revenue
Product Part
Jun 2015 Jul 2015 Aug 2015 Sep 2015 Oct 2015 Nov 2015
1 DVD-0080A Annual Plan $18,543,448 $6,147,334 $6,150,057 $6,315,533 $6,372,255 $6,614,694
2 Active Plan $7,997,369 $7,540,787 $7,540,958 $7,540,958 $7,311,589 $7,311,547
3 Proposed Plan $7,452,086 $7,502,045 $7,617,789 $7,989,906 $8,033,871
4 Event Adjustment -$3,464,670 -$547,684 -$978,099 -$1,267,884 -$1,861,026 -$1,150,698
5 Adjusted Plan $4,532,699 $6,904,402 $6,523,946 $6,349,906 $6,128,880 $6,883,174
6 Difference -$636,385 -$1,017,012 -$1,191,052 -$1,182,708 -$428,374
7 Cumulative Difference -$636,385 -$1,653,397 -$2,844,449 -$4,027,158 -$4,455,531
8 Cumulative Difference in % -8.44% -10.96% -12.57% -13.45% -11.96%
9 Actual
10 Assumptions 1 1 2 2 2
Product
<blank>
Bar & Sheet Stock
Bearing
Brake
Bundle
Cables
Card Assy
Case Assy
Cell-Phones
Cell Phone
Cellular
Computers
Connectors and Jumpers
CRUISER
Cruisers
Custom
Displays
Drive
DVD
DVD-0080A
DVD-0160A
DVDS-Europe
PDVD-0824
PDVD-1053
Electronic Components
Final
FinishedGood
Frame
Hardware
HDTVs
HW
ICs and Processors
KITS
KSELL
Laptop
LIGHTS
Manuals
Memory
Midsize Car
MOUNTAIN
$7,500,000
$5,000,000
$2,500,000
$0
-$2,500,000
Jun 2014 Aug 2014 Oct 2014 Dec 2014 Feb 2015 Apr 2015 Jun 2015 Aug 2015 Oct 2015 Dec 2015
Product Part: DVD-0080A
Active Plan
Proposed Plan
Event Adjustm
Applied Events
Event Phase
Event
Name Type Description Planner Site Code Creation Date Copy
1 Competitor Release e Patent End Competitor releasing generic cor Canton PB 07-05-16
2 End of Life EOL Product to be replaced SOPMassMar DP001 07-04-16
3 Marketing Event Marketing Campa Major marketing cycle SOPMassMar DP001 07-04-16
4 NPI NPI Product to be introduced SOPMassMar DP001 07-04-16
5 Pricedown Sales Promotion Price reduction only Canton PB 07-04-16
6 Sale Sales Promotion Spring Sales Promotion on HDTV SOPMassMar DP001 07-04-16

4.5.9. S&OP Sales Forecast

Workbook Name	S&OP Sales Forecast
Purpose	Use this workbook to: • Review or modify the sales forecast (by units or revenue). • Compare the optimistic marketing plans to the pessimistic sales plans. This option is available depending on your company's processes. • Evaluate the differences between the annual or active plans to the proposed plan. • View the number of assumptions for a given period. • Create, modify and apply events to the forecast
Is Standard PDR?	Yes
Key Data Elements	Annual Plan, Active Plan, Proposed Plan (editable if user is in Sales group), Difference and Cumulative Difference, Assumptions.
Key Calculations	The differences are a comparison between the Proposed Plan and either the Annual Plan or the Active Plan.

This workbook is used by the Sales planner to enter maintain the Sales Forecast. If the user belongs to the Sales group, then the Proposed Plan row in this report is editable. This is the primary resource used in the Task Flow called “Input Forecast – Sales”. The planner can either use this workbook with units or with their chosen currency.

The reported Annual Plan consists of ForecastDetail records where the category is either $DP_SalesVolumeTarget (SalesVolumeTarget if editing in units) or $DP_SalesValueTarget (SalesValueTarget if editing in units).



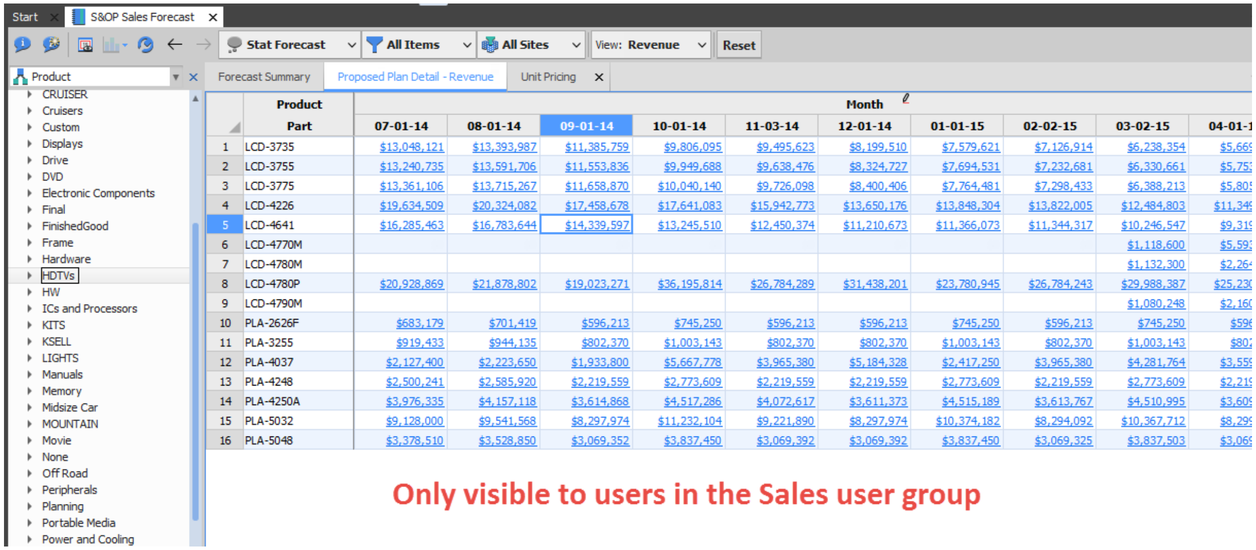
The Active Plan consists of the set of HistoricalDemandSeriesDetail records with a category of $DP_SalesForecastCategory (SalesForecast) and the last Series on the last AsOfDate.

The Proposed Plan consists of ForecastDetail records with the same category of $DP_SalesForecastCategory (SalesForecast). This is the editable row (either units or revenue) that the Sales planner will be maintaining.

There is a control on the toolbar allowing the planner to compare the “Proposed Plan” to either the “Active Plan” or the “Annual Plan” (as defined above). This comparison row has a special background color. Values in the “Difference” row are highlighted in red if they deviate from the selected plan by more than the specified tolerance threshold. This “tolerance threshold” is specified in the Data Settings dialog.

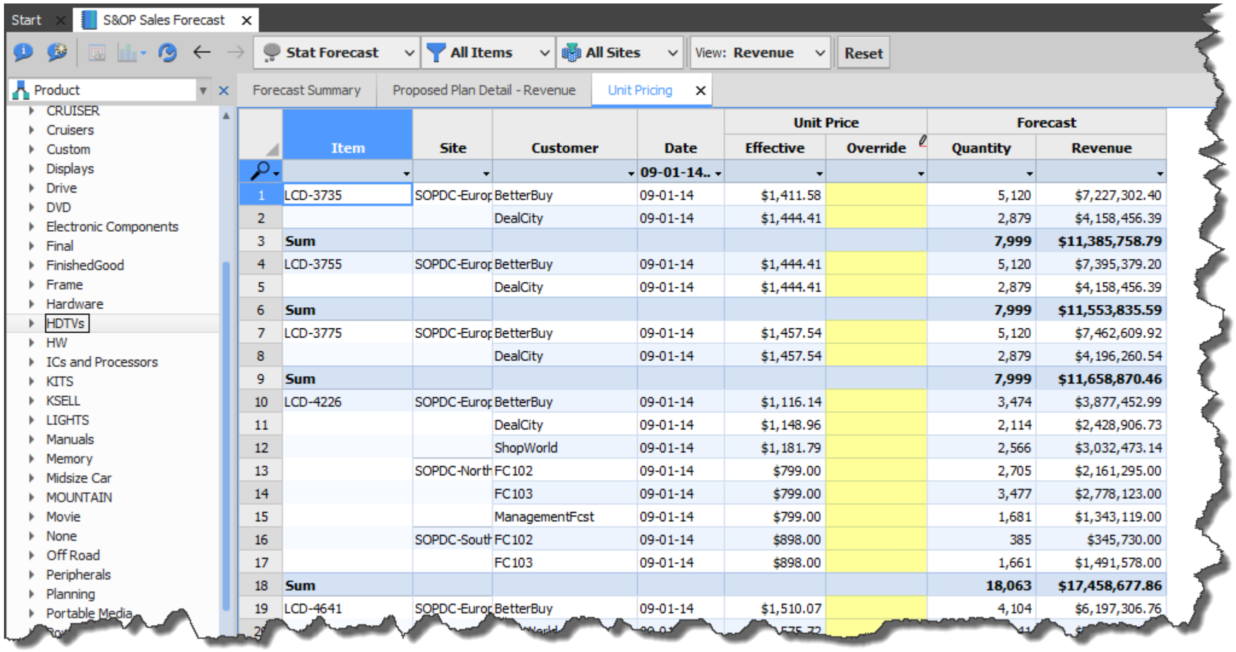
Finally, the number displayed in the “Assumptions” row indicate the total number of assumptions effective for the period. Clicking on this number will open a worksheet listing those specific assumptions.

If the Sales planner is editing in either revenue or units there will be a second worksheet available to them in the top pane that will report a bucketed detailed view of the Sales Plan broken out by one level below the selected hierarchy level.



The intent of this worksheet is to provide an easy way to edit the details more directly. This has been designed to be able to import the Sales Plan from a similarly formatted Excel worksheet. There is a version of this for units and for revenue.

If the Sales planner is working with revenue rather than units, there is also the opportunity to set the effective price for Part/Customers for specific date ranges. Clicking on any of the Proposed Plan revenue values will open the Unit Pricing worksheet.

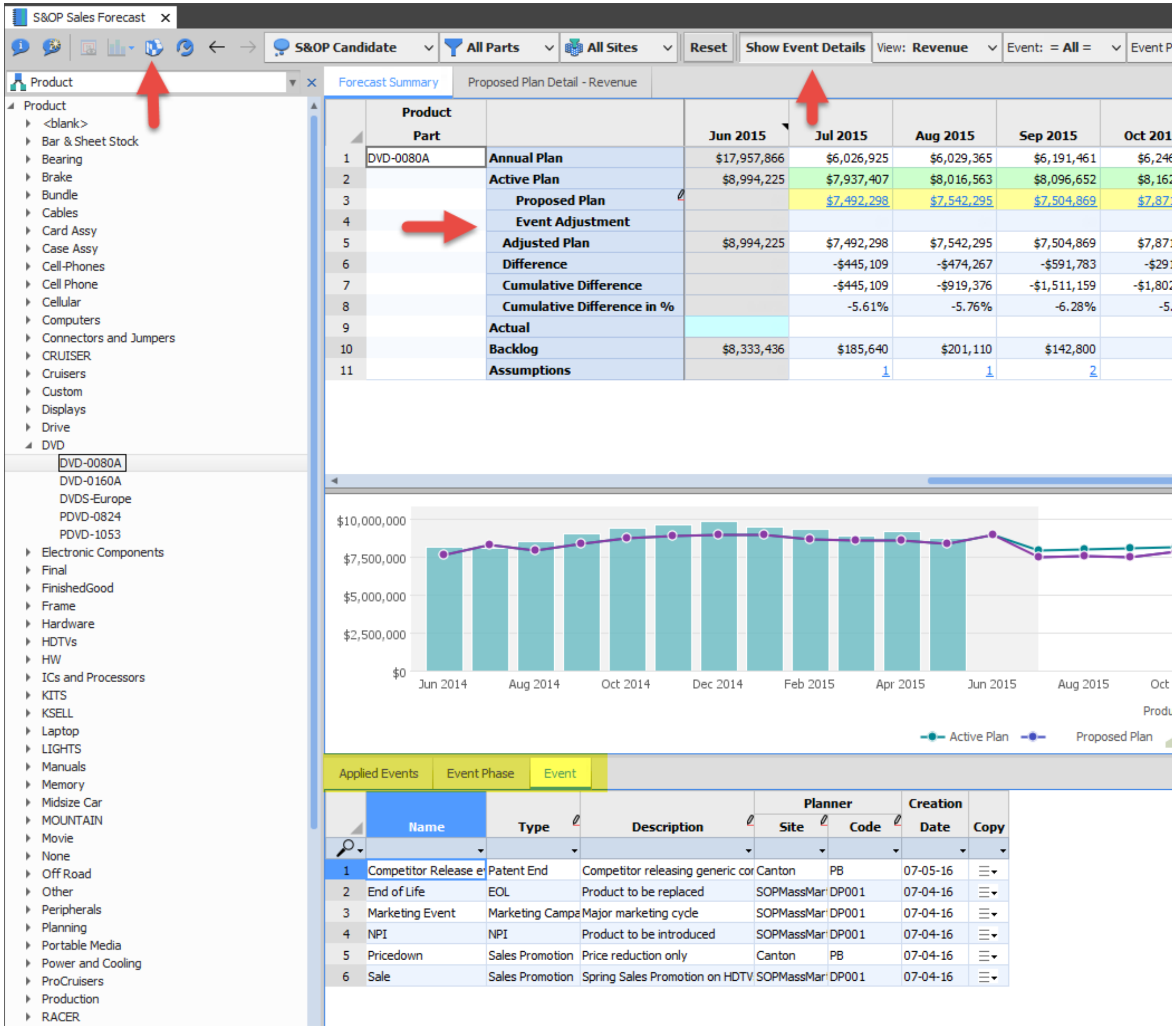


	Item	Site	Customer	Date	Unit Price Effective	Unit Price Override	Forecast Quantity	Forecast Revenue
				09-01-14..				
1	LCD-3735	SOPDC-Europ	BetterBuy	09-01-14	$1,411.58		5,120	$7,227,302.40
2			DealCity	09-01-14	$1,444.41		2,879	$4,158,456.39
3	**Sum**						**7,999**	**$11,385,758.79**
4	LCD-3755	SOPDC-Europ	BetterBuy	09-01-14	$1,444.41		5,120	$7,395,379.20
5			DealCity	09-01-14	$1,444.41		2,879	$4,158,456.39
6	**Sum**						**7,999**	**$11,553,835.59**
7	LCD-3775	SOPDC-Europ	BetterBuy	09-01-14	$1,457.54		5,120	$7,462,609.92
8			DealCity	09-01-14	$1,457.54		2,879	$4,196,260.54
9	**Sum**						**7,999**	**$11,658,870.46**
10	LCD-4226	SOPDC-Europ	BetterBuy	09-01-14	$1,116.14		3,474	$3,877,452.99
11			DealCity	09-01-14	$1,148.96		2,114	$2,428,906.73
12			ShopWorld	09-01-14	$1,181.79		2,566	$3,032,473.14
13		SOPDC-North	FC102	09-01-14	$799.00		2,705	$2,161,295.00
14			FC103	09-01-14	$799.00		3,477	$2,778,123.00
15			ManagementFcst	09-01-14	$799.00		1,681	$1,343,119.00
16		SOPDC-South	FC102	09-01-14	$898.00		385	$345,730.00
17			FC103	09-01-14	$898.00		1,661	$1,491,578.00
18	**Sum**						**18,063**	**$17,458,677.86**
19	LCD-4641	SOPDC-Europ	BetterBuy	09-01-14	$1,510.07		4,104	$6,197,306.76

The current pricing is reported in this worksheet along with the ability to set new prices for the Part/Customer starting on an effective date.

If Event Planning is enabled, new options become available in order to allow creating, modifying and applying events. Two new rows are added **Event Adjustment** and **Adjusted Plan.** There will be an option to Show Event Details, which will make three new worksheets available in a lower pane

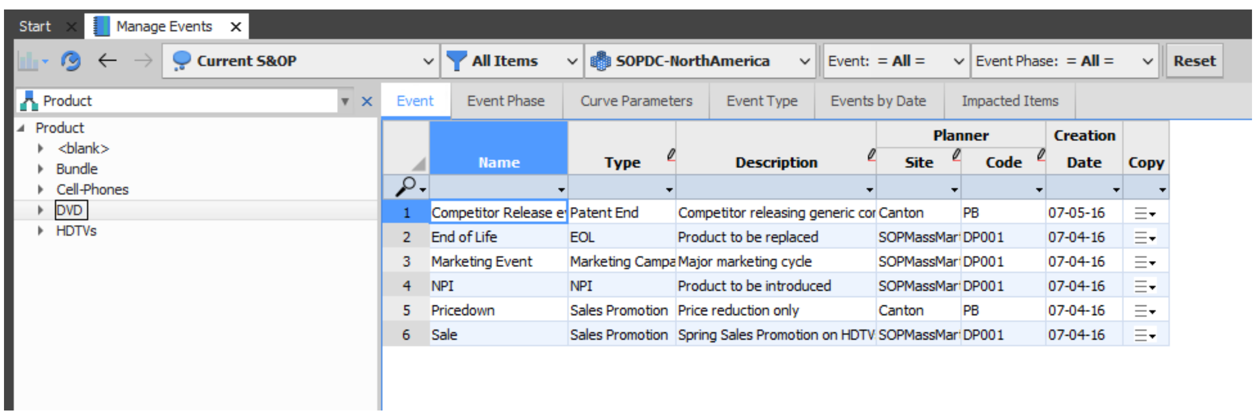
Events Phases can be applied from either the **Event Phase** or **Event worksheet**. The Apply Event icon in the toolbar will become enabled when the cursos is in any cell of these worksheets. Only members of the Sales group will be able to apply events. The **Applied Events** worksheet will list those events currently applied at the selected hierarchy level.


S&OP Sales Forecast
S&OP Candidate
All Parts
All Sites
Reset
Show Event Details
View: Revenue
Event: = All =
Product
Forecast Summary
Proposed Plan Detail - Revenue
Product Part
DVD-0080A
Jun 2015
Jul 2015
Aug 2015
Sep 2015
Annual Plan $17,957,866 $6,026,925 $6,029,365 $6,191,461
Active Plan $8,994,225 $7,937,407 $8,016,563 $8,096,652
Proposed Plan $7,492,298 $7,542,295 $7,504,869
Event Adjustment
Adjusted Plan $8,994,225 $7,492,298 $7,542,295 $7,504,869
Difference -$445,109 -$474,267 -$591,783
Cumulative Difference -$445,109 -$919,376 -$1,511,159
Cumulative Difference in % -5.61% -5.76% -6.28%
Actual
Backlog $8,333,436 $185,640 $201,110 $142,800
Assumptions 1 1 2
Bar & Sheet Stock
Bearing
Brake
Bundle
Cables
Card Assy
Case Assy
Cell-Phones
Cell Phone
Cellular
Computers
Connectors and Jumpers
CRUISER
Cruisers
Custom
Displays
Drive
DVD
DVD-0080A
DVD-0160A
DVDS-Europe
PDVD-0824
PDVD-1053
Electronic Components
Final
FinishedGood
Frame
Hardware
HDTVs
HW
ICs and Processors
KITS
KSELL
Laptop
LIGHTS
Manuals
Memory
Midsize Car
MOUNTAIN
Movie
None
Off Road
Other
Peripherals
Planning
Portable Media
Power and Cooling
ProCruisers
Production
RACER
$10,000,000
$7,500,000
$5,000,000
$2,500,000
$0
Jun 2014
Aug 2014
Oct 2014
Dec 2014
Feb 2015
Apr 2015
Jun 2015
Aug 2015
Active Plan
Proposed Plan
Applied Events
Event Phase
Event
Name
Type
Description
Planner Site
Planner Code
Creation Date
Copy
Competitor Release Patent End Competitor releasing generic Canton PB 07-05-16
End of Life EOL Product to be replaced SOPMassMar DP001 07-04-16
Marketing Event Marketing Campa Major marketing cycle SOPMassMar DP001 07-04-16
NPI NPI Product to be introduced SOPMassMar DP001 07-04-16
Pricedown Sales Promotion Price reduction only Canton PB 07-04-16
Sale Sales Promotion Spring Sales Promotion on HDTV SOPMassMar DP001 07-04-16

4.5.10. Manage Events

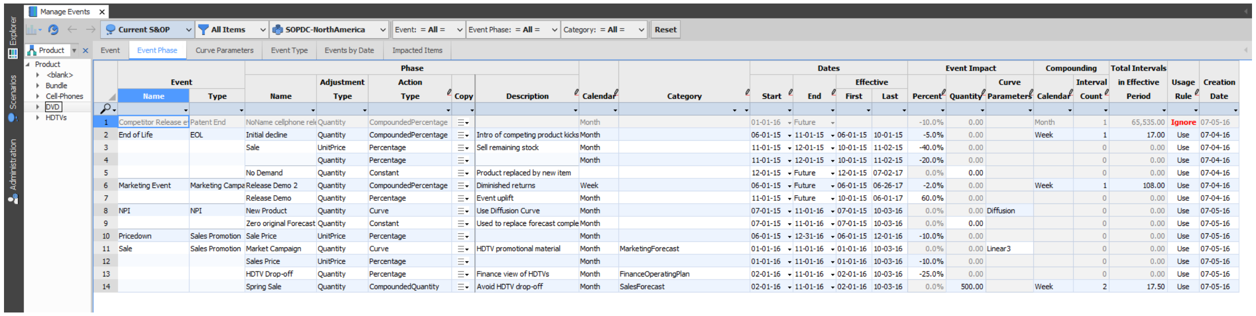
Workbook Name	Manage Events
Purpose	Use this workbook to: Review or modify Events, Event Phases, Curve Parameters and Event Types. Review Events by Date (Gantt Chart) and Impacted Items (items affected by an Event Phase) It's also used as source workbook to fetch the event details worksheets found in the Input Forecast workbooks (S&OP Sales Forecast , S&OP Marketing Forecast and S&OP Finance Operating Plan)
Is Standard PDR?	Yes
Key Data Elements	Events, Event Phases, Curve Parameters and Event Types
Key Calculations	Effective Dates to Event Phases, Impacted Items

This workbook is used by the Demand planner to enter/maintain Events. This is the primary resource to maintain Curve Parameters and Event Types and to look at Events by Date. The Event and Event Phase worksheets can be fetched by the input forecast workbooks.



The **Event** tab allows creating/modifying events. Events are used to group Event Phases (which eventually will be the ones adjusting the forecast). You can assig an event type (a way to group similar events), a description and a designated planner.

Warning: Please exercise caution if modifying this worksheet, as this tab is fetched by the input forecast workbooks.



	Event Name	Event Type	Phase Name	Phase Adjustment Type	Phase Action Type	Copy	Description	Calendar	Category	Dates Start	Dates End	Dates Effective First	Dates Effective Last	Event Impact Percent	Event Impact Quantity	Event Impact Curve Parameters	Compounding Calendar	Compounding Interval Count	Total Intervals in Effective Period	Usage Rule	Creation Date
1	Competitor Release e	Patent End	NoName cellphone rele	Quantity	CompoundedPercentage			Month		01-01-16	Future			-10.0%	0.00		Month	1	65,535.00	Ignore	07-05-16
2	End of Life	EOL	Initial decline	Quantity	CompoundedPercentage		Intro of competing product kicks	Month		06-01-15	11-01-15	06-01-15	10-01-15	-5.0%	0.00		Week	1	17.00	Use	07-04-16
3			Sale	UnitPrice	Percentage		Sell remaining stock	Month		11-01-15	12-01-15	10-01-15	11-02-15	-40.0%	0.00			0	0.00	Use	07-04-16
4				Quantity	Percentage			Month		11-01-15	12-01-15	10-01-15	11-02-15	-20.0%	0.00			0	0.00	Use	07-04-16
5			No Demand	Quantity	Constant		Product replaced by new item			12-01-15	Future	12-01-15	07-02-17	0.0%	0.00			0	0.00	Use	07-04-16
6	Marketing Event	Marketing Campa	Release Demo 2	Quantity	CompoundedPercentage		Diminished returns	Week		06-01-15	Future	06-01-15	06-26-17	-2.0%	0.00		Week	1	108.00	Use	07-04-16
7			Release Demo	Quantity	Percentage		Event uplift	Month		11-01-15	Future	10-01-15	06-01-17	60.0%	0.00			0	0.00	Use	07-04-16
8	NPI	NPI	New Product	Quantity	Curve		Use Diffusion Curve	Month		07-01-15	11-01-16	07-01-15	10-03-16	0.0%	0.00	Diffusion		0	0.00	Use	07-05-16
9			Zero original Forecast	Quantity	Constant		Used to replace forecast comple	Month		07-01-15	11-01-16	07-01-15	10-03-16	0.0%	0.00			0	0.00	Use	07-05-16
10	Pricedown	Sales Promotion	Sale Price	UnitPrice	Percentage			Month		06-01-15	12-31-16	06-01-15	12-01-16	-10.0%	0.00			0	0.00	Use	07-05-16
11	Sale	Sales Promotion	Market Campaign	Quantity	Curve		HDTV promotional material	Month	MarketingForecast	01-01-16	11-01-16	01-01-16	10-03-16	0.0%	0.00	Linear3		0	0.00	Use	07-05-16
12			Sales Price	UnitPrice	Percentage			Month		01-01-16	11-01-16	01-01-16	10-03-16	-10.0%	0.00			0	0.00	Use	07-05-16
13			HDTV Drop-off	Quantity	Percentage		Finance view of HDTVs	Month	FinanceOperatingPlan	02-01-16	11-01-16	02-01-16	10-03-16	-25.0%	0.00			0	0.00	Use	07-05-16
14			Spring Sale	Quantity	CompoundedQuantity		Avoid HDTV drop-off	Month	SalesForecast	02-01-16	11-01-16	02-01-16	10-03-16	0.0%	500.00		Week	2	17.50	Use	07-05-16

The **Event Phase** tab allows creating/modifying event phases. Phases are the cornerstone of Event Planning, as these define how the forecast will eventually be adjusted. Adjustment types allows us to select the target of this phase, and there are two types: **Quantity,** for adjusting forecast quantities and **Unitprice,** for adjusting forecasted unit prices.

Action Types allow us to define *how* the adjustment is going to take place. Each of the action types have certain relevant parameters, those which are not relevant will appear greyed out. There are currently six Action types:

Quantity (default): Forecasts (Quantities or Unit Prices) will be adjusted by adding a fix amount (Event Impact Quantity) to them in the range specified by FirstEffectiveDate and LastEffectiveDate.

CompoundedQuantity: Forecasts will be adjusted by the compounded increment of Event Impact Quantity. Starting from FirstEffectiveDate and in the increments of a compounding bucket (defined by Compounding Interval Count and Calendar), we will add a multiple of EventPhase.Quantity to every forecast quantity. The first compounding bucket will use a single multiple of `Quantity`, the 2nd bucket will use `2*EventPhase.Quantity`, ... , the n-th bucket will use `n* Quantity`, until LastEffectiveDate is reached.

Percentage: Forecasts will be adjusted by the percentages defined in Event Impact Percent. The percentage of every forecast quantity will be added directly to the forecast quantity in the range specified by FirstEffectiveDate and LastEffectiveDate.

CompoundedPercentage: Forecasts will be adjusted by the compounded percentage increment using defined in Event Impact Percent. Starting from FirstEffectiveDate and in the increments of a compounding bucket (defined by Compounding Interval Count and Calendar), we will multiply the forecast quantity by `(1+ Percent/100) * i` where i is the sequence number of a compounding bucket. In the first compounding bucket we will multiply forecast quantities by `(1+ Percent/100) * 1,` in the 2nd bucket will multiply by `(1+ Percent/100) * 2`, ... , the n-th bucket we will multiply by `(1+ Percent/100) *` n, until we reach LastEffectiveDate

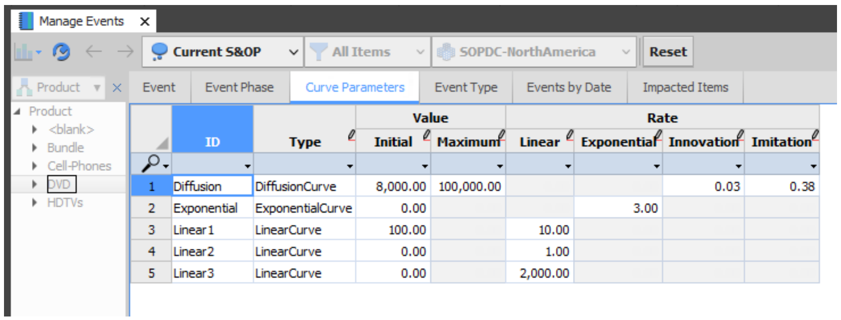
Constant: Forecasts will be set the constant value specified in Event Impact Quantity in the range between FirstEffectiveDate and LastEffectiveDate (both inclusive).

Curve: Forecasts will be adjusted by <u>adding</u> the generated curve values (points) to the existing forecast quantities on the time markers of the effective calendar. A curve needs to be selected using the Curve Parameters field.

Category field is used to limit an event phase to a particular forecast category. In case an event phase is expected to be applied to more than one, this field needs to be left blank.

A **Usage Rule** is provided with optional values of “Use” or “Ignore”, this provides an option to disable a phase without the need to delete it

Warning: Please exercise caution if modifying this worksheet, as this tab is fetched by the input forecast workbooks.



			Value		Rate			
	ID	Type	Initial	Maximum	Linear	Exponential	Innovation	Imitation
1	Diffusion	DiffusionCurve	8,000.00	100,000.00			0.03	0.38
2	Exponential	ExponentialCurve	0.00			3.00		
3	Linear1	LinearCurve	100.00		10.00			
4	Linear2	LinearCurve	0.00		1.00			
5	Linear3	LinearCurve	0.00		2,000.00			

The **Curve Paramentes** tab allows for managing curves to be applied to event phases. The parameters of the curve to be generated are provided here. We have three types: LinearCurve, ExponentialCurve and DiffusionCurve.

For Linear curves, the slope (Linear Rate) is required. The adjustment will take the form of `Initial Value + LinearRate*time`

For Exponential curves the the base of the exponent (Exponential Rate) is required. The adjustment will take the form of `Initial Value +` $\texttt{ExponentialRate}^{\texttt{time}}$

For Diffusion curves, a Bass Diffusion model is used. The Diffusion Model was proposed by Frank Bass in 1969. It models the simplified behavior of consumers introduced to a new product. A diffusion curve depicts the rate of the new product's adoption in the consumers' population. Consumers are divided into 2 groups: *innovators* and *imitators*. Innovators adopt (buy) new products right away. Imitators need someone else to buy a new product beforehand, in order to buy it themselves. Therefore, there are three main parameters in this curve:

Innovation Rate: (let's name it p) The proportion of innovators in the consumers' population (the ones buying new products for the first time without depending on the prior adoptions of other consumers), default 0. This is critical parameter of the diffusion curve (if not specified or equals zero, the resulting diffusion curve will be flat and constant = InitialValue). Valid range of values is [0.0, 1.0]. In the case user specifies values outside the range, they are forced by the calculation to be at least 0.0 and at most 1.0.

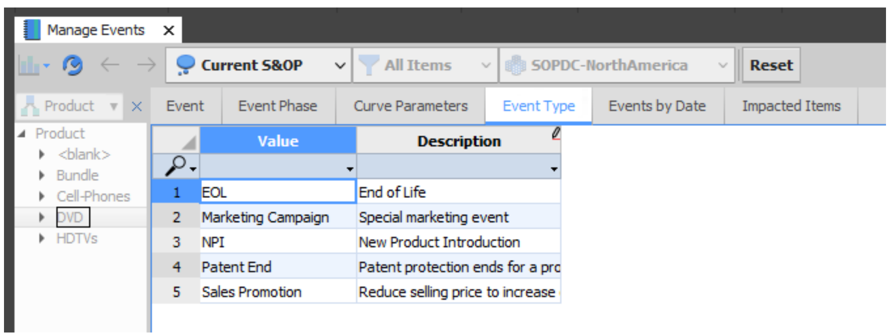
Imitation Rate: (let's name it q) the proportion of imitators in the consumers' population (the ones buying new products for the first time only if other consumers already bought the product), default 0. Valid range of values is [0.0, 1.0]. In the case user specifies values outside the range, they are forced by the calculation to be at least 0.0 and at most 1.0.

Population Size: (let's name it m) - the potential market size (or maximum possible demand for a product). This is represented in the data model by the range between **Initial** and **Maximum Values** If population size is 0, the curve is a flat constant equal to the Initial Value.

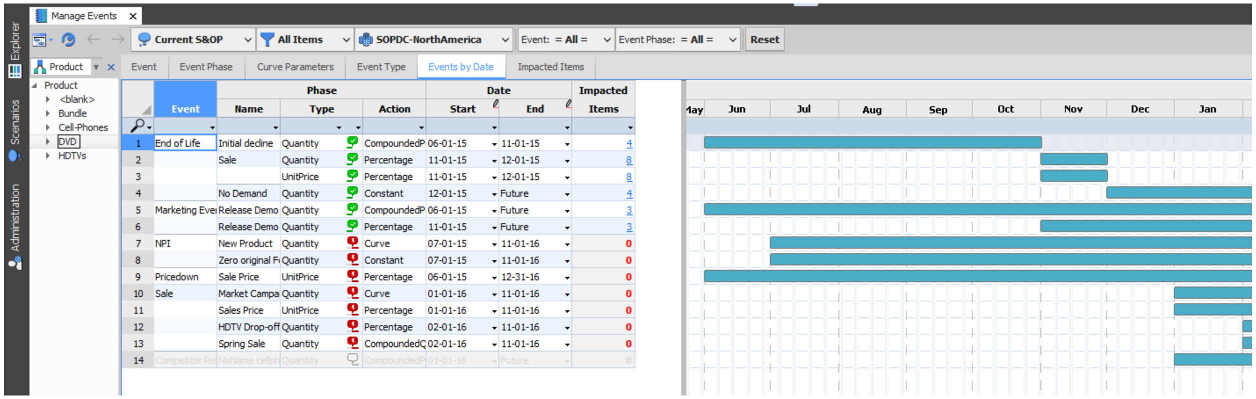
The actual adjustment values will be shifted up by Initial Value as follow: `adjustment(t) = Initial Value+ S(p,q,m,t)`, where `t` is a discrete time axis.

$$S(p,q,m,t) = m \cdot \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p} \cdot e^{-(p+q)t}}$$

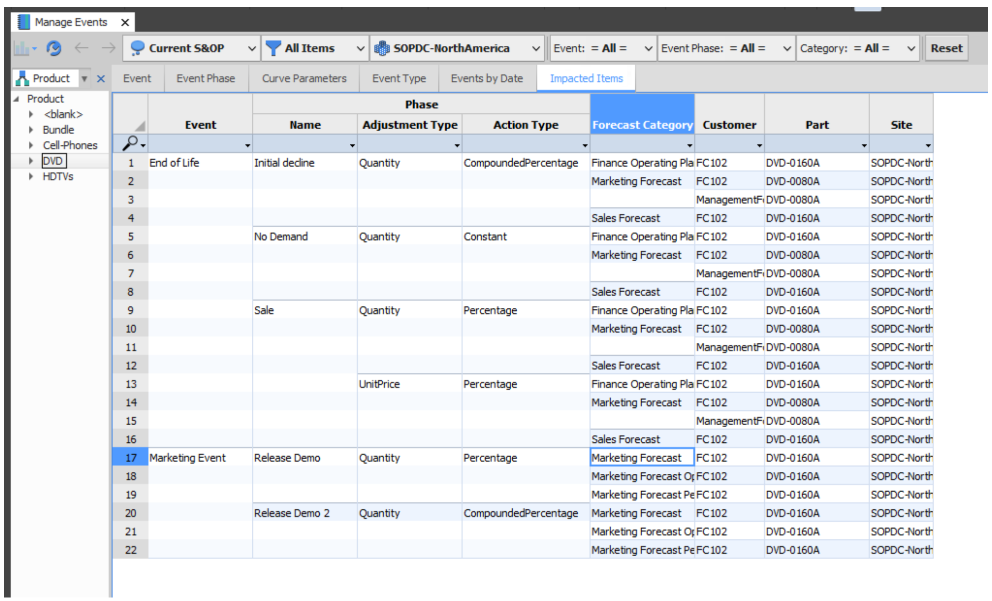
The intent of this worksheet is to provide an easy way to edit the details more directly. This has been designed to be able to import the Sales Plan from a similarly formatted Excel worksheet. There is a version of this for units and for revenue.



The **Event Type** tab allows managing event types. Event types provide a way to group similar events for reference.



The **Events by Date** tab allows the user to review existing events in a Gantt chart format. Start/End dates can be modified here, either at the columns or by dragging the chart. It provides visual cues on whether the event phases are active (applied and/or not applied), or inactive. Also, it's possible to see how many items each header is impacted by a phase. Please note this report will be dependent on the hierarchy used.



The **Impacted Items** tab allows the user to review existing event phase headers EPH (a combination of an event phase and a forecast header). This report also provides the option to delete EPHs, therefore it can be used for fine tuning or resetting event applications.

Note: Creating an event phase does not modify any forecast by itself. In order for an event phase to have an impact, it needs to be **applied** to a forecast. This can be accomplished from the **Input Forecast workbooks**